RQF LEVEL 3



NITWI301

# NETWORK AND INTERNET TECHNOLOGIES

# Wireless Network Indoor Installation

## TRAINEE'S MANUAL

*October, 2024*

# WIRELESS NETWORK INDOOR INSTALLATION

KOICA
Korea International Cooperation Agency

TQUM
TVET Quality Management Project

2024

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## COORDINATION TEAM

RWAMASIRABO Aimable

MARIA Bernadette M. Ramos

MUTIJIMA Asher Emmanuel

## PRODUCTION TEAM

### Authoring and Review

TUYISABE Annoncee

Karera Pièrre

### Validation

HABIMANA Sixbert

MANISHIMWE Yves

### Conception, Adaptation and Editorial works

HATEGEKIMANA Olivier

GANZA Jean Francois Regis

HARELIMANA Wilson

NZABIRINDA Aimable

DUKUZIMANA Therese

NIYONKURU Sylvestre

NYINAWUMUNTU Gaudence

### Formatting, Graphics, Illustrations, and infographics

YEONWOO Choe

SUA Lim

SAEM Lee

SOYEON Kim

WONYEONG Jeong

KAYUMBA Fiston

### Financial and Technical support

KOICA through TQUM Project

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# ACRONYMS

**ACL:** Access Control List

**AP:** Access Point

**Blueprint:** a design of a network

**CBT/A**: Competent Based Training/Assessment

**IBSS:** Independent Basic Service Set

**IDE:** Integrated Environment Development

**IEEE:** Institute of Electrical Electronics Engineering

**IOT:** Internet of Things

**MU-MIMO:** Multi-User Multiple Input Multiple Output

**NIC:** Network Interface Card

**OFDMA:** Orthogonal Frequency-Division Multiple Access

**POE:** Power over Internet

**PPE:** Personal Protective Equipment

**QoS:** Quality of Service

**RFID:** Radio Frequency Identification

**RQF:** Rwanda Qualification Framework

**RTB**: Rwanda TVET Board

**TOS:** Table of Specification

**TQUM Project:** TVET Quality Management Project

# INTRODUCTION

This trainee's manual includes all the knowledge and skills required in Networking and Internet Technologies specifically for the module of **"Wireless Network Indoor Installation".** Trainees enrolled in this module will engage in practical activities designed to develop and enhance their competencies. The development of this training manual followed the Competency-Based Training and Assessment (CBT/A) approach, offering ample practical opportunities that mirror real-life situations.

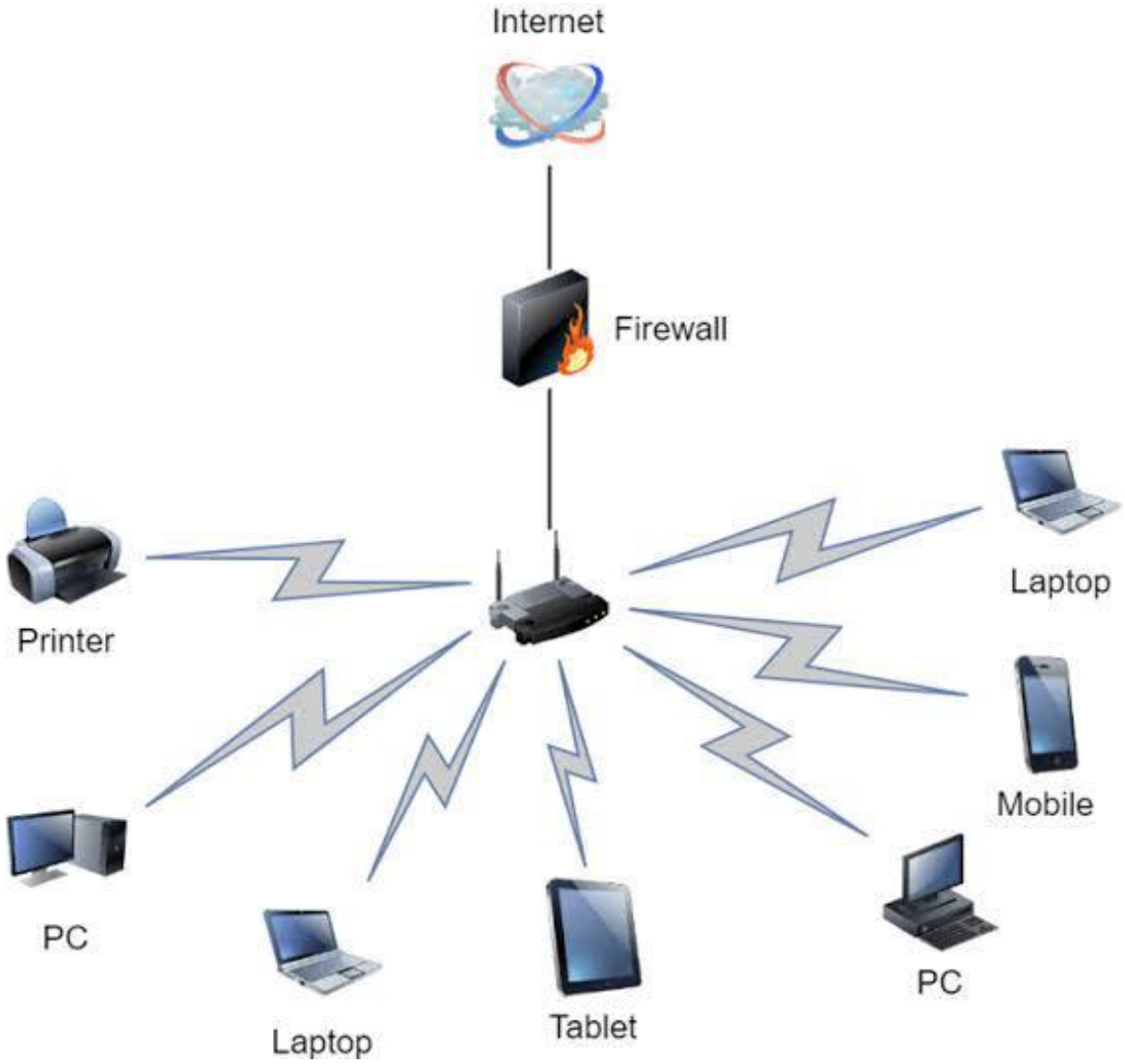
The trainee's manual is organized into Learning Outcomes, which is broken down into indicative content that includes both theoretical and practical activities. It provides detailed information on the key competencies required for each learning outcome, along with the objectives to be achieved.

As a trainee, you will start by addressing questions related to the activities, which are designed to foster critical thinking and guide you towards practical applications in the labor market. The manual also provides essential information, including learning hours, required materials, and key tasks to complete throughout the learning process.

All activities included in this training manual are designed to facilitate both individual and group work. After completing the activities, you will conduct a formative assessment, referred to as the end learning outcome assessment. Ensure that you thoroughly review the key readings and the 'Points to Remember' section.

# MODULE CODE AND TITLE: NITWI301 WIRELESS NETWORK INDOOR INSTALLATION

| |
|---|
| **Learning Outcome 1: Apply wireless network Concepts** |
| **Learning Outcome 2: Plan wireless network indoor installation** |
| **Learning Outcome 3: Deploy wireless network indoor** |
| **Learning Outcome 4: Maintain wireless network indoor** |


Internet
Firewall
Printer
Laptop
Mobile
PC
Laptop
Tablet
PC

| Indicative contents |
|---|
| **1.1 Introduction to Wireless network** |
| **1.2 Identification of wireless network Types** |
| **1.3 Identification of wireless network components** |
| **1.4 Application of Wireless technologies** |
| **1.5 Classification of Wireless network standards** |
| **1.6 Operation mode of Wireless network** |

## Key Competencies for Learning Outcome 1: Introduction to Wireless network

| Knowledge | Skills | Attitudes |
|---|---|---|
| • Description of wireless applications and their benefits<br>• Identification of wireless network indoor<br>• Description of wireless network types<br>• Identification of wireless network components<br>• Distinction of wireless network standards<br>• Differentiation of wireless network operations mode | • N/A | • Having Teamwork<br>• Being Problem Solver<br>• Being a Critical thinker while classifying wireless network standard |



**Duration: 15 hrs**

**Learning outcome 1 objectives**:



By the end of the learning outcome, the trainees will be able to:

1. Describe properly Wireless network application and its benefits based on network structure
2. Identify correctly wireless network types based on network range
3. Identify clearly wireless network components according to their functionalities
4. Apply correctly Wireless technologies based on network requirements
5. Classify clearly Wireless network standards according to network requirements
6. Apply correctly operation mode of Wireless network in accordance with network standard.



**Resources**

| Equipment | Tools | Materials |
|---|---|---|
| • Router<br>• Access point<br>• Wireless antenna<br>• Wireless NIC<br>• Smart phone with NFC<br>• Bluetooth Radio receiver<br>• Bluetooth woofer - Remote control | • Software tools (packet tracer, GNS3, E-draw Max) | • Internet bundles |

## Indicative content 1.1: Introduction to Wireless Network

**Duration: 2 hr**

**Theoretical Activity 1.1.1: Introduction to wireless network**

**Tasks:**

1: In small groups, you are requested to answer the following questions related to the introduction of wireless network:

i. Define wireless network indoor?
ii. Identify the area of applications of wireless networks indoors?
iii. Describe the benefits of wireless network indoor?

2: Provide the answer for the asked questions and write them on papers.

3: Present the findings/answers to the whole class

4: For more clarification, read the key readings 1.1.1.

5: In addition, ask questions where necessary.

**Key readings 1.1.1.: Introduction to wireless network**

1. A wireless network indoor refers to a network setup that enables devices within an enclosed space, such as a home, office, or other building, to communicate with each other and access the internet without the need for physical cables. Instead of using wired connections, devices connect via wireless signals, typically through technologies like Wi-Fi, Bluetooth, or other radio frequency (RF) methods.

- Indoor wireless networks are often managed by a central device, such as a wireless router or access point, that broadcasts the wireless signal to which devices like smartphones, laptops, tablets, and smart home gadgets can connect. This type of network is designed to provide flexible and convenient access to network resources and the internet throughout an indoor environment.

  Wireless networks indoors have a wide range of applications, including:

1. **Home Networking**: Connecting devices such as smartphones, laptops, smart TVs, gaming consoles, and home automation systems (e.g., smart thermostats, security cameras).

2. **Office and Business Environments**: Facilitating communication, collaboration, and access to resources like printers, servers, and cloud services. Wireless networks enable flexible working arrangements, including hot-desking and conference room setups.
3. **Retail and Hospitality**: Providing customers with Wi-Fi access, enabling mobile point-of-sale (POS) systems, and supporting inventory management systems.
4. **Healthcare**: Connecting medical devices, patient monitoring systems, and electronic health records (EHR) systems. Wireless networks are crucial for mobile access to patient data and telemedicine.
5. **Education**: Supporting digital learning tools, online resources, and enabling seamless access to learning management systems (LMS) across classrooms, libraries, and other campus areas.
6. **Entertainment and Media**: Streaming services, gaming, and virtual reality (VR) experiences often rely on stable wireless connections for uninterrupted usage.
7. **Industrial and Warehouse Settings**: Supporting automation, inventory tracking, and the operation of wireless sensors and robotics within indoor environments.
8. **Public Spaces**: Airports, shopping malls, and convention centers use wireless networks to offer internet access, digital signage, and enhanced visitor experiences.

**1. The benefits of wireless network indoor are:**

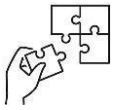
- **Increased Mobility:** Wireless networks allow mobile users to access real-time information so they can roam around your company's space without getting disconnected from the network. This increases teamwork and productivity company-wide that is not possible with traditional networks.
- **Installation Speed and Simplicity:** Installing a wireless network system reduces cables, which are cumbersome to setup and can impose a safety risk, should employees' trip on them. It can also be installed quickly and easily, when compared to a traditional network.
- **Wider Reach of the Network:** The wireless network can be extended to places in your organization that are not accessible for wires and cables.
- **More Flexibility:** Should your network change in the future, you can easily update the wireless network to meet new configurations.
- **Reduced Cost of Ownership over Time:** Wireless networking may carry a slightly higher initial investment, but the overall expenses over time are lower. It also may have a longer lifecycle than a traditionally connected network.
- **Increased Scalability:** Wireless systems can be specifically configured to meet the needs of specific applications. These can be easily changed and scaled depending on your organization's needs.
- Easier to provide connectivity in areas that are difficult to lay cable.

- Portable or semi-permanent buildings can be connected using WLAN.



**Points to Remember**

- Wireless network indoor: is a type of computer network that uses wireless communication technology to connect various devices and allow them to communicate and share data without the need for physical cables inside a particular interior space.
- Applications of wireless network: Wireless networks can be applied in different areas such as health, education, business, home, Retail, Industrial and Warehouse Settings, public spaces,
- The benefits of wireless network are:
  - Increase Mobility
  - Easy to install
  - Wider Reach of the Network
  - More Flexibility.
  - Increased Scalability



**Application of learning 1.1.**

A TERABA school has traditionally relied on a wired network for all its computing needs, including internet access, file sharing, and communication between staff and students. However, with the increasing use of laptops, tablets, and mobile devices by students and staff, the school is now facing with different network limitations such as the lack of mobility, difficulty in scaling, and the challenges of connecting modern devices like tablets and smartphones. The staff want to know what should be done so that teachers and student will increasingly use mobile devices for educational activities and be able to move freely around the school with their devices without being disconnected from the network.

As the student who knows much about the network you are needed to do the following:

- Suggest which network that can be good for school so that the stated problem will be solved.
- Highlight what the school will benefit from the new network setup
- Mention where that network can be applied apart from the school

## Indicative content 1.2: Identification of Wireless Network Types

**Duration: 3**

### Theoretical Activity 1.2.1: Identifying wireless network Types

**Tasks:**

1. In small groups, you are requested to answer the following questions related to the identification of wireless network Types:
   1. Describe the wireless network types.
   2. What are the typical factors that distinguish different wireless networks?
2. Provide answers and write them on flipchart/paper
3. Discuss with others on provided answers and choose the correct answers.
4. Ask any clarification to the trainer
5. Read the types of wireless networks in the key reading 1.2.1 in the trainee manual

**Key readings 1.2.1.: Identifying wireless network Types**

- Wireless networks are indeed categorized based on their application area and signal range, allowing for a better understanding of how each type is used.

- A Wireless Body Area Network (WBAN) is a network of wearable computing devices that communicate with each other wirelessly. These devices, which can be placed on the body, embedded in the body, or positioned near the body, monitor various physiological parameters and provide real-time data for health and wellness applications

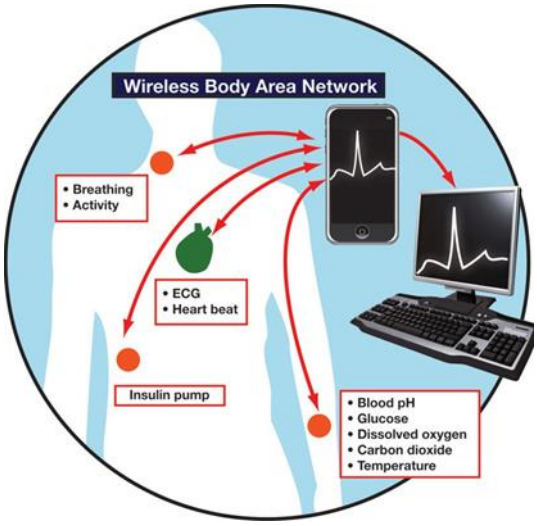
  **Key Components of WBAN:**

  - ✓ Sensors: These are the primary devices in a WBAN, used to monitor physiological data such as heart rate, body temperature, blood pressure, glucose levels, and physical activity. They can be wearable devices like smartwatches or fitness bands, or implantable devices like pacemakers and glucose monitors.
  - ✓ Communication Modules: These enable wireless data transmission between the sensors and other devices like smartphones, tablets, or dedicated healthcare monitoring systems. Common wireless communication technologies used in WBANs include Bluetooth, Zigbee, and Wi-Fi.

- ✓ Data Processor: This component processes the data collected by the sensors. It can be a part of the sensor device or a separate device such as a smartphone or a computer that aggregates and analyses the data.
- ✓ Power Supply: Since WBAN devices are typically small and portable, they require efficient power sources. This can include rechargeable batteries, energy harvesting techniques, or other low-power technologies.

**Some application of Wireless Body Area Network:**

- ✓ Healthcare Monitoring: WBANs are extensively used for continuous monitoring of patients' vital signs, enabling early detection of potential health issues and providing data for remote diagnosis and treatment. They are particularly useful for chronic disease management, post-operative care, and elderly care.
- ✓ Sports and Fitness: Athletes and fitness enthusiasts use WBANs to monitor their physical performance, track workout progress, and optimize their training routines. Devices can measure parameters such as heart rate, oxygen levels, and muscle activity.
- ✓ Rehabilitation: WBANs assist in rehabilitation by monitoring the progress of patients recovering from injuries or surgeries. They provide real-time feedback to both patients and healthcare providers, facilitating better and more personalized treatment plans.
- ✓ Military and Emergency Response: WBANs can be used to monitor the physiological status of soldiers and first responders, ensuring their safety and readiness. They can provide critical information during missions or emergencies.




- Wireless Personal-Area Networks (WPAN)

  **Signal Range**: Typically, within a few meters (up to 10 meters).

  **Applications**: WPANs are designed for very short-range communication between devices that are close to a user, such as Bluetooth connections between

smartphones, wireless headsets, and fitness trackers. Other examples include infrared (IR) communication and Zigbee for smart home devices.

**Examples**: Bluetooth, Zigbee, Infrared (IR).




- Wireless Local-Area Networks (WLANs)
- **Signal Range**: Typically covers a building or a group of buildings (up to 100 meters indoors, more outdoors).
- **Applications**: WLANs provide wireless network connectivity within a localized area like a home, office, or campus. The most common use of WLANs is for internet access via Wi-Fi routers and access points, allowing multiple devices to connect to the internet and to each other within the network.
- **Examples**: Wi-Fi (IEEE 802.11 standards).




- **Wireless Metropolitan-Area Networks (WMANs)**

  **Signal Range**: Covers a city or large metropolitan area.

  **Applications**: WMANs are designed to connect multiple WLANs within a city or metropolitan area. They are often used to provide broadband wireless access across large areas, such as connecting various buildings in a city to the internet or providing municipal wireless networks. They can also be used for connecting multiple branch offices of a business within a city.

**Examples**: WiMAX (Worldwide Interoperability for Microwave Access), municipal Wi-Fi networks.

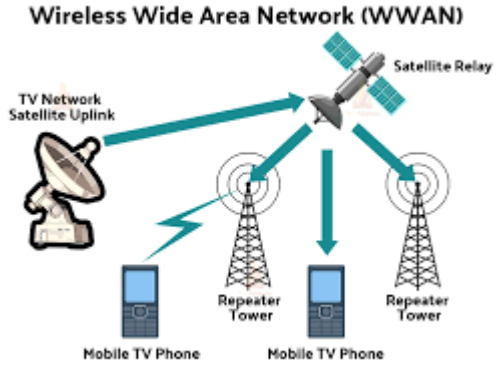



- Wireless Wide-Area Networks (WWANs)

  **Signal Range**: Covers large geographical areas, ranging from a country to entire continents.

  **Applications**: WWANs are used for long-distance communication across wide areas. They provide internet access and communication services over cellular networks or satellite connections. WWANs are essential for mobile phone networks, enabling voice, data, and text services across regions and countries. They also support IoT devices that require long-range connectivity.

  **Examples**: Cellular networks (3G, 4G, 5G), satellite communication




2. Wireless networks can indeed be differentiated based on several factors, each impacting their performance, application, and suitability for different environments. Here's how they differ across the specified criteria:

- **Distance Coverage**

  **WBAN**: Very short range, typically a few centimeters to a few meters, focused around the human body.

  **WPAN**: Short range, up to 10 meters, ideal for personal devices.

  **WLAN**: Medium range, usually up to 100 meters indoors (can be extended with outdoor setups).

  **WMAN**: Large range, covering several kilometers, suitable for cities or large metropolitan areas.

**WWAN**: Extremely large range, covering regions, countries, or even continents, depending on the technology used (e.g., cellular, satellite).

- **Types of Equipment They Use**

  **WBAN**: Wearable sensors, implantable medical devices, body-worn transmitters, smartphones, and medical monitoring devices.

  **WPAN**: Bluetooth-enabled devices, Zigbee modules, USB dongles, personal hubs, and infrared devices.

  **WLAN**: Wireless routers, access points, network cards, antennas, and repeaters.

  **WMAN**: Base stations, WiMAX antennas, repeaters, and modems.

  **WWAN**: Cellular towers, satellite dishes, modems, and mobile devices (smartphones, tablets).

- **Frequency Band**

  **WBAN**: Typically operates in the 2.4 GHz ISM band, or specific medical frequency bands (e.g., MICS at 402–405 MHz).

  **WPAN**: Uses 2.4 GHz or sub-GHz bands (e.g., 868 MHz in Europe, 915 MHz in North America for Zigbee).

  **WLAN**: Primarily uses 2.4 GHz and 5 GHz bands (with Wi-Fi 6E also using 6 GHz).

  **WMAN**: Often operates in licensed bands, such as 2.3 GHz, 2.5 GHz, or 3.5 GHz (e.g., WiMAX).

  **WWAN**: Utilizes a wide range of frequencies, including 700 MHz to 2.6 GHz for cellular networks, and higher frequencies for satellite communications.

- **Network Topology**

  **WBAN**: Typically star or point-to-point, with a central device (e.g., a smartphone) collecting data from sensors.

  **WPAN**: Star or peer-to-peer, with devices connecting directly to each other or through a central hub.

  **WLAN**: Usually star topology, with all devices connecting to a central access point.

  **WMAN**: Point-to-multipoint or mesh topology, connecting multiple buildings or areas to a central network.

  **WWAN**: Hierarchical or mesh topology, with a network of base stations or satellites providing widespread coverage.

- **Use Case and Application**

  **WBAN**: Health monitoring, fitness tracking, medical diagnostics, wearable technology.

  **WPAN**: Personal device communication, short-range data transfer, home automation.

  **WLAN**: Home and office networking, internet access, corporate networks, educational institutions.

  **WMAN**: City-wide internet access, connecting multiple LANs, public Wi-Fi, and municipal services.

**WWAN**: Mobile communication, rural broadband, global IoT connectivity, and satellite-based services.

- **Scalability and Capacity**

  **WBAN**: Limited to a small number of devices, usually focused on one individual.

  **WPAN**: Scalable for personal or small office setups, with limited device capacity.

  **WLAN**: Can support many devices within a localized area, scalable with additional access points.

  **WMAN**: Can support thousands of users across a city, scalable with more base stations.

  **WWAN**: Highly scalable, designed to support millions of users and devices across large regions.

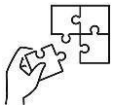
- **Data Transfer Speed**
  - ✓ **WBAN**: Typically low to moderate speeds, sufficient for transmitting physiological data (e.g., 250 kbps to 1 Mbps).
  - ✓ **WPAN**: Moderate speeds, depending on the technology (e.g., Bluetooth 5.0 up to 2 Mbps, Zigbee up to 250 kbps).
  - ✓ **WLAN**: High speeds, varying with Wi-Fi standards (e.g., Wi-Fi 5 up to 3.5 Gbps, Wi-Fi 6 up to 9.6 Gbps).
  - ✓ **WMAN**: Moderate to high speeds, depending on the technology (e.g., WiMAX up to 1 Gbps for fixed stations).**\11**
  - ✓ **WAN**: Varies widely; cellular networks (e.g., 4G LTE up to 100 Mbps, 5G up to 10 Gbps), satellite networks typically slower.



**Points to Remember**

- Wireless networks can be classified into four specific groups according to the area of application and the signal range:
  - Wireless Body Area Network (WBAN):
  - Wireless Personal-Area Networks (WPAN),
  - Wireless Local-Area Networks (WLANs),
  - Wireless Metropolitan-Area Networks (WMAN),
  - And Wireless Wide-Area Networks (WWANs).
- All wireless networks differ from:
  - distance coverage
  - types of equipment they use
  - Frequency Band
  - Network Topology
  - Use Case and Application
  - Scalability and Capacity

- Data Transfer Speed.



**Application of learning 1.2.**

Greenfield High School has been facing significant challenges with its current network infrastructure, which has become increasingly outdated as the number of students, staff, and connected devices has grown. This wired network is no longer able to support the modern demands of digital learning, such as online classes, collaborative projects, and the use of personal devices like laptops, tablets, and smartphones. The school administration has recognized the need to upgrade to a wireless network to support the growing demand for digital tools and enhance the overall learning experience. However, they are unsure which type of wireless network would best suit the school's needs. The administration has asked you to help them in understanding the different wireless networks that are mainly used and provide a recommendation on which will be the best for this school

The staffs to be connected to the network wirelessly in the building are dispatched as follow:

- Computer room: they will use network for research, online classes and collaborative projects
- Office for staff: they will use network for communication, file sharing, sensor system and online printing

## Indicative content 1.3: Identification of Wireless Network Components

**Duration:**

**Theoretical Activity 1.3.1: Identifying wireless network components**

**Tasks:**

1. In small groups, you are requested to answer the following questions related to the identification of wireless network components
   i. List wireless network components
   ii. What is the role of a Wireless Network Interface Card (NIC) in a wireless network, and how does it differ from a traditional wired network card?
   iii. How does a wireless router/home router contribute to the functionality of a wireless network, and what are its primary components and features?
   iv. What are the functions of an access point and a wireless controller in managing wireless connections within a network?
   v. How do wireless antennas and wireless bridges enhance the coverage and connectivity of a wireless network, and in what scenarios would each be utilized?
1. Write your answers on flipchart /paper
2. Discuss with other trainees on provided answers and choose the correct answers
3. Listen to trainer's clarifications and ask questions if necessary
4. Do the application of learning 1.3 and read the key readings 1.3.1

**Key readings 1.3.1.: Identifying wireless network components**

1. List of common wireless network components:

**Wireless Network Interface Card (NIC)** - Allows a device to connect to a wireless network.



- **Wireless Router/Home Router** - Provides both routing functions and wireless access to the network.



**• Access Point (AP) - Extends the wireless network's range and provides connectivity for wireless clients.**



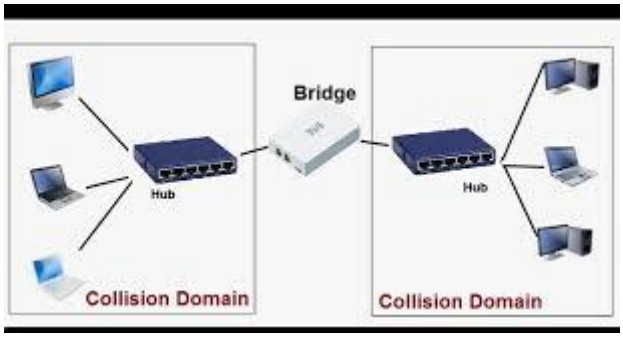
**• Wireless Antennas - Enhance the transmission and reception of wireless signals.**



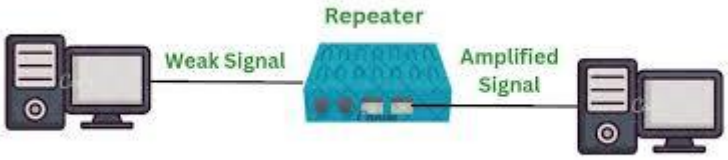
- **Wireless Controller - Manages multiple access points and oversees network performance and configuration.**



- **Wireless Bridges - Connects two or more physically separated networks wirelessly.**




- **Repeater/Range Extender - Amplifies and retransmits the wireless signal to extend coverage.**




- **Wireless Gateway - Combines routing and access point functions with additional features like firewall protection.**



- **Client Devices** - Includes laptops, smartphones, tablets, and other devices that connect to the wireless network.
- **Modem** - Converts digital data from the network into a signal suitable for transmission over phone lines or other media (often integrated with routers in home setups).2. Role of a 2. Wireless Network Interface Card (NIC) and Its Differences from a Wired Network Card
- **Role of a Wireless NIC**: A Wireless Network Interface Card (NIC) is a hardware component that enables a computer or device to connect to a wireless network. It acts as an interface between the device and the wireless network, allowing the device to send and receive data over radio waves rather than through a physical wired connection.
- **Key Functions:**
- **Wireless Communication:** It translates data between the computer and the wireless network using radio frequencies.
- **Network Access:** It enables the device to detect available wireless networks and connect to them.
- **Data Transmission:** It manages the transmission and reception of data packets over the wireless network.
- **Differences from a Traditional Wired Network Card:**
- **Connection Medium:** A wired NIC connects to the network via Ethernet cables, while a wireless NIC uses radio signals.
- **Mobility:** Devices with wireless NICs can connect to the network without being tethered to a physical location, unlike wired NICs.
- **Hardware:** Wireless NICs include antennas and radio transceivers, whereas wired NICs use physical connectors (like RJ-45) and do not require antennas.

**3. Contribution of a Wireless Router/Home Router to a Wireless Network**

- **Role and Contribution:** A wireless router, or home router, is a key component of a wireless network that provides connectivity to devices within its range. It combines the functions of a router and a wireless access point.
- **Primary Components and Features:**
- **Router Functionality:** Directs data packets between devices on the local network and external networks, such as the internet. It assigns IP addresses, manages network traffic, and handles routing protocols.
- **Wireless Access Point:** Allows wireless devices to connect to the network by broadcasting and receiving radio signals.
- **Built-in Modem (optional):** Some routers have an integrated modem to connect directly to the internet service provider.
- **Security Features:** Includes encryption protocols (e.g., WPA2, WPA3) to protect the network from unauthorized access and attacks.

- **Configuration Options:** Allows users to configure network settings, such as SSID (network name), password, and security settings, often through a web-based interface.

**4. Functions of an Access Point and a Wireless Controller**

- **Access Point (AP):**
- **Function:** Extends the range of a wireless network by creating a new network segment. It acts as a bridge between the wired network and wireless clients.
- **Role:** Provides wireless clients with network access and facilitates communication with the wired network. It helps manage traffic and can be used to expand coverage in areas with weak signal strength.
- **Wireless Controller:**
- **Function:** Centralizes management of multiple access points within a network. It allows administrators to configure, monitor, and control APs from a single interface.
- **Role:** Manages AP settings, firmware updates, security policies, and network performance. It can also handle load balancing, optimize wireless coverage, and provide seamless roaming for clients moving between APs.

**5. Enhancement of Coverage and Connectivity by Wireless Antennas and Wireless Bridges**

- **Wireless Antennas:**
- **Function:** Enhance the transmission and reception of radio signals between wireless devices and access points. They can increase the range and quality of the wireless signal.
- **Types:** Includes omnidirectional antennas (which broadcast in all directions) and directional antennas (which focus the signal in a specific direction).
- **Utilization:** Omnidirectional antennas are used for general coverage in a broad area, while directional antennas are used to focus the signal in a specific direction to reach distant locations or to overcome obstacles.
- **Wireless Bridges:**
- **Function:** Connect two or more physically separated networks by extending the wireless network across distances where direct connection is not feasible.
- **Types:** Includes point-to-point bridges (connecting two locations directly) and point-to-multipoint bridges (connecting multiple locations to a central point).
- **Utilization:** Used to link buildings, connect remote areas, or extend network coverage where running cables is impractical. They are particularly useful for connecting separate buildings in a campus environment or for extending coverage in large outdoor areas.

**Points to Remember**

- Wireless network is made of different components such as Wireless NICs, Wireless Router/Home router, Access point, Wireless Antennas, Wireless controller and Wireless bridges
- Each component among these fulfils its own function in wireless networks.

**Application of learning 1.3.**

XYZ is a Telecommunication Company located in Kigali city in the IMBUTO Plaza building of 3 floors, it has 50 staff dispatched in their offices. The CEO of the company need to deploy wireless network so that the employees will be able to communicate each other.

The staffs to be connected to the network wirelessly in the building are dispatched as follow:

- 1st floor: 20
- 2nd floor: 20
- 3rd floor: 10

As a wireless network technician, according to the above information, propose the components to be used in deploying this wireless network and justify your proposal.

## Indicative content 1.4: Application of Wireless Technologies

**Duration: 3 hrs**

**Theoretical Activity 1.4.1: Identification of wireless technologies application**

**Tasks:**

1. In small groups, you are requested to answer the following questions related to the identification of wireless technology applications
    i. Differentiate short-range/low-power from large-range wireless technologies in terms of their applications
    ii. What is an example of a short-range, low-power wireless technology and its typical use case?
    iii. How do short-range wireless technologies like Bluetooth and Zigbee differ from large-range technologies like Wi-Fi and cellular networks?
    iv. What are the advantages of using large-range wireless technologies in rural or remote areas?
    v. How does power consumption compare between short-range, low-power wireless technologies and large-range wireless technologies?
    vi. In what types of applications would large-range wireless technologies be preferred over short-range, low-power options?
2. Write your findings on reserved place
3. Discuss with others the provided answers and choose the correct answers
4. Listen to trainer's clarifications and ask questions if necessary
5. Do the application of learning 1.4 and read the key readings 1.4.1

**Key readings 1.4.1.: Identification of wireless technologies application**

Differentiation between short-range/low-power and large-range wireless technologies in terms of their applications:

- **Short-Range/Low-Power Wireless Technologies**
    - ✓ **Characteristics:**
        - **Range:** Typically covers a small area, usually up to 10 meters (e.g., Bluetooth) to a few hundred meters (e.g., Zigbee).
        - **Power Consumption:** Designed for minimal power use, extending battery life in devices.

- **Data Rate:** Generally, supports lower data transfer rates compared to large-range technologies.
- **Device Communication:** Often used for connecting devices within close proximity.

✓ **Applications:**

- **Personal Area Networks (PANs):** Connecting devices like wireless keyboards, mice, and headphones to a computer or smartphone.
- **Smart Home Devices:** Zigbee and Bluetooth are commonly used in smart home systems for devices like smart thermostats, lights, and locks.
- **Wearable Technology:** Bluetooth is used in devices like fitness trackers and smartwatches to sync data with smartphones.
- **IoT (Internet of Things):** Short-range technologies facilitate communication between sensors and controllers in various IoT applications, such as home automation.

- **Large-Range Wireless Technologies**

✓ **Characteristics:**

- **Range:** Covers larger distances, from several hundred meters (e.g., Wi-Fi) to several kilometers or more (e.g., cellular networks).
- **Power Consumption:** Generally, consumes more power due to the need for greater coverage and higher data rates.
- **Data Rate:** Supports higher data transfer speeds, suitable for more data-intensive applications.
- **Coverage:** Provides connectivity over broader areas and is suited for environments where devices are not in close proximity

✓ **Applications**

- **Internet Access:** Wi-Fi provides internet connectivity in homes, offices, and public areas, while cellular networks offer internet access over large geographic areas, including rural and remote regions.
- **Public and Enterprise Networks:** Wi-Fi is used in public hotspots, cafes, and businesses for broad network coverage. Cellular networks support mobile communication and data services over vast areas.
- **Rural and Remote Connectivity:** Large-range technologies like satellite and cellular networks are used to provide internet and communication services in areas where wired infrastructure is not feasible.
- **Wide Area Networks (WANs):** Cellular networks, satellite, and other large-range technologies are used to connect geographically dispersed networks and devices.

1. **What is an example of a short-range, low-power wireless technology and its typical use case?**

Example: **Bluetooth is a short-range, low-power wireless technology**. It is commonly used for connecting devices like wireless headphones, keyboards, and mice to computers or smartphones. Its typical use case includes personal area networks where devices need to communicate over short distances (typically up to 10 meters) with minimal power consumption.

2. **How do short-range wireless technologies like Bluetooth and Zigbee differ from large-range technologies like Wi-Fi and cellular networks**?

Differences:

- **Range**: Short-range technologies like Bluetooth and Zigbee have limited coverage, typically ranging from a few meters to about 100 meters. In contrast, large-range technologies like Wi-Fi can cover larger areas (up to several hundred meters indoors) and cellular networks provide coverage over several kilometers to hundreds of kilometers.
- **Power Consumption**: Short-range technologies are designed to be low-power, extending battery life in devices. Large-range technologies generally consume more power due to the need to cover greater distances and handle higher data rates.
- **Data Rate:** Short-range technologies usually support lower data transfer rates compared to large-range technologies, which can handle higher data speeds for tasks like streaming and large file transfers.

3. **What are the advantages of using large-range wireless technologies in rural or remote areas?**

**Advantages:**

- **Extended Coverage**: Large-range technologies like cellular networks and satellite communications can cover vast areas, providing connectivity in locations where traditional wired infrastructure is not feasible.
- **Connectivity Improvement**: They offer better connectivity options for rural or remote communities, allowing for internet access, communication, and data transfer where other technologies might be impractical.
- **Reduced Infrastructure Costs**: Large-range wireless technologies can reduce the need for extensive physical infrastructure, making it more cost-effective to provide connectivity over large areas.

4. **How does power consumption compare between short-range, low-power wireless technologies and large-range wireless technologies?**

**Comparison:**

- **Short-Range, Low-Power Technologies:** These technologies, such as Bluetooth and Zigbee, are optimized for minimal power consumption. They

use low-energy protocols to extend battery life in devices, making them suitable for battery-operated devices that do not require constant communication.

- **Large-Range Technologies**: These typically consume more power due to their need to transmit signals over longer distances and handle higher data rates. For instance, cellular networks and Wi-Fi require more energy to maintain connections over greater distances and support higher data transfer speeds.

5. **In what types of applications would large-range wireless technologies be preferred over short-range, low-power options?**

**Applications:**

- **Rural Internet Access:** Large-range technologies like cellular networks and satellite internet are preferred for providing internet access in rural or remote areas where short-range technologies would be insufficient.
- **Public Wi-Fi Networks:** Wi-Fi is used in public areas such as cafes, airports, and offices where a broader coverage area is needed.
- **Data-Intensive Applications**: Large-range technologies are suitable for applications requiring high data transfer rates, such as video streaming, online gaming, and large file transfers, where short-range technologies would not provide adequate performance.

**Differences between Short Range/Low-Power Wireless Technologies and Large Range Wireless Technologies**

| Aspect | Short Range/Low-Power Wireless Technologies | Large Range Wireless Techn |
|---|---|---|
| Communication Distance | Typically, short-range (up to 100 meters) | Long-range (can range from to kilometers) |
| Power Consumption | Low power consumption for battery-operated devices | Can be higher power, espec longer ranges |
| Use Cases | Ideal for IoT, home automation, wearables, and sensor networks | Suitable for wide-area r deployments, such as networks |
| Network Topology | Typically used in personal area networks | Designed for wide area n (WAN) |

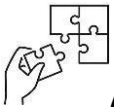
| | | |
|---|---|---|
| | (PAN) and local area networks (LAN) | |
| Frequency Bands | Commonly operate in unlicensed ISM bands (e.g., 2.4 GHz, 868 MHz, 915 MHz) | Use licensed or unlicensed bands, depending on the technology (e.g., cellular networks use licensed bands) |
| Data Rate | Lower data rates, typically below 1 Mbps | Higher data rates, ranging from Kbps to Gbps |
| Security | May have basic security features, but often rely on higher layers for security | Incorporate robust security measures, such as encryption and authentication |
| Interference Susceptibility | Susceptible to local interference and congestion in crowded areas | Generally, more robust in dealing with interference and congestion |
| Device Density | Supports a higher density of connected devices in a localized area | Supports a broad range of device densities, including high-density areas |
| Cost of Deployment | Lower cost for infrastructure and devices | Can be higher in terms of infrastructure deployment and operational costs |
| Protocols and Standards | Common standards include Zigbee, Bluetooth, and Z-Wave | Includes standards like Wi-Fi, 4G/5G cellular, and LPWAN technologies (e.g., LoRaWAN, Sigfox) |
| Latency | Lower latency, suitable for real-time and low-latency applications | Varies based on the specific technology but can be higher than short-range technologies |
| Mobility | May not be designed for high mobility; more suited for stationary or limited mobility applications | Designed to handle high mobility scenarios, such as in vehicular or cellular networks |

| | | | |
|---|---|---|---|
| Coverage Area | Suitable for localised coverage within a building or limited area | Designed to provide extensive geographic coverage, including urban and rural areas | |



**Points to Remember**

- **Short-Range/Low-Power:** Ideal for applications requiring minimal power consumption and short-range connectivity, such as personal devices and home automation.
- **Large-Range:** Suitable for applications needing extensive coverage and higher data rates, including internet access over large areas and connectivity in rural or remote regions.



**Application of learning 1.4.**

A large university is planning to upgrade its networking infrastructure to support both indoor and outdoor connectivity across its expansive campus. The university's existing network consists of a mix of wired connections in buildings and some outdated wireless access points, leading to inconsistent coverage and connectivity issues. This network is insufficient to meet the growing demands for seamless wireless connectivity both indoors and across the campus. With the increasing number of mobile devices used by students, faculty, and staff, the administration has decided to implement a more robust wireless network.

As the network technician, you are hired to help the staff in selecting the best technologies to use according to these criteria:

- Technology that provides reliable, high-speed wireless connectivity within classrooms, libraries, dormitories, and administrative offices. The network must support a high density of users and devices in confined areas, such as lecture halls and common spaces.
- Technology that ensures seamless wireless connectivity across the entire campus, including outdoor areas like courtyards, sports fields, parking lots, and pathways between buildings. The network must allow students and staff to stay connected while moving across the campus.

## Indicative content 1.5: Classification of Wireless Network Standards

**Duration: 2 hrs**

**Theoretical Activity 1.5. : Classification of Wireless network standards**

**Tasks:**

1. In small groups, you are requested to answer the following questions related to the classification of wireless network standard
    1. Define wireless network standard
    2. What are the factors to consider when classifying wireless network standard?
    3. Classify the main standards that are currently used by their typical use cases and ranges
    4. What distinguishes short-range wireless standards like Bluetooth from large-range standards like Cellular Networks in terms of their typical use cases?
    5. How do sub-GHz frequency bands used by Z-Wave and LoRaWAN differ in application from the 2.4 GHz and 5 GHz bands used by Wi-Fi and Bluetooth?
    6. What are the key differences in data rates between low-data-rate standards like Zigbee and high-data-rate standards like Wi-Fi and 5G?
    7. In what scenarios would you use a personal area network (PAN) technology like Bluetooth compared to a wide area network (WAN) technology like WiMAX?
    8. Which standardization bodies are responsible for defining the standards for Wi-Fi, Bluetooth, and 5G, and why is their role important?
2. Present your findings in front of a whole class
3. Take short notes for what is being clarified and ask questions when necessary.
4. Read key readings 1.5.1
5. Do the application of learning 1.5

**Key readings 1.5.1.: Classification of Wireless network standards**

1. A wireless network standard is a set of agreed-upon technical specifications and protocols that define how wireless communication systems and devices operate and interact. These standards are established to ensure compatibility, interoperability, performance, and safety across different devices and systems within a wireless network.
2. When classifying wireless network standards, several factors are considered to determine their suitability, performance, and application. These factors help in understanding the capabilities, limitations, and appropriate use cases of different wireless technologies.

- Here are the primary factors to consider:
    1. **Range and Coverage Area**: The geographical area over which the wireless network can effectively operate.
        - **Considerations:**
        - **Short-Range**: Technologies designed for small areas, such as personal area networks (PANs) and local area networks (LANs).
        - **Medium-Range:** Technologies for broader coverage, like metropolitan area networks (MANs).
        - **Long-Range**: Technologies for extensive coverage, such as wide area networks (WANs) and global communication.
    2. **Data Transmission Speed:** The rate at which data can be transmitted over the network, usually measured in bits per second (bps), megabits per second (Mbps), or gigabits per second (Gbps).
        - **Considerations:**
            - **Low-Speed**: Suitable for applications with low data requirements, such as simple device connections**.**
            - **High-Speed:** Required for bandwidth-intensive applications, such as video streaming and large data transfers.
    3. **Frequency Bands:** The specific range of electromagnetic frequencies used for communication.
        - **Considerations:**
            - **GHz Band:** Common for many short-range technologies like Wi-Fi and Bluetooth.
            - **5 GHz Band:** Offers higher speeds and less interference for Wi-Fi.
            - **Sub-GHz Bands:** Used for long-range, low-power technologies like LoRaWAN.
    4. **Modulation Techniques:** Methods used to encode data onto carrier waves for transmission.
        - **Considerations:**

**OFDM (Orthogonal Frequency Division Multiplexing):** Used in Wi-Fi and LTE for high data rates and efficient use of spectrum.

- **DSSS (Direct Sequence Spread Spectrum):** Used in older Wi-Fi standards and some other wireless technologies.

5. **Power Consumption:** The amount of energy required by devices operating on the network.
   - **Considerations:**
     - **Low Power:** Suitable for battery-operated devices and applications that need long battery life.
     - **Higher Power:** Used in systems where power is less of a concern, such as fixed installations.
6. **Security Features:** The protocols and methods used to protect data and ensure secure communication.
   - **Considerations:**
     - **Encryption:** Methods like WPA2 and WPA3 for Wi-Fi to secure data.
     - **Authentication:** Techniques to verify device and user identities, such as Bluetooth pairing and network passwords**.**
7. **Interoperability:** The ability of devices from different manufacturers to work together within the same network.
   - **Considerations:**
     - **Standards Compliance:** Adherence to established standards ensures devices can connect and communicate effectively.
     - **Backward Compatibility:** Ability of newer standards to work with older devices.

**Quality of Service (QoS):** Mechanisms to prioritize certain types of traffic to ensure performance for critical applications.

Considerations:

**Traffic Management:** Techniques to handle network congestion and prioritize high-priority traffic.

**Latency and Jitter:** Management of delays and variation in packet delivery.

3 List of major wireless network standards, classified by their typical use cases and ranges:

**1. Personal Area Network (PAN) Standards:**

**Bluetooth:**

**Standards:** Bluetooth 1.0 to 5.2

**Use Case:** Short-range communication between devices such as headsets, keyboards, and smartphones.

**NFC (Near Field Communication):**

**Standards:** ISO/IEC 14443, ISO/IEC 18092

**Use Case:** Very short-range communication for contactless payments and data exchange.

**2. Local Area Network (LAN) Standards:**

**Wi-Fi (Wireless LAN):**

**Standards:** IEEE 802.11a/b/g/n/ac/ax

**Use Case:** Wireless internet access in homes, offices, and public hotspots.

Details:

802.11a: 5 GHz band, up to 54 Mbps

802.11b: 2.4 GHz band, up to 11 Mbps

802.11g: 2.4 GHz band, up to 54 Mbps

802.11n: 2.4 GHz and 5 GHz bands, up to 600 Mbps

802.11ac: 5 GHz band, up to 3.5 Gbps

802.11ax (Wi-Fi 6): 2.4 GHz and 5 GHz bands, up to 9.6 Gbps

Zigbee:

**Standards:** IEEE 802.15.4

**Use Case:** Low-power, low-data-rate applications such as home automation and industrial control.

**3. Metropolitan Area Network (MAN) Standards:**

WiMAX (Worldwide Interoperability for Microwave Access):

**Standards:** IEEE 802.16

**Use Case:** Broadband wireless access over larger areas, such as cities.

LoRaWAN (Long Range Wide Area Network):

**Standards:** LoRaWAN Specification

**Use Case:** Long-range, low-power IoT applications and sensor networks.

**4. Wide Area Network (WAN) Standards:**

Cellular Networks:

**Standards:**

**2G (GSM):** Global System for Mobile Communications

**3G (UMTS):** Universal Mobile Telecommunications System

**4G (LTE):** Long-Term Evolution

**5G NR (New Radio):** Fifth-generation wireless technology

**Use Case:** Mobile telephony and internet access over large geographic areas.

**Satellite Communication:**

**Standards:** Various protocols and standards for satellite communication

**Use Case:** Global communication, remote areas, and weather monitoring.

**5. Specialized and Emerging Standards:**

**NB-IoT (Narrowband IoT):**

**Standards:** 3GPP Release 13 and beyond

**Use Case:** Low-power, wide-area network for IoT applications with limited bandwidth needs.

**5G NR (New Radio):**

**Standards:** 3GPP Release 15 and beyond

**Use Case:** Enhanced mobile broadband, ultra-reliable low latency communication, and massive IoT.

**WLAN (Wireless Local Area Network) Standards Beyond Wi-Fi:**

**Standards:** IEEE 802.11ah (HaLow) for sub-1 GHz communication

**Use Case:** Long-range communication for IoT devices in local networks.

**6. Radar and Sensing Standards:**

**ISM Band Standards:**

**Standards:** Various protocols for industrial, scientific, and medical (ISM) bands.

**Use Case:** Includes radar and sensing technologies that operate in the ISM bands (e.g., 24 GHz).

**7. Network and Device Management Standards:**

**SNMP (Simple Network Management Protocol):**

**Standards:** RFC 1157 and updates

**Use Case:** Network management and monitoring, including wireless networks.

4. **What distinguishes short-range wireless standards like Bluetooth from large-range standards like Cellular Networks in terms of their typical use cases?**

**Short-range wireless standards like Bluetooth** are designed for communication over short distances, typically within 10 meters. They are commonly used for connecting personal devices such as smartphones, headsets, keyboards, and fitness trackers. The focus of Bluetooth is on low power consumption and ease of pairing, making it ideal for applications where close proximity between devices is expected.

**Large-range standards like Cellular Networks (e.g., 4G LTE, 5G)**, on the other hand, are designed to provide wide-area coverage, enabling communication over several kilometers. These networks are essential for mobile phones, tablets, and IoT devices that need to stay connected while moving across large geographic areas. Cellular networks support high data rates and robust connectivity, making them suitable for applications like voice calls, mobile internet access, and streaming services.

5. **How do sub-GHz frequency bands used by Z-Wave and LoRaWAN differ in application from the 2.4 GHz and 5 GHz bands used by Wi-Fi and Bluetooth?**

**Sub-GHz frequency bands** (around 900 MHz) used by standards like Z-Wave and LoRaWAN are characterized by longer range and lower power consumption compared to higher frequency bands. These properties make them ideal for IoT applications in home automation, smart cities, and agriculture, where devices need to communicate over long distances with minimal power usage. The lower frequencies penetrate obstacles like walls and vegetation better than higher frequencies.

**2.4 GHz and 5 GHz bands used by Wi-Fi and Bluetooth** offer higher data rates but over shorter distances. Wi-Fi in these bands provides fast internet access and networking within homes and offices, while Bluetooth facilitates short-range communication between devices like smartphones and wireless peripherals. The higher frequencies are more prone to interference and have less range compared to sub-GHz bands, but they support faster data transmission.

**6. What are the key differences in data rates between low-data-rate standards like Zigbee and high-data-rate standards like Wi-Fi and 5G?**

**Low-data-rate standards like Zigbee** offer data rates up to 250 kbps. They are designed for low-power, low-bandwidth applications such as sensors, smart lighting, and industrial control systems. The focus of Zigbee is on energy efficiency and simplicity, making it ideal for applications where large amounts of data do not need to be transmitted quickly.

**High-data-rate standards like Wi-Fi and 5G** support data rates ranging from hundreds of Mbps to several Gbps. Wi-Fi is commonly used for high-bandwidth applications like

video streaming, online gaming, and large file transfers within homes and businesses. 5G takes this further, offering ultra-fast data rates, low latency, and massive connectivity, which are essential for advanced applications like virtual reality (VR), autonomous vehicles, and smart cities.

7. **In what scenarios would you use a personal area network (PAN) technology like Bluetooth compared to a wide area network (WAN) technology like WiMAX?**

**Personal Area Network (PAN) technology like Bluetooth** is used in scenarios where devices need to communicate over short distances. This includes pairing smartphones with headphones, connecting smartwatches to phones, or enabling wireless input devices like keyboards and mice. Bluetooth is preferred in situations where low power consumption, ease of use, and proximity between devices are critical.

**Wide Area Network (WAN) technology like WiMAX** is used for providing broadband internet access over large areas, often in rural or underserved regions. WiMAX can connect multiple users to the internet over distances of several kilometers, making it suitable for providing last-mile connectivity where traditional wired infrastructure is not feasible. It is ideal for applications requiring wide-area coverage and higher data rates than PAN technologies.

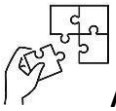
8. **Which standardization bodies are responsible for defining the standards for Wi-Fi, Bluetooth, and 5G, and why is their role important?**

- **Wi-Fi standards** are defined by the **IEEE (Institute of Electrical and Electronics Engineers)**, specifically under the IEEE 802.11 family. The IEEE is crucial because it ensures that Wi-Fi devices from different manufacturers are compatible, promoting widespread adoption and interoperability.
- **Bluetooth standards** are also developed by the **IEEE**, specifically under the IEEE 802.15.1 standard, and are further managed by the **Bluetooth Special Interest Group (SIG)**. The SIG ensures that Bluetooth technology evolves to meet new market needs while maintaining backward compatibility.
- **5G standards** are defined by the **3GPP (3rd Generation Partnership Project)**, a collaboration between telecommunications standards organizations. 3GPP's role is critical because it ensures that 5G networks are globally standardized, allowing seamless communication across different countries and networks, and supporting a wide range of applications from mobile broadband to IoT.



**Points to Remember**

- When classifying wireless network standards, consider factors such as range, data transmission speed, frequency bands, modulation techniques, power consumption, security features, interoperability, Quality of Service
- Network standards cover a range of applications from short-range personal area networks to wide area networks and specialized communication needs. Each standard is designed to address specific requirements related to range, speed, power consumption, and application types.
- **Range-Based Classification**: Wireless standards are categorized as short-range (e.g., Bluetooth for personal devices), medium-range (e.g., Wi-Fi for home and office networks), and large-range (e.g., Cellular Networks like 4G and 5G for wide-area coverage).
- **Frequency Bands**: Standards are classified based on the frequency bands they use, such as sub-GHz (e.g., Z-Wave for IoT), 2.4 GHz and 5 GHz (e.g., Wi-Fi for faster data rates), and millimeter-wave bands (e.g., 5G for ultra-high-speed communication).
- **Data Rate**: Wireless standards are divided by data rates, with low-data-rate standards (e.g., Zigbee for low-bandwidth IoT applications) and high-data-rate standards (e.g., Wi-Fi and 5G for high-speed internet and multimedia).
- **Application-Based Classification**: Standards are also categorized by their application areas, such as PAN (e.g., Bluetooth for personal area networks), LAN (e.g., Wi-Fi for local networks), WAN (e.g., WiMAX for wide-area networks), and IoT networks (e.g., LoRaWAN for long-range, low-power IoT).



**Application of learning 1.5.**

The city of Tech ville is undergoing a major upgrade to its network infrastructure as part of a new smart city initiative. The goal is to integrate various technologies to improve urban management, enhance public services, and offer better connectivity to residents and businesses. The project involves deploying a range of wireless technologies across different areas of the city.

As a student who specialized in network standardization, you are tasked with selecting and classifying the appropriate wireless network standards used for Body Area Networks (BAN), Personal Area Networks (PAN), Local Area Networks (LAN), Metropolitan Area Networks (MAN), and Wide Area Networks (WAN) segments of the smart city project so that each network segment uses the most suitable standard to meet its specific requirements, including coverage area, data transmission speed, power consumption, and application needs.

## Indicative content 1.6: Operation Mode of Wireless Network

**Duration: 2 hrs**

**Theoretical Activity 1.6.1: Identify the operation mode of Wireless network**

**Tasks:**

1. In small groups, you are requested answer the following question.
   How do different wireless network operation modes, such as Infrastructure, Ad-Hoc, Mesh, Repeater/Extender, Bridge, and Monitor Mode, differ in terms of their network structure, communication methods, and typical use cases?"
2. Write your answers on flipchart/paper
3. Discuss with others about the answers and choose the correct answers
4. Listen to the clarification of trainer and take short note
5. Ask questions if necessary
6. For more clarification, read the key readings 1.6.1.

**Key readings 1.6.1.: Identify the operation mode of Wireless network**

The operation mode of a wireless network refers to the way in which devices connect and communicate with each other within the network. There are several modes of operation, each suited to different networking needs and environments. Here are the primary operation modes

**1. Infrastructure Mode**

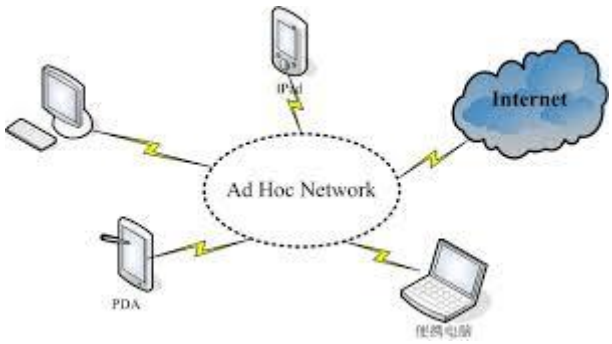
- **Description**: Infrastructure mode is the most common wireless network configuration. In this mode, all wireless devices communicate through a central device, usually a wireless router or an access point.
- **How It Works**: Devices like laptops, smartphones, and tablets connect to the wireless router or access point, which acts as a central hub. The router connects to the internet and manages network traffic between devices.

  **Use Cases**: Home and office networks, public Wi-Fi hotspots, and any environment where centralized control and internet access are needed.




2. **Ad-Hoc Mode**

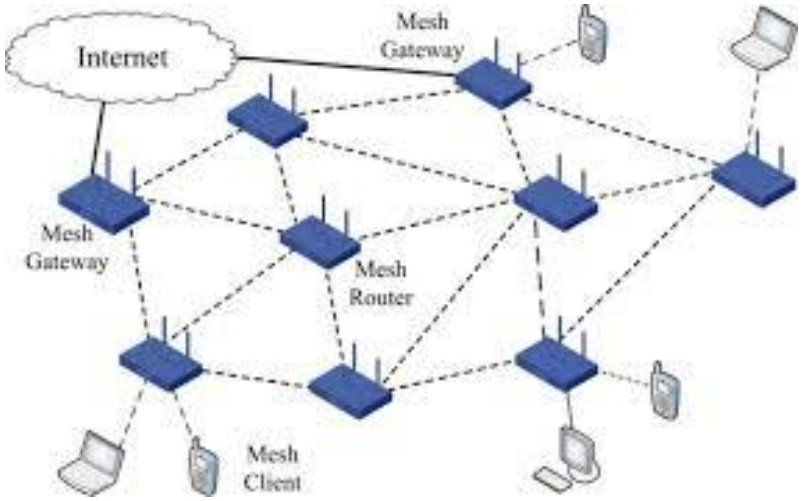
- **Description**: Ad-hoc mode (also known as peer-to-peer mode) allows wireless devices to connect directly to each other without using a central device like a router or access point.
- **How It Works**: Devices within range of each other form a temporary network, where each device communicates directly with the others. This is a decentralized network, meaning there is no single point of control.
- **Use Cases**: Small, temporary networks, such as file sharing between laptops in a meeting room or gaming between devices without an internet connection.




3. **Mesh Mode**

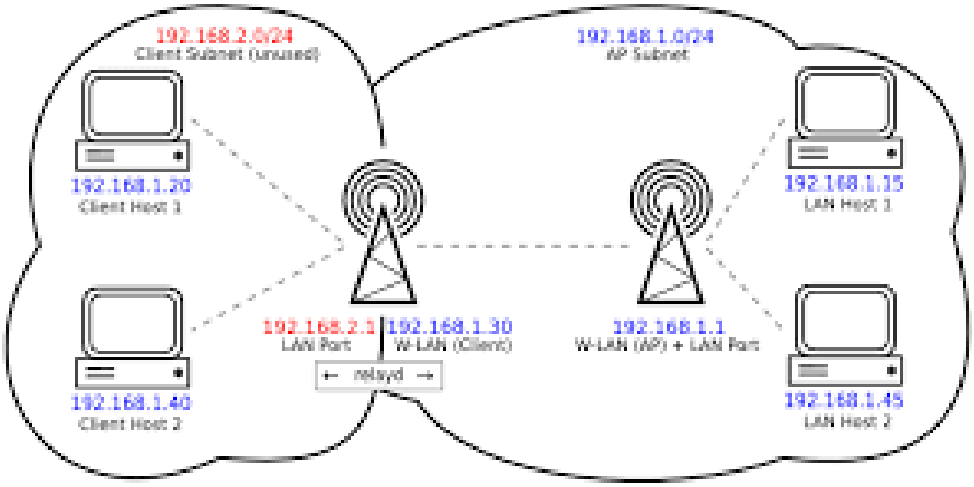
- **Description**: In mesh mode, multiple access points (or mesh nodes) work together to form a single, seamless network. Each node in a mesh network communicates with its neighbors, creating a network with multiple paths for data to travel.
- **How It Works**: Data can be routed through different nodes to reach its destination, which enhances network reliability and coverage. If one node fails, data can still travel through alternative paths.

- **Use Cases**: Large areas requiring extensive coverage, such as campuses, large homes, outdoor areas, and rural settings where traditional infrastructure is challenging.




4. **Repeater/Extender Mode**

- **Description**: Repeater or extender mode is used to extend the coverage of an existing wireless network by amplifying and retransmitting the signal.
- **How It Works**: A wireless repeater or range extender receives the signal from the main wireless router and rebroadcasts it, effectively increasing the coverage area.
- **Use Cases**: Extending Wi-Fi coverage in large homes, offices, or areas where the main router's signal is weak or obstructed.




5. **Bridge Mode**

- **Description**: Bridge mode allows two or more networks to be connected together over a wireless link, effectively creating a single network from multiple segments.
- **How It Works**: A wireless bridge connects two different network segments, allowing devices on separate networks to communicate as if they were on the same network.
- **Use Cases**: Connecting different buildings in a campus, integrating wired and wireless networks, and linking remote offices.

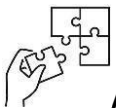
**6. Monitor Mode**

- **Description**: Monitor mode is a special operation mode used primarily for network analysis and security monitoring. In this mode, a wireless device can capture all packets on a network without connecting to it.
- **How It Works**: The device listens to all wireless traffic within range, capturing data packets for analysis. This mode is often used in conjunction with network analysis tools.
- **Use Cases**: Network troubleshooting, security auditing, and wireless penetration testing.



**Points to Remember**

- Each Operation mode of Wireless network serves different purposes, and the choice of mode depends on the specific requirements of the wireless network, such as coverage area, number of devices, network management needs, and security considerations.
- Infrastructure Mode: Devices connect through a central router or access point.
- Ad-Hoc Mode: Devices connect directly to each other without a central device.
- Mesh Mode: Multiple nodes work together to form a seamless, extended network.
- Repeater/Extender Mode: Amplifies and retransmits an existing network's signal to increase coverage.
- Bridge Mode: Connects separate network segments to form a unified network.
- Monitor Mode: Captures all wireless traffic for analysis without connecting to the network.



**Application of learning 1.6.**

You are an IT consultant for a small startup company, Innovatech, which has just moved into a new office space. The office is a single floor with an open-plan layout, and the company needs a wireless network that can support about 20 employees, each with a laptop and a mobile device. Additionally, there are several IoT devices such as smart thermostats and security cameras. Innovatech's CEO has asked for your recommendation on the best wireless network operation mode to ensure reliable connectivity, easy management, and scalability as the company grows. Based on the given situation, suggest the most suitable operation mode for Innovatech's wireless network and justify your recommendation.

## Learning outcome 1 end assessment

**Theoretical assessment**

**I. Circle the right answer which correspond to the asked question**

1. Which of the following is not a benefit of a wireless network?

a. Increased mobility and convenience
b. Easier and faster installation
c. Lower susceptibility to interference
d. Flexibility in network expansion

2. Which wireless networking standard provides the highest maximum data rate?
   a. 802.11b
   b. 802.11g
   c. 802.11n
   d. 802.11ac
3. In a wireless network, what does the term 'SSID' stand for?
   a. Secure Service Internet Device
   b. Service Set Identifier
   c. Secure Set Internet Device
   d. Service Security Identifier
4. What is a primary disadvantage of using a wireless network compared to a wired network?
   a. Higher installation cost
   b. Increased security risks
   c. Lower mobility
   d. Higher latency
5. In which operation mode do all wireless devices communicate through a central access point, such as a router?
   a. Ad-hoc mode
   b. Infrastructure mode
   c. Monitor mode
   d. Mesh mode

**II. Match each frequency range to the corresponding wireless network or technology**

| | | |
|---|---|---|
| 1. ……. | 1. 2.4 GHz | a. Wi-Fi 5 (802.11ac) |
| 2. …….. | 2. 5 GHz | b. WiGig (802.11ad) |
| 3. …….. | 3. 60 GHz | c. Zigbee |
| 4. …….. | 4. 900 MHz | d. Wi-Fi 4 (802.11n) |
| 5. ……. | 5. 1.8 GHz | e. Wi-Fi 4 (802.11n) |

**III. Answer by T if the answer is true or F if the answer is false.**

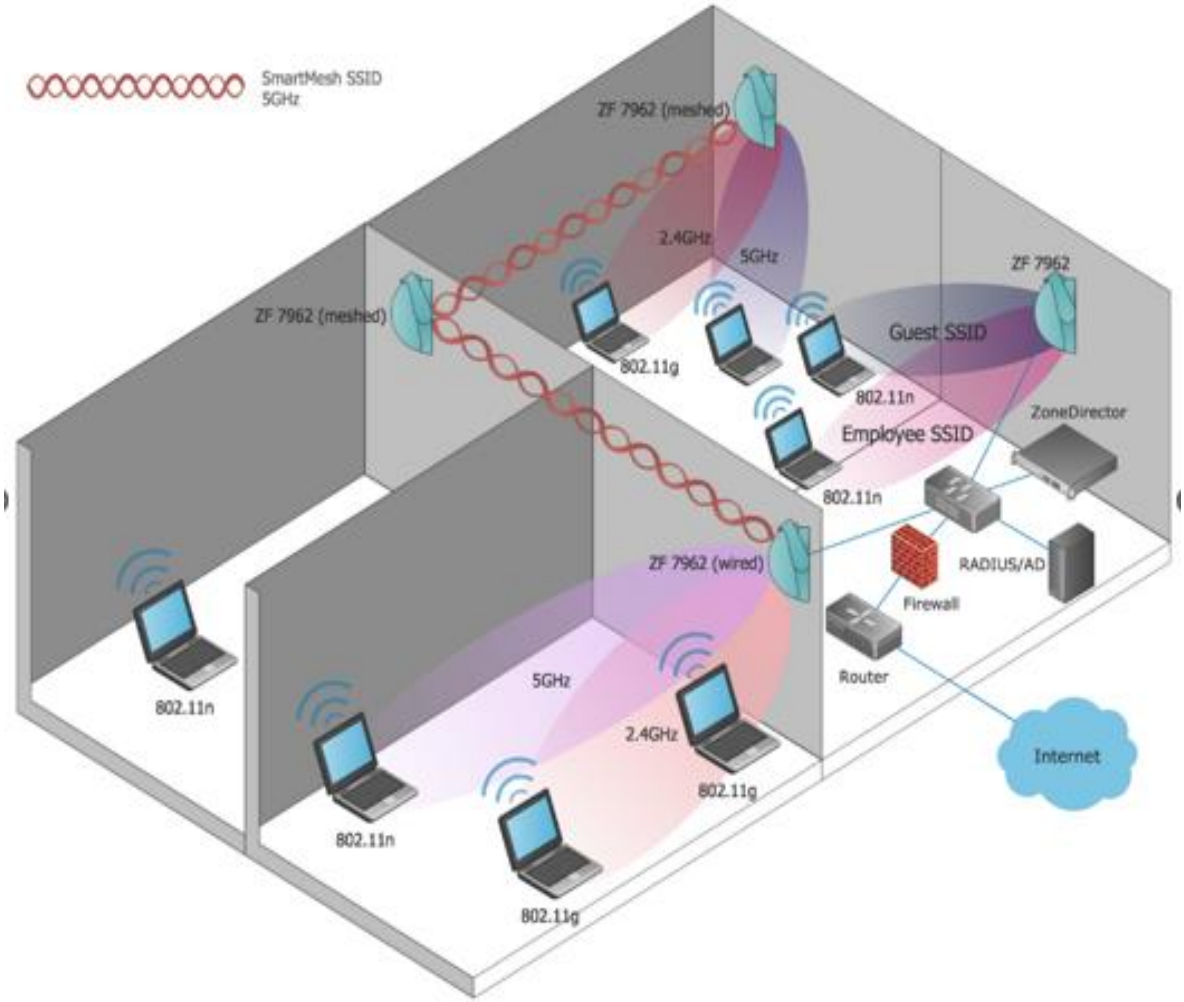
a. An access point connects wireless devices to a wired network
b. An access point functions as a router to manage traffic between devices on a network
c. An access point requires a physical connection to a router or switch to function
d. Wireless Personal Area Networks (WPANs) are typically used for connecting devices within a very short range, such as Bluetooth devices.
e. A Wireless Local Area Network (WLAN) is commonly used to connect devices in a large geographical area, such as an entire city
f. Wireless Wide Area Networks (WWANs) can cover large geographic areas and are often used for mobile phone networks.
g. Wireless Metropolitan Area Networks (WMANs) are designed to provide wireless connectivity across multiple buildings or a campus within a city.
h. Wireless Body Area Networks (WBANs) are designed to connect devices placed on, around, or within a human body.

**END**



## References


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3. https://www.akibia.com/what-factors-to-consider-when-selecting-network-cabling/
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5. Smith, J. A. (2020). Apply wireless network concepts. Tech Publishers.
6. Stakeholder Technology Branch. (2007, October). *Wireless local area network (WLAN) best practices guide*.


SmartMesh SSID
5GHz
ZF 7962 (meshed)
2.4GHz
5GHz
ZF 7962
ZF 7962 (meshed)
802.11g
Guest SSID
802.11n
ZoneDirector
Employee SSID
802.11n
ZF 7962 (wired)
RADIUS/AD
Firewall
Router
802.11n
5GHz
2.4GHz
802.11g
Internet
802.11n
802.11g

| Indicative contents |
|---|
| **2.1 Identification of wireless network indoor installation requirements** |
| **2.2 Designing Wireless Network Blueprint** |
| **2.3 Estimation of wireless network indoor installation cost** |

## Key Competencies for Learning Outcome 2: Plan Wireless network indoor installation

| Knowledge | Skills | Attitudes |
|---|---|---|
| • Enumeration of different ways of conducting physical environment visit<br>• Identifications of requirements for wireless network indoor installation<br>• Description of physical and logical topologies of wireless network<br>• Identification of wireless network designing tools | • using site survey equipment's<br>• designing topologies<br>• designing network topology<br>• estimating installation cost | • having team work<br>• being Attentive<br>• being Questioner and critical thinker |



**Duration: 10 hrs**

**Learning outcome 2 objectives**:



By the end of the learning outcome, the trainees will be able to:

1. Conduct properly site survey according to user requirements
2. Identify clearly wireless network indoor installation requirements based on site survey findings
3. Design correctly Wireless Network Blueprint in accordance with site condition
4. Estimate correctly network installation cost based on its requirements.



**Resources**

| Equipment | Tools | Materials |
|---|---|---|
| • Laptop<br>• Smart phones<br>• Access point<br>• Switch | • Spectrum analyser<br>• Cisco packet tracer<br>• GNS3<br>• Edraw max<br>• Tape measure | • Internet bundles<br>• Pencil/pen<br>• Notebook/papers |

## Indicative content 2.1: Identification of Wireless Network Indoor Installation Requirements

**Duration: 4**

### Theoretical Activity 2.1.1: Description of wireless network site survey concepts

**Tasks:**

1. In small groups, you are requested answer the following questions
    i. What is a wireless network site survey and its main objectives?
    ii. Identify main points to consider when making a site survey
    iii. What is the physical site survey?
    iv. Identify the different types of wireless network site survey
    v. Mention the ways of recording site survey
    vi. Illustrate tools and equipment used in site survey by their functions
2. Provide the answer for the asked questions and write them on papers.
3. Present your findings in front of the whole class.
4. Listen to the expert view from trainer and ask questions or clarification if any.
5. Read the key readings 2.1.1 in this manual

### Key readings 2.1.1.: Description of wireless network site survey concepts

**A wireless site survey** is the physical inspection of a site where a wireless radio frequency (RF) network will be installed. The survey assesses the environment to determine wireless coverage, data rates, network capacity, roaming capability, and quality of service and provides site survey insights via Ekahau, AirMagnet, iBwave, etc.

**The goal is to visualize the wireless coverage areas (generally through heatmaps) to know where your signal will be strongest and weakest.**

Also known as WLAN site survey, wireless network survey, RF site survey, or networking site survey

In other words, a survey is conducted to ensure that your business gets the most from your wireless network and saves you money (and frustration) in the long run.

Site surveys can be conducted in any environment: warehouse, office, hospital, hotel, school, etc.

**The two main objectives of a wireless site survey:**

- Finding areas of RF coverage and interference
- Determining the placement of access points (APs)

1. **Identifying requirements for wireless**
    - **network indoor installation is a crucial task that require paying attention: here are key factors to consider while identifying these requirements:**

✓ Site Layout and Environment
- Building Floor Plan: Detailed layout including rooms, partitions, and open spaces.
- Construction Materials: Impact of materials like metal, concrete, or glass on signal propagation.
- Obstructions: Potential barriers such as large furniture, walls, and shelving.

✓ Coverage and Signal Strength
- Device Density: Number of devices to be connected in each area (e.g., laptops, smartphones, IoT devices).
- Coverage Areas: Ensuring complete coverage throughout the building, including hard-to-reach areas.
- Signal Strength: Target signal strength levels for reliable connectivity.

✓ Interference Sources
- External Interference: Presence of other wireless networks or electronic devices that may cause interference.
- Internal Interference: Devices or materials within the building that could affect signal quality.

✓ Access Point Placement
- Optimal Locations: Strategically place access points to maximize coverage and minimize dead spots.
- Mounting Options: Decide between ceiling-mounted or wall-mounted access points based on the environment and design.

✓ Network Infrastructure
- Cabling: Type and routing of Ethernet cables from network switches to access points.
- Power Requirements: Considerations for Power over Ethernet (PoE) or other power solutions for access points.

✓ Performance Requirements
- Bandwidth Needs: Assess required bandwidth based on anticipated usage and application types.

- Wireless Standards: Ensure access points support the latest standards (e.g., Wi-Fi 6) for optimal speed and efficiency.

✓ Scalability
- Future Expansion: Plan for network growth by allowing easy addition of more access points or network devices.
- Flexibility: Design the network to accommodate changes in layout or increased device density.

✓ Security
- Encryption: Implement strong encryption protocols (e.g., WPA3) to protect network traffic.

**Physical Site Survey:** A physical site survey is an essential process for planning and deploying wireless networks. It involves a hands-on assessment of the physical environment where the wireless network will be installed. Here's an overview of what a physical site survey entail:

1. **Purpose**:
   - **Identify Optimal Access Point Locations**: Determine the best locations for placing wireless access points to ensure comprehensive coverage and minimize interference.
   - **Assess Physical Obstacles**: Evaluate the impact of walls, furniture, and other structures on wireless signal propagation.
2. **Process**:
   - **Site Walkthrough**: Conduct a detailed walkthrough of the site, noting key factors like room dimensions, building materials, and potential sources of interference.
   - **Signal Measurement**: Use tools to measure signal strength and quality at various points to identify coverage gaps and strong signal areas.
   - **Interference Analysis**: Identify and analyze sources of interference such as other wireless devices, electronic equipment, and neighboring networks.
3. **Tools and Equipment**:
   - **Spectrum Analyzers**: To detect and analyze wireless signals and identify sources of interference.
   - **Site Survey Software**: For mapping out signal strength and coverage visually.
   - **Signal Strength Meters**: To measure the strength of the wireless signal at different locations.
4. **Outcome**:
   - **Coverage Plan**: Develop a detailed plan showing the placement of access points and any necessary adjustments to optimize network performance.

- **Installation Recommendations**: Provide guidelines for the installation process to ensure the network meets coverage and performance requirements.

**different types of wireless network site surveys:**

- **Predictive Site Survey:**

A predictive site survey is performed without field measurements and relies on computer software, floor plans, and Wi-Fi standards. RF software simulates the building and predicts the number of APs and locations needed to meet the required coverage.

Predictive site surveys do not require an onsite visit, so they are typically used when deploying Wi-Fi in a new space. Predictive surveys are beneficial when budgeting but are only as reliable as the information provided. With this in mind, it's best to do a passive site survey once a site's built to arm the engineer or designer with as much information as possible.

Predictive Site Survey **Uses** software tools to model and simulate wireless coverage based on building plans and environmental factors. It predicts the performance and coverage of a wireless network before physical installation. It is used Initial planning and design of wireless networks.

**Benefits of Predictive Site Surveys**

- **Cost-Effective:** Predictive Site Surveys are less expensive than on-site surveys since they do not require travel or physical presence.
- **Time-Saving:** These surveys can be conducted quickly, making them ideal for tight project timelines.
- **Early Planning:** Predictive Site Surveys allow for early network design and planning, providing a preliminary layout that can be adjusted during on-site validation

**When to Use Predictive Site Surveys**

- **New Construction Projects**: For new buildings or renovations, Predictive Site Surveys provide a head start on network design before the physical space is available for on-site surveys.
- **Budget Constraints:** When budget constraints limit the ability to conduct on-site surveys, Predictive Site Surveys offer a viable alternative for preliminary network design.
- **Initial Planning:** During the initial planning stages of a network deployment, Predictive Site Surveys can provide valuable insights into the optimal number and placement of access points

- **Active Site Survey:** An active survey is similar to a passive survey, but the survey tools are connected to the network's access points. This allows for an array of information to be gathered, like network traffic, round-trip-time (RTT), throughput, access point locations, and upload/download speeds.

Active site surveys are usually **used to troubleshoot Wi-Fi networks and measure the real-world performance of your network.** The active site surveys are generally more expensive and need to be conducted during business hours for the most accurate reporting. Still, it collects information the other two survey types can't provide.

**Active Site Survey:** Involves deploying temporary access points and measuring the actual wireless performance in the environment. It gathers real-time data on signal strength, coverage, and interference. It validates and optimizes wireless network performance in a live environment.

**Benefits of Active Site Surveys**

- **Real-Time Performance Metrics:** Active Site Surveys provide real-time data on network performance, which is essential for identifying issues like low throughput, high latency, and packet loss.
- **User Experience Simulation:** By actively transmitting data, these surveys simulate actual user experiences, making them invaluable for assessing network performance under real-world conditions.
- **Detailed Analysis:** The data collected during Active Site Surveys allows for a detailed analysis of network performance, helping to pinpoint specific areas that require improvement.

**When to Use Active Site Surveys**

- **Post-Deployment Verification:** After installing a new wireless network, an Active Site Survey can verify that the network meets performance expectations and identify any areas that may need adjustments.
- **Troubleshooting:** When users report connectivity issues, an Active Site Survey can help diagnose the problem by providing detailed performance metrics.
- **Capacity Planning:** For environments with varying user loads, such as conference centers or stadiums, Active Site Surveys can help plan for capacity and ensure consistent performance during peak usage times

- **Passive Site Survey:**

With passive surveys, we do a physical survey of the building to collect data. A physical walk-through of the site ensures that all factors that may impact network performance are accounted for.

The surveying tool is not connected to the Wi-Fi network during the walk-through and only listens to the Wi-Fi environment. The survey software scans specific channels, Wi-Fi networks, and networks to measure signal strength, signal-to-noise ratios, rogue access points, and interference.

Passive site surveys are performed once a space is built, both before a network is operational and routinely after going live to monitor and assess performance.

**Passive Site Survey:** Involves measuring the existing wireless signals without transmitting data, typically using a spectrum analyser.
It assesses the current wireless environment, including interference and signal strength from existing networks.
It is used to identify potential sources of interference and assess the existing wireless environment before network changes.

**Benefits of Passive Site Surveys**

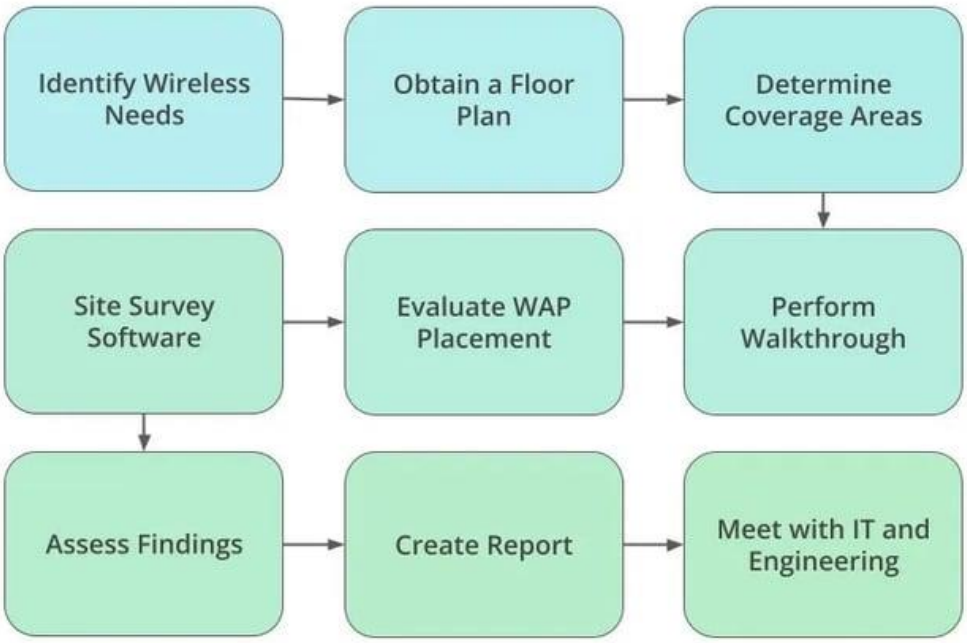
- **Comprehensive RF Analysis:** Passive Site Surveys provide a detailed view of the RF environment, including signal strength, interference sources, and channel utilization.
- **Non-Intrusive:** Since no data is transmitted during the survey, it does not interfere with the existing network, making it ideal for environments where minimal disruption is essential.
- **Baseline Assessment:** These surveys are excellent for establishing a baseline of the current RF environment, which can be used for future comparisons and troubleshooting.

**When to Use Passive Site Surveys**

- **Initial Network Design:** Before deploying a new wireless network, a Passive Site Survey can help design the network by identifying optimal access point locations and potential interference sources.
- **Periodic Audits:** Regular Passive Site Surveys can help maintain network performance by identifying changes in the RF environment, such as new sources of interference or changes in signal strength.
- **Compliance and Security:** For environments with strict compliance and security requirements, Passive Site Surveys can help ensure that the network meets regulatory standards and identify unauthorized access points.

**How is a Passive Site Survey Conducted?**

1. The survey engineer will identify your wireless network needs by asking questions like: How fast do you want your internet? How many clients/employees will access the network at once?
2. Obtain a floor plan.
3. Determine what areas need the most coverage.
4. Perform a walkthrough and assess the current infrastructure
5. Evaluate where wireless access points (WAPs) can be installed
6. Run the data collected from the site visit through the survey software of choice
7. Assess the software's findings and make adjustments as needed.
8. Record results and build a report that includes mounting locations, cable paths, the scope of work for installation, hardware required, configuration recommendations, licensing information, etc.
9. Pass report and findings along to the engineering or IT teams




**Post-Installation Site Survey:** Description: Conducted after the wireless network has been installed and configured.

Purpose: To verify that the network meets performance expectations, coverage requirements, and to identify any issues.

Use Case: To ensure the installed network is functioning as intended and to make any necessary adjustments.

**Validation Site Survey:** Similar to post-installation but may include more in-depth testing of specific applications or devices. It confirms that the network supports specific use cases, applications, and device types effectively.

It validates the network's ability to handle specific traffic loads and performance requirements.

**Heat Map Survey:** Uses specialized tools to create a visual representation of signal strength and coverage throughout the area. It provides a detailed visual map showing the wireless signal strength and coverage.

It is used to identify areas with weak or strong signals and to adjust the placement of access points for optimal coverage.

The cost of wireless survey will depend on the following Factors:

- Size of the building (square footage)
- Type of environment (warehouse, retail, office space, etc.)
- Where the site is located
- Client-specific requirements (turnaround time, after-hours surveying, etc.)

There are plenty of "low-cost" options for wireless site surveys, but you will get what you pay for. If a professional isn't handling your network, there's a greater chance that it won't be optimized to the fullest. Wireless site surveys give you the edge you need.

When your network signal is strong, your business is strong.

Ensure you're using a qualified Wi-Fi engineer with plenty of experience in the field to conduct a wireless site survey.

**Ways of Recording Site Information for Indoor Wireless Network Installation:**

- **Site Survey Report:**

**Written Documentation**: Prepare a detailed report documenting the site's layout, existing infrastructure, and network requirements. Include sections on signal strength, coverage areas, and any potential obstacles or issues.

**Recommendations:** Provide recommendations for access point placement, network equipment, and configuration settings based on the survey findings.

- **Floor Plans and Layout Diagrams:**

**Annotated Floor Plans:** Overlay survey data on building floor plans to indicate proposed locations for access points, coverage areas, and potential sources of interference.

**Coverage Maps:** Create visual maps showing expected signal strength and coverage patterns throughout the indoor space.

- **Photos and Videos:**

**Photographs:** Take photos of the installation environment, including existing network equipment, potential mounting locations for new devices, and areas of concern.

**Video Walkthroughs:** Record videos of the installation area to document physical conditions and context, which can be useful for planning and troubleshooting.

- **Site Survey Software**:

**Digital Tools:** Use specialized site survey software to capture and analyse signal strength, coverage, and interference in real-time. This software often includes features for generating heatmaps and detailed site analysis.
**Data Export:** Export survey data into formats compatible with network planning and design tools for further analysis.

- **Manual Logs and Notes:**

**Field Notes:** Maintain handwritten or digital notes during the site survey to record observations, measurements, and specific details about the site conditions and infrastructure.
**Log Sheets:** Use structured log sheets to record data such as signal strength readings, noise levels, and locations of network equipment.

- **Measurements and Readings:**

**Signal Strength Measurements:** Document signal strength readings at various points using tools like signal strength meters or Wi-Fi analyzers.
**Interference Analysis:** Record data on sources of interference and their impact on network performance using spectrum analyzers or similar tools.

**list of tools and equipment used in the site survey of wireless network indoor installations, along with their functions:**

- **Wi-Fi Analysers**

**Function:** Measure and analyse Wi-Fi signal strength, channel utilization, and overall network performance. They help identify coverage gaps and interference sources.
**Examples:** Net Spot, Ekahau Site Survey

- **Spectrum Analysers**

**Function:** Detect and analyse the electromagnetic spectrum to identify sources of interference (both Wi-Fi and non-Wi-Fi). They help in troubleshooting and optimizing network performance by revealing signals that may affect Wi-Fi.
**Examples:** Wi-Spy (MetaGeek), Anritsu MS2720T

- **Signal Strength Meters**

**Function:** Measure the strength of the Wi-Fi signal at different locations to determine optimal placement for access points and ensure adequate coverage.
**Examples:** Fluke Networks AirCheck G2, NetAlly LinkRunner

- **Floor Plan and Mapping Software**

**Function:** Create and annotate detailed floor plans and maps of the installation site. Useful for planning access point placement and visualizing coverage areas.
**Examples:** AutoCAD, Microsoft Visio

- **Site Survey Software**

**Function:** Collect, analyze, and record data on signal strength, coverage, and interference. Generate heatmaps and detailed reports to aid in network design and optimization.

**Examples:** Ekahau Pro, AirMagnet Survey

- **Distance Measuring Tools**

**Function:** Measure distances within the site to ensure accurate placement of network equipment and assess the layout of the installation environment.

**Examples:** Laser distance measurers

- **Laptops and Tablets**

**Function:** Run survey and analysis software, document survey results, and visualize network data. Essential for processing and recording survey findings.

**Examples:** Laptops with Wi-Fi and spectrum analysis software

- **Cables and Connectors**

**Function:** Connect measurement devices to network equipment and ensure proper connectivity during the survey.

**Examples:** Ethernet cables, coaxial cables

- **Cameras**

**Function:** Capture images of the site for documentation purposes, including equipment placement, potential obstacles, and site conditions.

**Examples:** Digital cameras, smartphones

- **Video Recording Devices**

**Function:** Record walkthroughs of the site to provide visual documentation of the environment, layout, and equipment placement.

**Examples:** Camcorders, smartphones



**Practical Activity 2.1.2: Conduct site survey and Record site**




**Task:**

1. Read carefully the given task:

Your school has recently decided to implement a wireless network to support digital learning in classrooms, allowing students and teachers to use tablets, laptops, and other devices for educational activities. The school administration wants to ensure that every classroom, as well as the library and teachers' lounge, has reliable Wi-Fi coverage.

**The school layout is as follow:**

- **Classrooms:** 10 classrooms, each with a teacher's desk, student desks, and a smartboard.
- **Library:** A central area with bookshelves, study tables, and computer stations.

- **Teachers' Lounge:** A space for teachers to relax, prepare lessons, and access the internet.
- **Administrative Offices:** Located near the entrance of the school, where the principal and administrative staff work.
- **Hallways:** Connect all the rooms and are used frequently by students and staff

Currently, the school has a basic Wi-Fi setup, but coverage is limited, and there are several areas where the signal is weak or drops out entirely. As a student learning how to conduct a site survey for wireless network installation, you have been tasked with conducting site visit and identifying the requirements to improve Wi-Fi coverage throughout the school

2. Listen to the instructions and explanation about the task.
3. Receive a floor plan from your trainer or draw your own floor plan based on the given information in the scenario
4. Follow the demonstration on how site survey tools and equipment are used in collecting specific information and applying it.
5. collecting all needed information under trainer guidance by doing the following:
    - Measure the current Wi-Fi signal strength in different areas
    - Note any areas where the Wi-Fi signal is weak or non-existent
    - Test Signal Coverage
    - Analyse Coverage
    - Determine Equipment Needs
6. Repeat the task as demonstrated by trainer
7. Ask questions for clarifications
8. Read the key readings 2.1.2
9. Perform the task provided in application of learning 2.1



**Key readings 2.1.2: Conduct site survey and Record site**

- Site record means documented observations undertaken as an aspect of systematic examination or controlled recovery or discovery of prehistoric or historic places or objects, including locational records, photographs, notes, charts, drawings and descriptions of situations and materials;
- If you record a piece of information or an event, you write it down, photograph it, or put it into a computer so that in the future people can refer to it.
- These are set of instructions you have to consider while conducting site survey and recording the site information:
    - ✓ Understand the Project Scope: Before you start the survey, ensure you have a clear understanding of the project's goals and requirements. What

type of wireless network is being installed (Wi-Fi, cellular, etc.) and what are the expected coverage and performance objectives?

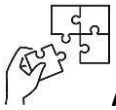
- ✓ Review Building Plans: Familiarise yourself with the building's architectural plans, including floor layouts and any construction materials that may affect wireless signal propagation. This will help you plan your survey effectively.
- ✓ Survey Equipment: Ensure you have the necessary survey equipment, including wireless access points, antennas, signal measurement tools, and a laptop with survey software. Familiarise yourself with their operation.
- ✓ Survey Methodology: Follow a systematic survey methodology, which typically includes walking through the building, taking signal strength measurements at specified intervals or in specific areas, and recording the data.
- ✓ Client Device Considerations: Take into account the types of devices that will connect to the network (e.g., laptops, smartphones, and IoT devices) and their specific requirements for signal strength and bandwidth.
- ✓ Reporting: Create a comprehensive site survey report that includes your findings, recommendations, and a plan for the wireless network installation
- ✓ Client Communication: Maintain open communication with the client or project manager throughout the survey process to ensure alignment with project goals and expectations.

- Remember that conducting a successful site survey for wireless network installation requires attention to detail and a thorough understanding of wireless technology and network design principles.



**Points to Remember**

- Selection and use of tools and equipment as per site survey tool kit and the intended use.
- Site record means documented observations undertaken as an aspect of systematic examination or controlled recovery or discovery of prehistoric or historic places or objects, including locational records, photographs, notes, charts, drawings and descriptions of situations and materials;
- If you record a piece of information or an event, you write it down, photograph it, or put it into a computer so that in the future people can refer to it.



**Application of learning 2.1.**

Bluebell Café is a popular neighbourhood coffee shop located in a single-story building with a comfortable interior that attracts many customers throughout the day. The café recently decided to offer free Wi-Fi to its patrons to enhance their experience. However, the café's owner has received complaints from customers about weak or non-existent Wi-Fi signals in certain areas of the shop, particularly near the back and in the outdoor seating area.

**Café Layout:**

- **Main Indoor Seating Area:** The largest part of the café, with tables and chairs spread throughout the space.
- **Counter and Barista Area:** Located at the front, where customers place their orders.
- **Back Seating Area:** A smaller section with a few cozy tables.
- **Outdoor Patio:** A popular spot for customers who prefer to sit outside.

As a student learning how to conduct a site survey and determine the requirements for improving the Wi-Fi coverage at Bluebell Café

## Indicative content 2.2: Designing Wireless Network Blueprint.

**Duration: 4**

### Theoretical Activity 2.2.1: Describe Wireless Network Topologies

**Tasks:**

1. Observe the blueprint of a computer network and Listen to the explanation from trainer about that blueprint.
2. Then after, answer the following questions:
   i. Define wireless network topology
   ii. Classify wireless network topologies
   iii. Identify types of wireless network topologies
3. Write your findings on reserved place
4. Present your findings in front of a whole class
5. Listen to the clarification from trainer and take short note.
6. Ask questions if necessary
7. Read the key readings and do the application of Learning 2.2

### Key readings 2.2.1.: Describe Wireless Network Topologies

1. **Wireless network topology** refers to the layout and arrangement of devices and their connections within a wireless network. It represents the structure in which devices communicate with one another, either through physical links or logical paths.
2. **Wireless network topologies can be classified into two main categories:**

**Physical Topology: The** actual physical layout of devices, including how they are connected via hardware like cables, routers, and wireless access points.

**Logical Topology:** The logical arrangement that determines how data flows through the network, representing the communication paths between devices, regardless of their physical layout.

**Difference between Physical and Logical Topology**

- Physical topology indicates the arrangement of different elements of a network. It reflects the physical layout of devices and cables to form a connected network. It is concerned with the essentials of the network ignoring minute details like transfer of data and device type. The pattern of arrangement of nodes (computers) and network

cables depends on the ease of installation and setup of the network. It affects cost and bandwidth capacity based on a solution of devices. It takes into account the placement of nodes and the distance between them. Devices can be arranged to form a ring (Ring Topology) or linearly connected in a line called Bus Topology.

**Types of Physical Topology**

- The arrangement of a network that comprises nodes and connecting lines via sender and receiver is referred to as **Network Topology**. On the basis of arrangements, Physical Topology has several types:

  - P2P Topology
  - Bus Topology
  - Ring Topology
  - Star Topology
  - Tree Topology
  - Mesh Topology
  - Hybrid Topology

**Advantages of Physical Topology**

- **Reliability:** Some physical topologies, including the mesh topology, provide redundancy, making sure that community connectivity is maintained although a connection or device fails.
- **Tangible Representation:** Physical topology offers a concrete representation of how devices are physically linked, aiding in troubleshooting and maintenance responsibilities.
- **Scalability:** Certain physical topologies, like the star topology, provide scalability, allowing for smooth expansion by means of including extra devices.

**Disadvantages of Physical Topology**

- **Vulnerability to Physical Damage:** Physical topologies are liable to disruptions resulting from physical damage to cables or networking hardware.
- **Limited Flexibility:** Changing a network's physical architecture can be difficult and involve a significant amount of reconfiguration.
- **Costly Implementation:** Some physical topologies can be quite expensive to build, especially if they include a lot of cables or redundant hardware.

- Logical Topology reflects the arrangement of devices and their communication. It is the transmission of data over physical topology. It is independent of physical topology, irrespective of the arrangements of nodes. It is concerned with intricate details of network like type of devices (switches, routers) chosen and their quality, which affects rate and speed of data packets delivery. The logical topology ensures optimal flow

control that can be regulated within network. The data can either flow in a linear pattern called Logical bus or in form of a circle Logical ring.

**Types of Logical Topology**

Logical topology refers to the manner in which the data is transported and the network device structure is displayed. On the basis of data transfer, logical topology has below mentioned types:

- **Ethernet:** Ethernet is the most widely used LAN technology and is defined under IEEE standards 802.3. The reason behind its wide usability is that Ethernet is easy to understand, implement, and maintain, and allows low-cost network implementation. Also, Ethernet offers flexibility in terms of the topologies that are allowed. Ethernet generally uses a bus topology. Ethernet operates in two layers of the OSI model, the physical layer and the data link layer. For Ethernet, the protocol data unit is a frame since we mainly deal with DLLs. In order to handle collisions, the Access control mechanism used in Ethernet is CSMA/CD.

Although Ethernet has been largely replaced by wireless networks, wired networking still uses Ethernet more frequently. Wi-Fi eliminates the need for cables by enabling users to connect their smartphones or laptops to a network wirelessly. The 802.11ac Wi-Fi standard offers faster maximum data transfer rates when compared to Gigabit Ethernet. However, wired connections are more secure and less susceptible to interference than wireless networks. This is the main justification for why so many companies and organizations continue to use Ethernet.

**TCP/IP:** The TCP/IP model is a fundamental framework for computer networking. It stands for Transmission Control Protocol/Internet Protocol, which are the core protocols of the Internet. This model defines how data is transmitted over networks, ensuring reliable communication between devices. It consists of four layers: the Link Layer, the Internet Layer, the Transport Layer, and the Application Layer. Each layer has specific functions that help manage different aspects of network communication, making it essential for understanding and working with modern networks.

TCP/IP was designed and developed by the Department of Defence (DoD) in the 1960s and is based on standard protocols. The TCP/IP model is a concise version of the OSI model. It contains four layers, unlike the seven layers in the OSI model. In this article, we are going to discuss the TCP/IP model in detail.

TCP/IP model was developed alongside the creation of the ARPANET, which later became the foundation of the modern internet. It was designed with a focus on the practical aspects of networking at the time. The lower-level hardware details and physical transmission medium were largely abstracted away in favour of higher-level networking protocols.

**What Does TCP/IP Do?**

**The main work of TCP/IP** is to transfer the data of a computer from one device to another. The main condition of this process is to make data reliable and accurate so that the receiver will receive the same information which is sent by the sender. To ensure that, each message reaches its final destination accurately, the TCP/IP model divides its data into packets and combines them at the other end, which helps in maintaining the accuracy of the data while transferring from one end to another end. The TCP/IP model is used in the context of the real-world internet, where a wide range of physical media and network technologies are in use. Rather than specifying a particular Physical Layer, the TCP/IP model allows for flexibility in adapting to different physical implementations.

**OSI: OSI stands for Open Systems Interconnection**, where open stands to say non-proprietary. It is a 7-layer architecture with each layer having specific functionality to perform. All these 7 layers work collaboratively to transmit the data from one person to another across the globe. The OSI reference model was developed by **ISO – 'International Organization for Standardization** ', in the year 1984.

The OSI model provides a **theoretical foundation** for understanding **network communication**. However, it is usually not directly implemented in its entirety in real-world **networking hardware** or **software**. Instead, **specific protocols** and **technologies** are often designed based on the principles outlined in the **OSI model** to facilitate efficient data transmission and networking operations

**Advantages of Logical Topology**

- **Flexibility:** Logical topology allows community engineers to design and configure networks based totally on logical necessities, without being constrained by using physical limitations.
- **Efficient Resource Utilization:** Using logical topology to define efficient information paths and routing configurations, the most efficient use of resources is made possible.
- **Ease of Implementation:** Making changes or improvements to a logical topology is frequently less difficult and disruptive than making changes to a community's physical layout.

**Disadvantages of Logical Topology**

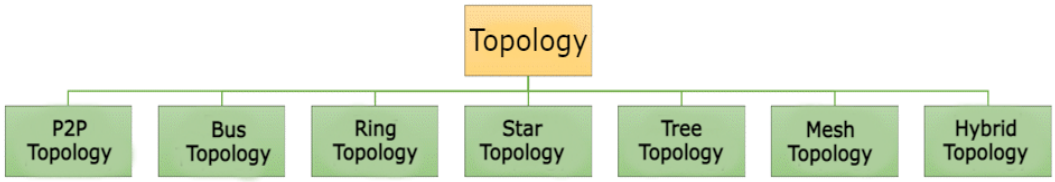
- **Complexity:** Technical expertise and understanding may also be needed to comprehend and manage the complexities of logical topology, which includes protocols and routing configurations.
- **Dependency on Protocols:** Logical topology is contingent upon the proper functioning of network protocols, making networks prone to issues arising from protocol failures or misconfigurations.

- **Potential Security Risks:** Misconfigurations or vulnerabilities in logical topology components, which include routing protocols, may leads to security risk

**Difference between Physical and Logical Topology**

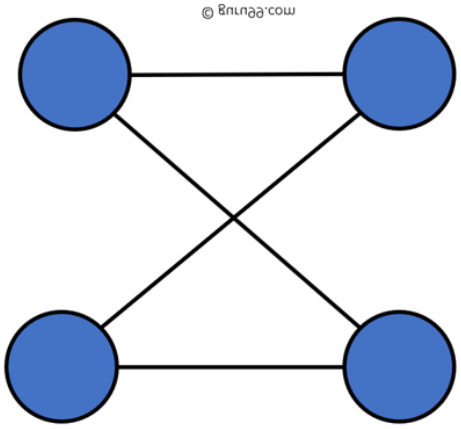
| Physical Topology | Logical Topology |
|---|---|
| Depicts physical layout of network. | Depicts logistics of network concerned with transmission of data. |
| The layout can be modified based on needs. | There is no interference and manipulation involved here. |
| This has major impact on cost, scalability and bandwidth capacity of network based on selection and availability of devices. | This has major impact on speed and delivery of data packets. It also handles flow control and ordered delivery of data packets. |
| It is actual route concerned with transmission. | It is a high-level representation of data flow. |
| Physical connection of the network. | Data path followed of the network. |

**8. Types of Wireless Network Topologies**




**Point to Point (P2P)**

**1. Point-to-point topology** is the easiest of all the network topologies. In this method, the network consists of a direct link between two computers.



P2P Topology Diagram

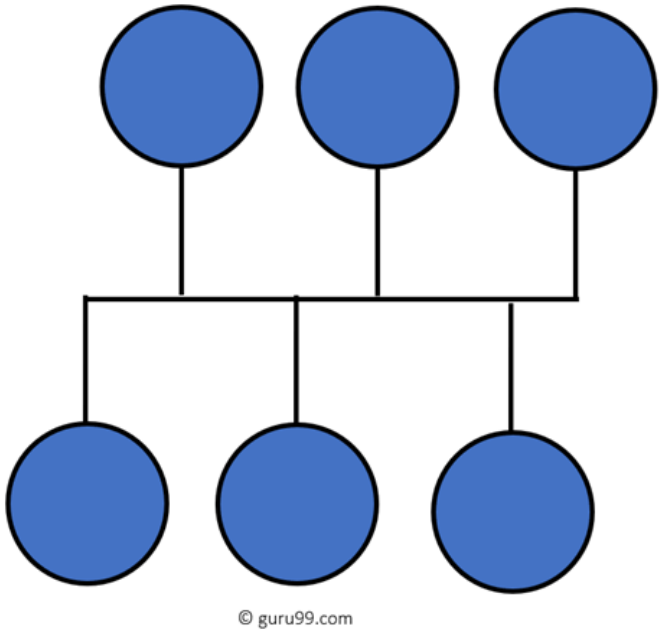
**Advantages**

- This is faster and highly reliable than other types of connections since there is a direct connection.
- No need for a network operating system
- Does not need an expensive server as individual workstations are used to access the files
- No need for any dedicated network technicians because each user sets their permissions

**Disadvantages**

- The biggest drawback is that it only be used for small areas where computers are in close proximity.
- You can't back up files and folders centrally
- There is no security besides the permissions. Users often do not require to log onto their workstations.

2. **Bus Topology**




Bus Topology Diagram

Bus topology uses a single cable which connects all the included nodes. The main cable acts as a spine for the entire network. One of the computers in the network acts as the computer server. When it has two endpoints, it is known as a linear bus topology.

**Advantages**

Here are pros/benefits of using a bus topology:

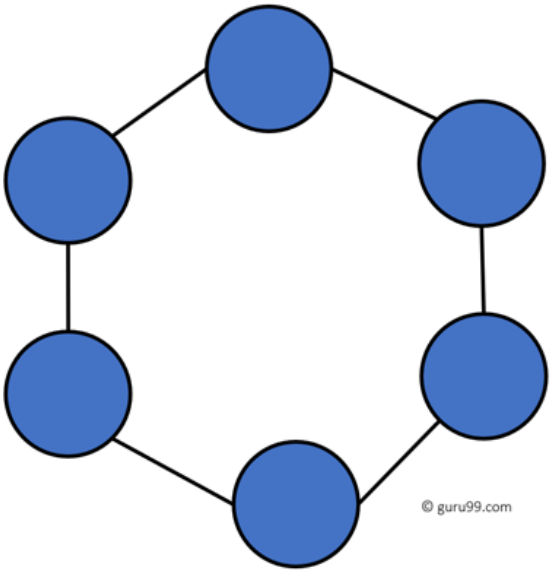
- Cost of the cable is very less as compared to other topology, so it is widely used to build small networks.
- Famous for LAN network because they are inexpensive and easy to install.
- It is widely used when a network installation is small, simple, or temporary.
- It is one of the passive topologies. So computers on the bus only listen for data being sent, that are not responsible for moving the data from one computer to others.

**Disadvantages**

The cons/drawbacks of bus topology:

- In case if the common cable fails, then the entire system will crash down.
- When network traffic is heavy, it develops collisions in the network.
- Whenever network traffic is heavy, or nodes are too many, the performance time of the network significantly decreases.
- Cables are always of a limited length.

### 3. Ring Topology




Ring Topology Diagram

In a ring network, every device has exactly two neighboring devices for communication purpose. It is called a ring topology as its formation is like a ring. In this topology, every computer is connected to another computer. Here, the last node is combined with a first one.

This topology uses token to pass the information from one computer to another. In this topology, all the messages travel through a ring in the same direction.

**Advantages**

Here are pros/benefits of ring topology:

- Easy to install and reconfigure.
- Adding or deleting a device in-ring topology needs you to move only two connections.
- The troubleshooting process is difficult in a ring topology.
- Failure of one computer can disturb the whole network.
- Offers equal access to all the computers of the networks
- Faster error checking and acknowledgment.

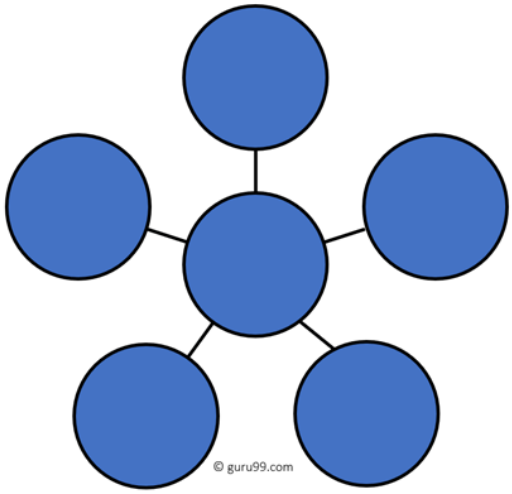
**Disadvantages**

Here are drawbacks/cons of ring topology:

- Unidirectional traffic.
- Break in a single ring can risk the breaking of the entire network
- Modern days high-speed LANs made this topology less popular.

- In the ring, topology signals are circulating at all times, which develops unwanted power consumption.
- It is very difficult to troubleshoot the ring network.
- Adding or removing the computers can disturb the network activity.

### 4. Star Topology




Star Topology Diagram

In the star topology, all the computers connect with the help of a hub. This cable is called a central node, and all other nodes are connected using this central node. It is most popular on LAN networks as they are inexpensive and easy to install.

**Advantages**

Here are pros/benefits of start topology:

- Easy to troubleshoot, set up, and modify.
- Only those nodes are affected, that has failed. Other nodes still work.
- Fast performance with few nodes and very low network traffic.
- In Star topology, addition, deletion, and moving of the devices are easy.

**Disadvantages**

Here are cons/drawbacks of using Star:

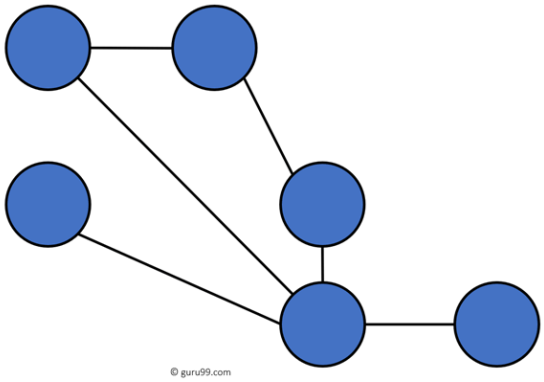
- If the hub or concentrator fails, attached nodes are disabled.
- Cost of installation of star topology is costly.
- Heavy network traffic can sometimes slow the bus considerably.
- Performance depends on the hub's capacity
- A damaged cable or lack of proper termination may bring the network down.

### 5. Mesh Topology

The mesh topology has a unique network design in which each computer on the network connects to every other. It is develops a P2P (point-to-point) connection between all the devices of the network. It offers a high level of redundancy, so even if one network cable fails, still data has an alternative path to reach its destination.

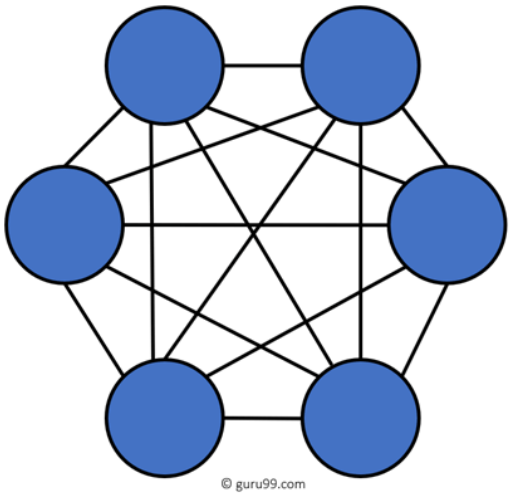
**Types of Mesh Topology**

- **Partial Mesh Topology:** In this type of topology, most of the devices are connected almost similarly as full topology. The only difference is that few devices are connected with just two or three devices.




Partially Connected Mesh Topology

- **Full Mesh Topology:** In this topology, every nodes or device are directly connected with each other.




Fully Connected Mesh Topology

**Advantages**

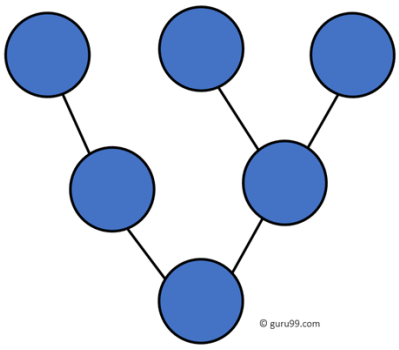
Here, are pros/benefits of Mesh topology

- The network can be expanded without disrupting current users.
- Need extra capable compared with other LAN topologies.
- No traffic problem as nodes has dedicated links.
- Dedicated links help you to eliminate the traffic problem.
- A mesh topology is robust.
- It has multiple links, so if any single route is blocked, then other routes should be used for data communication.
- P2P links make the fault identification isolation process easy.
- It helps you to avoid the chances of network failure by connecting all the systems to a central node.
- Every system has its privacy and security.

**Disadvantages**

- Installation is complex because every node is connected to every node.
- It is expensive due to the use of more cables. No proper utilization of systems.
- Complicated implementation.
- It requires more space for dedicated links.
- Because of the amount of cabling and the number of input-outputs, it is expensive to implement.
- It requires a large space to run the cables.

### 6. Tree Topology




Tree Topology

Tree topologies have a root node, and all other nodes are connected which form a hierarchy. So it is also known as hierarchical topology. This topology integrates various star topologies together in a single bus, so it is known as a Star Bus topology. Tree topology is a very common network which is similar to a bus and star topology.

**Advantages**

Here are pros/benefits of tree topology:

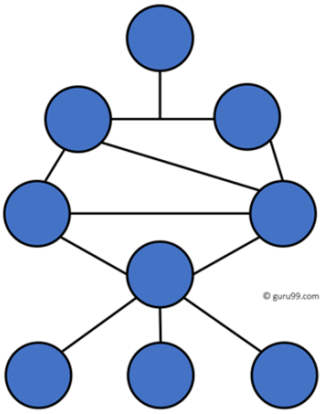
- Failure of one node never affects the rest of the network.
- Node expansion is fast and easy.
- Detection of error is an easy process
- It is easy to manage and maintain

**Disadvantages**

Here are cons/drawback of tree topology:

- It is heavily cabled topology
- If more nodes are added, then its maintenance is difficult
- If the hub or concentrator fails, attached nodes are also disabled.

### 7. Hybrid Topology




Hybrid Topology

Hybrid topology combines two or more topologies. You can see in the above architecture in such a manner that the resulting network does not exhibit one of the standard topologies.

For example, as you can see in the above image that in an office in one department, Star and P2P topology is used. A hybrid topology is always produced when two different basic network topologies are connected.

**Advantages**

The advantages/pros using Hybrid topology:

- Offers the easiest method for error detecting and troubleshooting
- Highly effective and flexible networking topology

- It is scalable so you can increase your network size

**Disadvantages**

- The design of hybrid topology is complex
- It is one of the costliest processes

**How to select a Network Topology?**

Here are some important considerations for selecting the best topology to create a network in your organization:

- Bus topology is surely least expensive to install a network.
- If you want to use a shorter cable or you planning to expand the network is future, then star topology is the best choice for you.
- Fully mesh topology is theoretically an ideal choice as every device is connected to every other device.
- If you want to use twisted pair cable for networking, then you should build star topologies.




**Theoretical Activity 2.2.2: Description of Cisco packet tracer as designing tool**

**Tasks:**

1. Observe the image showed by trainer and listen to his/her clarifications
2. In small groups, discuss on the following question related to Wireless Network Designing Tools.

    i. Explain different considerations in designing a wireless network blueprint and deploy wireless network
    ii. Describe Cisco packet tracer as designing tool

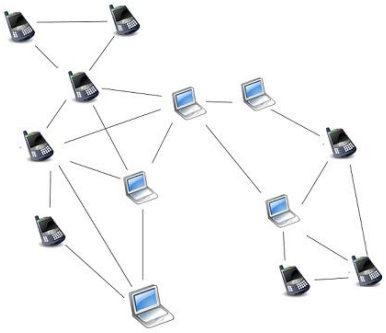
3. Discuss with your members, sharing ideas from different expert groups about the given tasks.
4. Listen to the expert view from trainer, take short note and ask clarification if necessary.
5. In addition, read the key readings2. 2.2 In this manual.

**Key readings 2.2.2.: Identification of wireless technologies application**

**A wireless network** is unbounded and does not suffer from the overhead of wiring. Wireless networks are robust and flexible and lack the need for protection and physical path routing. It is normal behavior for designers to ''copy-paste'' the design of the existing wired networks when designing wireless networks, but this approach needs a little revision as wireless networks require installation talents and less equipment than the wired networks. The following are the wireless network design considerations:

1. Site survey
2. Point-to-point bridging
3. WLAN roaming
4. Use of wireless networks
5. Number of network users
6. Construction materials
7. Types and capabilities of wireless client devices
8. Infrastructure devices in network

**An example of a wireless network set up is shown in the figure below:**

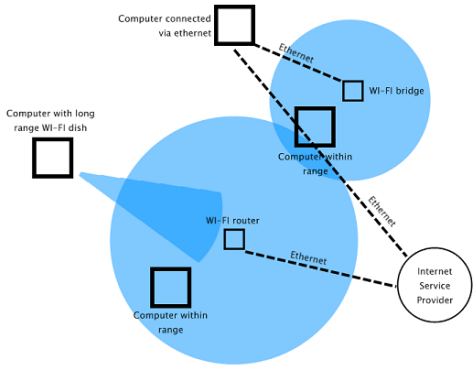


**Site Survey**

The site survey is the initial stage of the wireless design phase as it is very important to survey the site before a wireless network installation. The site survey objectives vary depending on the kind of business looking for a wireless installation. The major objective of the site survey is to identify the optimal sites for wireless network to reduce channel interference and maximizing ranges. The site survey also determines the size of the wireless network to be configured.

**Point-to-point Bridging**

When there are two wireless local area networks (WLANs) that are located at a minimum distance apart and are in the line of sight, it is always advisable to configure wireless bridges; running a network cable between the two buildings is not a good idea. Two Wireless Access Points (WAPs) are needed for creating one two-port bridge. WAPs are configured nodes or hardware devices configured on the local area network which allow wired networks and wireless devices to connect through some wireless standard like Bluetooth or Wi-Fi. The configured wireless access points operate in a dedicated bridge mode-point-to-point, and no longer seem like a wireless access point mode for clients.




**Wireless Local Area Network (WLAN) Roaming**

The cost of installation of a wireless local area network (WLAN) is cheaper than wired networks. The throughput is determined in a wireless local area network by the proximity of the WAPs. Overlapping WAPs are installed by network managers to increase throughput, thereby decreasing the cost. The major requirement is that the overlapping WAP signals should be placed inside non-overlapping channels. These scenarios need WLAN roaming. The WLAN roaming plan considers that when the network users move away from one particular WAP and lose signal strength, their connections must seamlessly move to a stronger-signal WAP.

**Use of Wireless Networks**

In order to identify the number of wireless access points that are needed to configure a wireless network, it is necessary to find the intended use of that network. For example, if you are establishing a wireless network for a casual application like a cybercafé, Wi-Fi is required for social media, internet browsing, checking mails, etc. A much more configuration with multiple wireless access points is needed in the case of larger establishments that run high-end applications including graphics design, database programs and electronic imaging.

**Number of Network Users**

The number of users and their networking devices (laptop, PDA, etc.,) have an impact on the total wireless access points count. Most of the companies have an estimation of three to four networking devices per user. Thus, when a greater number of users access a wireless network, there is a need to configure more wireless access points, based on the situation.

**Construction Materials**

The construction materials and the age of the building to be configured must be determined during site surveys as these two factors have an impact on signal quality. An older building needs a higher number of access points due to the type of construction materials present in both exterior and interior walls.

**Types and Capabilities of Wireless Client Devices**

It is important to identify the types of networking devices to be connected to the wireless network. Certain devices like smartphones and tablets possess specific functionalities that need to be matched to the network requirements. The surveyor identifies the kinds of networking devices by interviewing the users of the network during site surveys.

**These devices may include:**

1. Laptops (Mac/PC)
2. Smartphones & Tablets (Android/iOS)
3. IoT devices
4. Barcode scanners
5. Lighting & Environmental Controls
6. Security Cameras
7. Audio-Visual Conferencing Equipment

**Infrastructure Devices in Network**

During the design of a wireless network, it is crucial to identify whether the company has the equipment and space for installing the new wireless network. Location of wiring or telecom closets and the number of existing Ethernet ports are examined before installing the wireless network.

**Cisco Packet Tracer:**

**Cisco Packet Tracer** is a network emulator developed by Cisco Systems. It is primarily targeted towards students and instructors who are learning about networking. Packet Tracer allows you to create simple to complex network topologies using a drag-and-drop interface and supports a range of Cisco devices, such as routers, switches, firewalls, and wireless access points. Some of the key features of Cisco Packet Tracer are:

- Easy to use drag-and-drop interface
- Support for Cisco devices and NOS images
- Built-in tutorials and labs for learning
- Integration with Cisco NetAcad (an online learning platform)

**Pros:**

- It provides a user-friendly interface for beginners to learn about networking.
- Supports Cisco devices and configurations, providing a high degree of accuracy and detail.
- Provides a built-in assessment tool for testing and verifying network knowledge.

**Cons:**

- It only supports Cisco devices and configurations, limiting its versatility compared to other tools.
- It is an emulator and may not provide the same detail and accuracy as real-world devices.
- Limited to a pre-defined set of features and protocols.
- It may not be suitable for advanced network simulations.
- Trainers and students who are over-rely on packet tracers will have the illusion that the actual config is easy as no actual IOS is installed

**Key Features and Functionalities**

**Educational focus**: Designed with education in mind, Packet Tracer includes guided simulations and tutorials which are crucial for learning fundamental networking concepts.

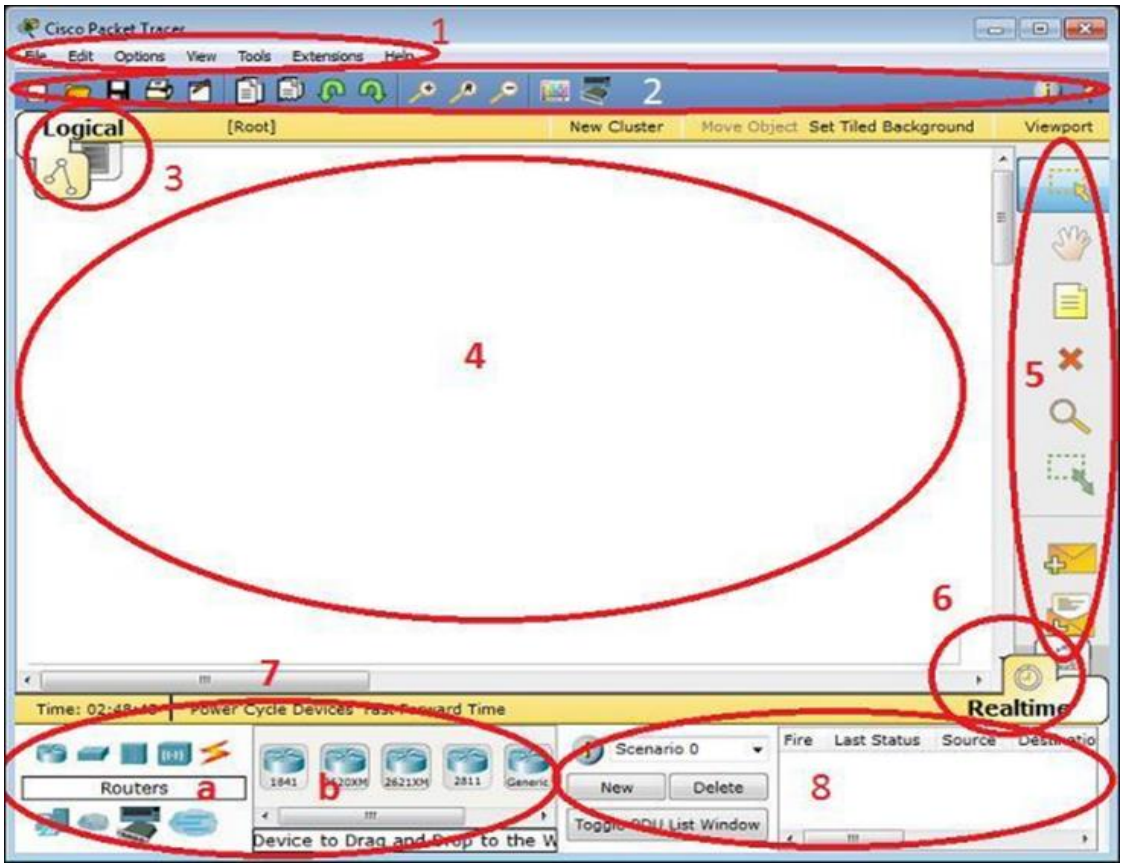
**User-friendly interface:** It offers a straightforward and intuitive interface that helps beginners grasp complex concepts without the overhead of configuring real equipment.

**Scenario-based learning:** Users can practice with pre-designed scenarios that demonstrate various networking challenges and solutions.

**Typical Use Cases in Educational Environments**

**Academic instruction:** Widely used in educational institutions as part of networking courses and Cisco's Networking Academy, providing a foundational understanding of networking.

**Entry-level certification preparation:** Excellent for preparing for Cisco's entry-level certifications, such as CCENT and CCNA, where basic concepts and troubleshooting are tested.




- **The components of the Packet Tracer interface are as follows:**

✓ **Area 1:** Menu bar – This is a common menu found in all software applications; it is used to open, save, print, change preferences, and so on.

✓ **Area 2:** Main toolbar – This bar provides shortcut icons to menu options that are commonly accessed, such as open, save, zoom, undo, and redo, and on the right-hand side is an icon for entering network information for the current network.

✓ **Area 3:** Logical/Physical workspace tabs – These tabs allow you to toggle between the Logical and Physical work areas.

✓ **Area 4:** Workspace – This is the area where topologies are created and simulations are displayed.

✓ **Area 5:** Common tools bar – This toolbar provides controls for manipulating topologies, such as select, move layout, place note, delete, inspect, resize shape, and add simple/complex PDU.

- ✓ **Area 6:** Realtime/Simulation tabs – These tabs are used to toggle between the real and simulation modes. Buttons are also provided to control the time, and to capture the packets.
- ✓ **Area 7:** Network component box – This component contains all of the network and end devices available with Packet Tracer, and is further divided into two areas:
- ✓ **Area 7a:** Device-type selection box – This area contains device categories
- ✓ **Area 7b:** Device-specific selection box – When a device category is selected, this selection box displays the different device models within that category
- ✓ **Area 8:** User-created packet box – Users can create highly-customized packets to test their topology from this area, and the results are displayed as a list.

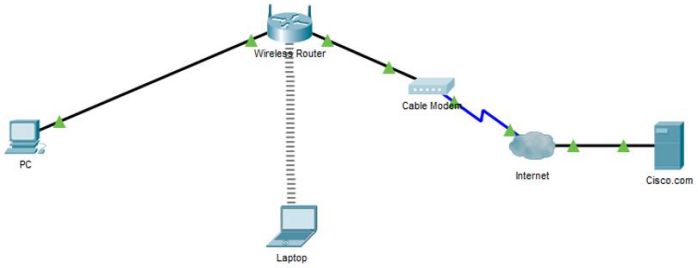





**Practical Activity 2.2.3: Design Wireless network using Cisco packet tracer**

**Task:**

1. Go in the computer lab and design a given image in Packet Tracer from scratch and then save the network as a Packet Tracer Activity File (.pkt).




2. Follow the demonstration on how to design wireless networks in packet tracer
3. Based on the similar task given, repeat the procedure and steps as demonstrated by trainer to perform the task.
4. After that, ensure that all process is followed based on founded output.
5. Read key readings 2.2.3
6. Perform the task provided in application of learning 2.2

**Key readings 2.2.3: Design Wireless network using Cisco packet tracer**

**Create Wireless network Blueprint**

**Part1: design a Simple Network in the Logical Topology Workspace using a packet tracer**

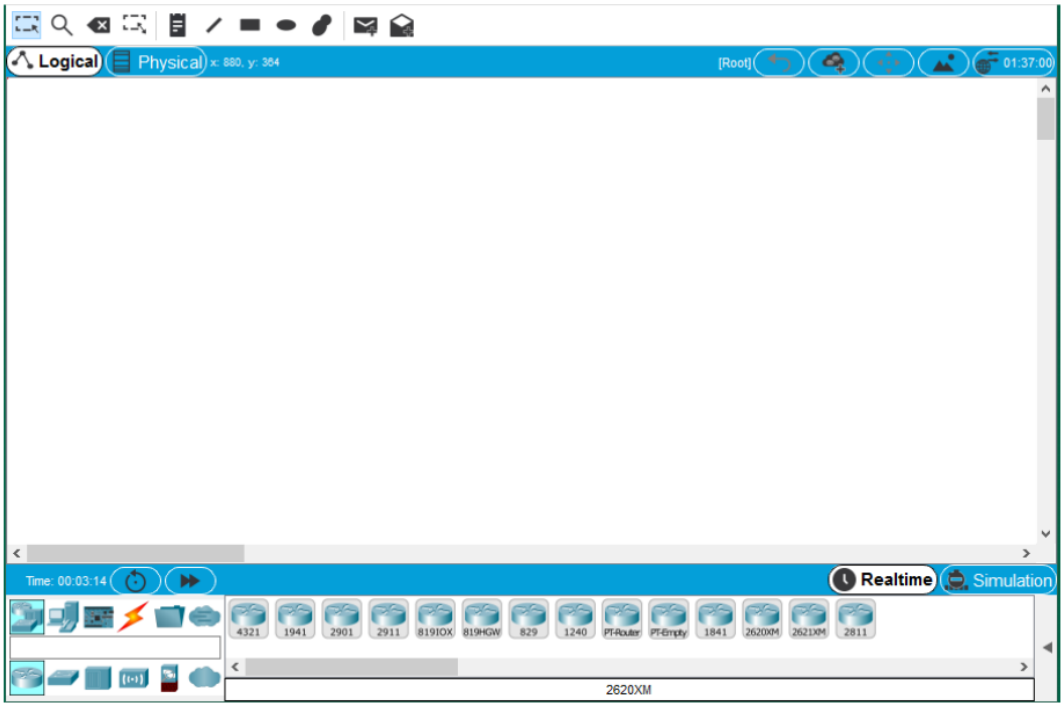
**Step 1: Launch Packet Tracer.**

**a. Launch Packet Tracer on your PC or laptop computer**

Double click on the Packet Tracer icon on your desktop or navigate to the directory that contains the

Packet Tracer executable file and launch Packet Tracer. Packet Tracer should open with a blank default

Logical topology workspace as shown in the figure




**Step 2: Build the topology**

**a. Add network devices to the workspace**

Using the device selection box, add the network devices to the workspace as shown in the topology diagram.

To place a device onto the workspace, first choose a device type from the Device-Type Selection box. Then, click on the desired device model from the Device-Specific Selection box. Finally, click on a location in the workspace to put your device in that location. If you want to

cancel your selection, click the **Cancel** icon for that device. Alternatively, you can click and

drag a device from the **Device-Specific Selection** box onto the workspace.

**b. Add network devices to the workspace.**

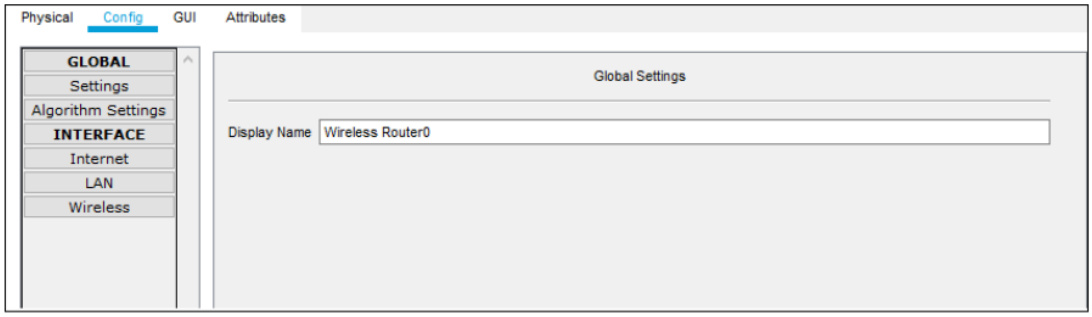
Using the device selection box, add the network devices to the workspace as shown in the topology diagram

To place a device onto the workspace, first choose a device type from the **Device-Type** Selection box. Then, click on the desired device model from the **Device-Specific Selection** box. Finally, click on a location in the workspace to put your device in that location. If you want to cancel your selection, click the **Cancel** icon for that device. Alternatively, you can click and drag a device from the Device-Specific

**Selection** box onto the workspace.

**c. Change display names of the nework devices.**

To change the display names of the network devices click on the device icon on the Packet Tracer**. Logical** workspace, then click on the **Config** tab in the device configuration window. Type the new name of the device into the **Display Name** box as show in the figure below.




d. Add the physical cabling between devices on the workspace

Using the device selection box, add the physical cabling between devices on the workspace as shown in the topology diagram.

The PC will need a copper straight-through cable to connect to the wireless router. Select the copper straight-through cable in the device selection box and attach it to the FastEthernet0 interface of the PC and the Ethernet 1 interface of the wireless router

The wireless router will need a copper straight-through cable to connect to the cable modem. Select the copper straight-through cable in the device-selection box and attach it to the Internet interface of the wireless router and the Port 1 interface of the cable modem.

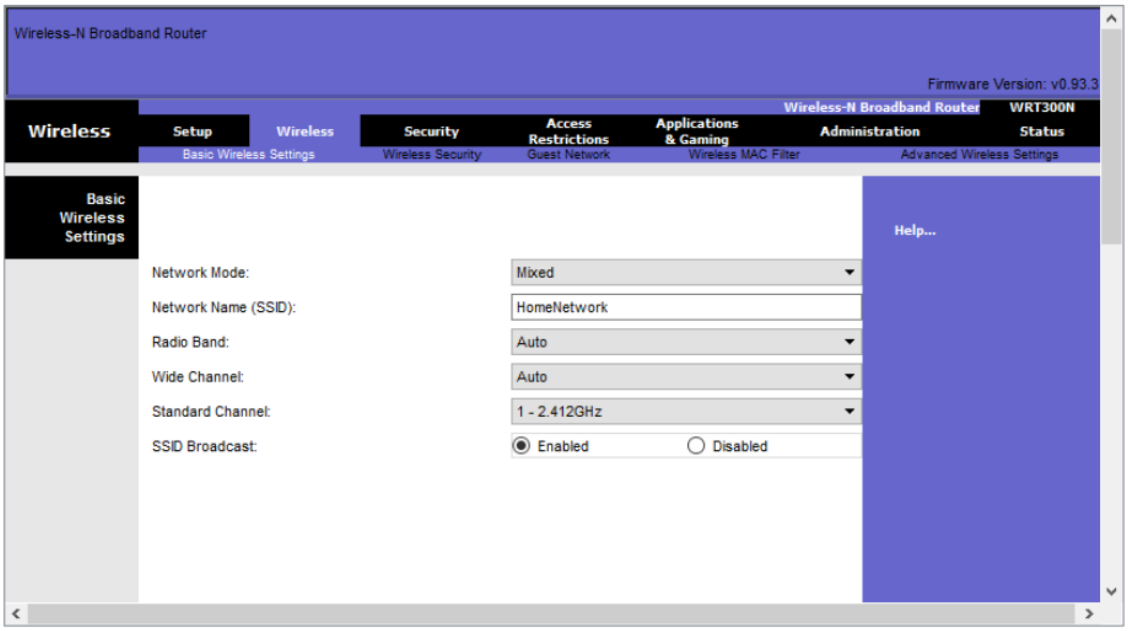
The cable modem will need a coaxial cable to connect to the Internet cloud. Select the coaxial cable in the device-selection box and attach it to the Port 0 interface of the cable modem and the coaxial interface of the Internet cloud. The Interne cloud will need copper straight-through cable to connect to the Cisco.com server. Select the copper straight-through cable in the device-selection box and attach it to the Ethernet interface of the Internet cloud and the FastEthernet0 interface of the Cisco.com server

**Part 2: Configure the Network Devices**

**Step 1: Configure the wireless router**

a. Create the wireless network on the wireless router

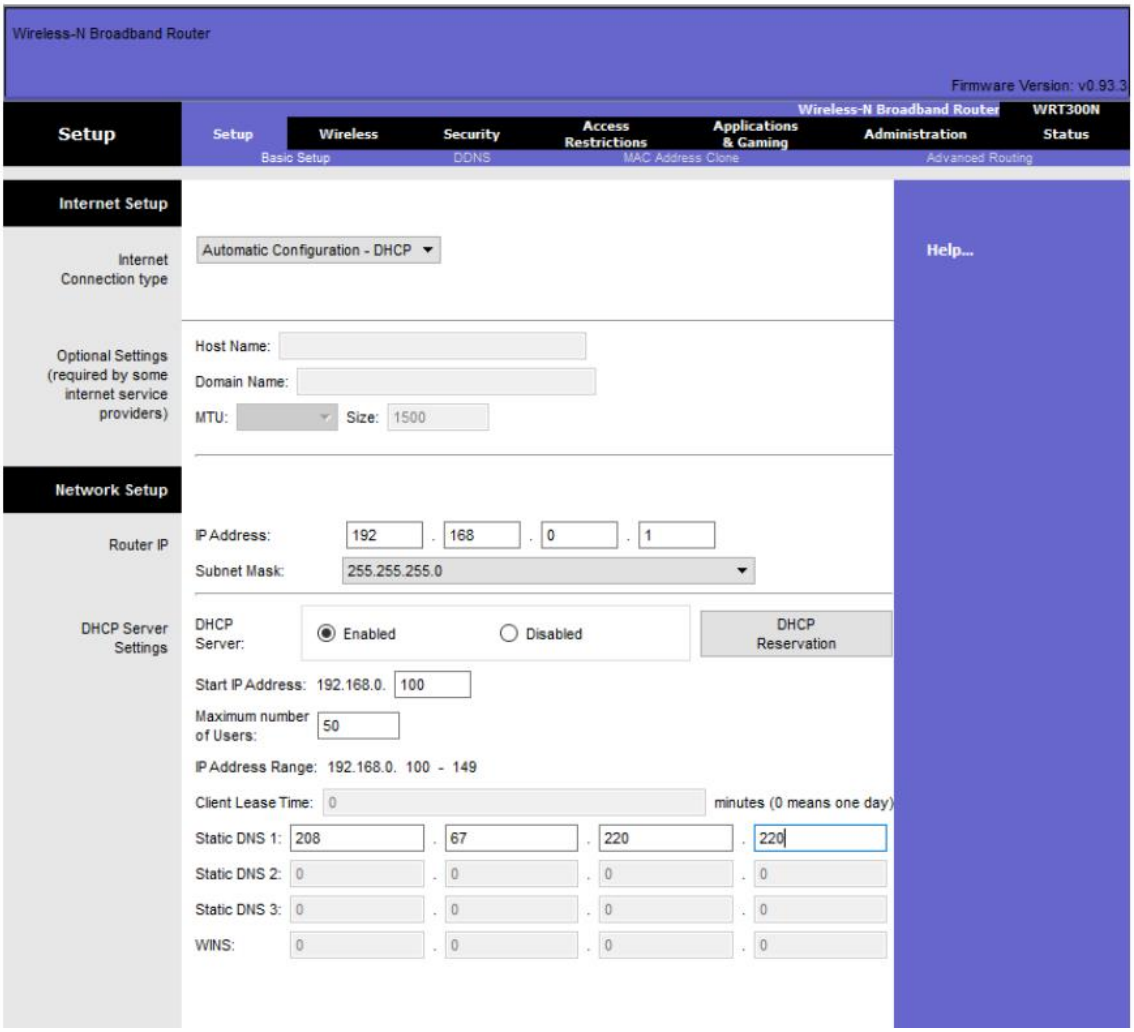
Click on the **Wireless Router** icon on the Packet Tracer **Logical** workspace to open the device configuration window. In the wireless router configuration window, click on the **GUI** tab to view configuration options for the wireless router. Next, click on the **Wireless** tab in the **GUI** to view the wireless settings. The only setting that needs to be changed from the defaults is the **Network Name (SSID).** Here, type the name "Home Network" as shown in the figure




Configure the Internet connection on the wireless router, Click on the **Setup** tab in the wireless router GUI.

In the **DHCP Server** settings verify that the **Enabled** button is selected and configure the static IP address of the DNS server as 208.67.220.220 as shown in the figure.

b. Click on the **Save Settings** tab.




**Step 2: Configure the laptop**

a. Configure the Laptop to access the wireless network

Click on the Laptop icon on the Packet Tracer **Logical** workspace and in the laptop configuration windows select the Physical tab. In the **Physical** tab you will need to remove the Ethernet copper module and replace it with the Wireless WPC300N module.

To do this, you first power the laptop off by clicking the power button on the side of the laptop. Then remove the currently installed Ethernet copper module by clicking on the module on the side of the laptop and dragging it to the **MODULES** pane on the left of the laptop window. Then install the Wireless WPC300N module by clicking on it in the **MODULES** pane and dragging it to the empty module port on the side of the laptop. Power the laptop back on by clicking on the laptop power button again. With the wireless module installed, the next task is to connect the laptop to the wireless network.

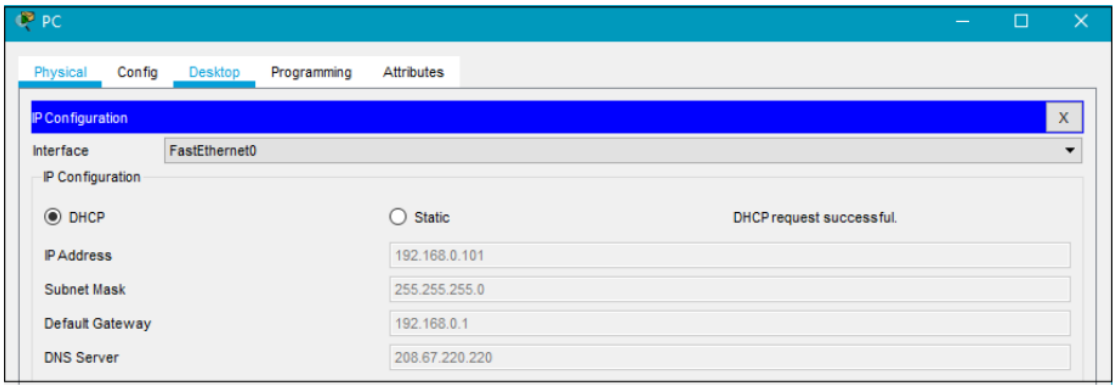
Click on the **Desktop** tab at the top of the Laptop configuration window and select the **PC Wireless** icon. Once the Wireless-N Notebook Adapter settings are visible, select the **Connect** tab. The wireless network "Home Network" should be visible in the list of wireless networks as shown in the figure. Select the network, and click on the **Connect** tab found below the **Site Information pane**.



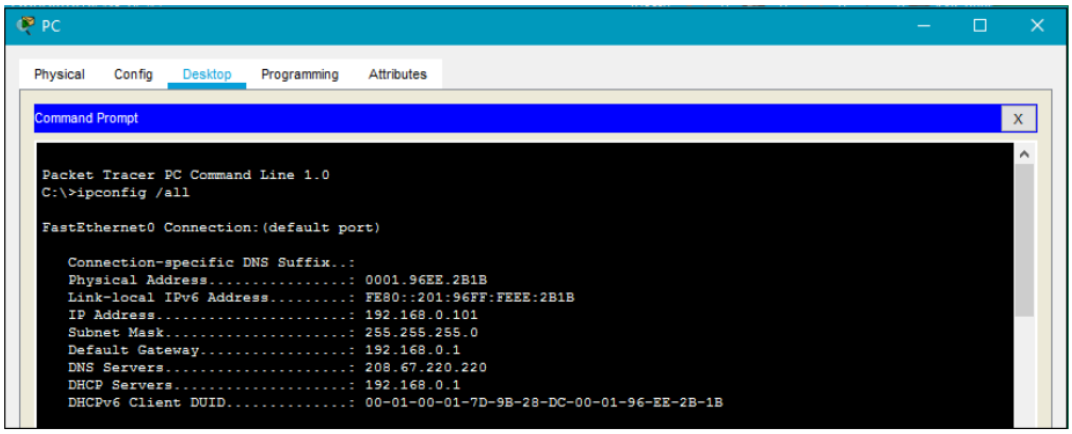

**Step 3: Configure the PC**

a. Configure the PC for the wired network

Click on the **PC** icon on the Packet Tracer **Logical** workspace and select the **Desktop** tab and then the IP Configuration icon. In the **IP Configuration** window, select the DCHP radio button as shown in the figure so that the PC will use **DCHP** to receive an IPv4 address from the wireless router. Close the IP Configuration window.




Click on the Command Prompt icon. Verify that the PC has received an IPv4 address by issuing the **ipconfig /all** command from the command prompt as shown in the figure. The PC should receive an IPv4 address in the 192.168.0.x range.



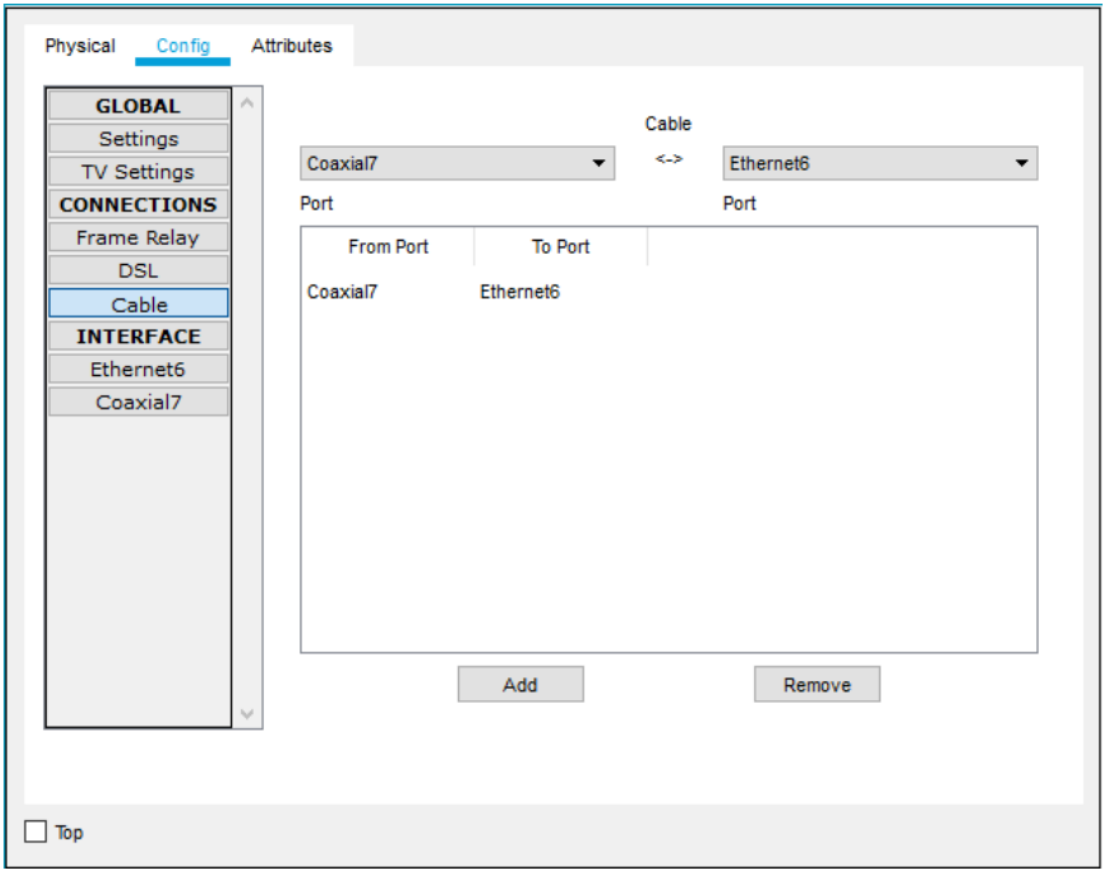

**Step 4: Configure the Internet cloud**

**a. Install network modules if necessary**

Click on the **Internet Cloud** icon on the Packet Tracer **Logical** workspace and then click on the Physical tab. The cloud device will need two modules if they are not already installed. The PT-CLOUD-NM-1CX which is for the cable modem service connection and the PT-CLOUD-NM-1CFE which is for a copper Ethernet cable connection. If these modules are missing, power off the physical cloud devices by clicking on the power button and drag each module to an empty module port on the device and then power the device back on.

**b. Identify the From and To Ports**

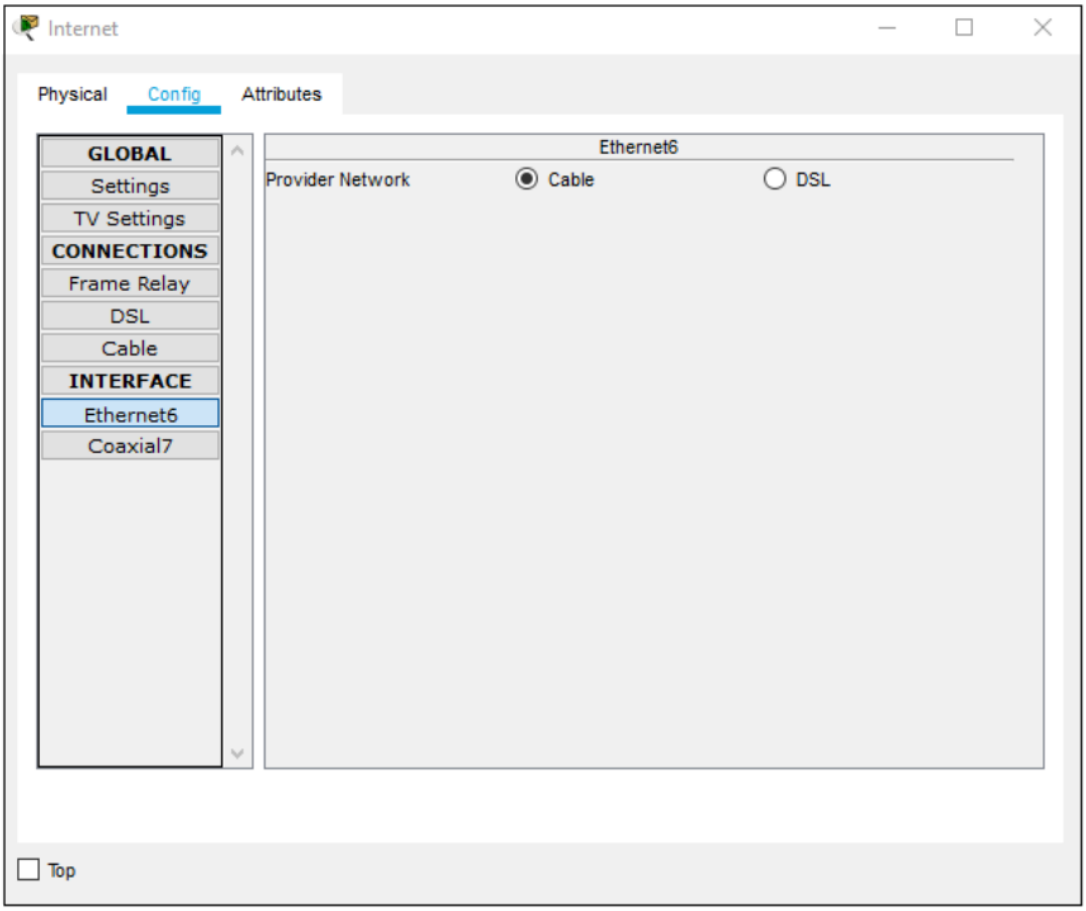
Click on the **Config** tab in the Cloud device window. In the left pane click on **Cable** under **CONNECTIONS.** In the first drop down box choose Coaxial and in the second drop down box choose Ethernet then click the Add button to add these as the **From Port** and **To Port** as shown in the figure.




c. Identify the type of provider

While still in the Config tab click Ethernet under INTERFACE in the left pane. In the Ethernet configuration window select Cable as the Provider Network as shown in the figure **c. Identify the type of provider**

While still in the **Config** tab click Ethernet under **INTERFACE** in the left pane. In the Ethernet configuration window select **Cable** as the Provider Network as shown in the figure




**Step 5: Configure the Cisco.com server**

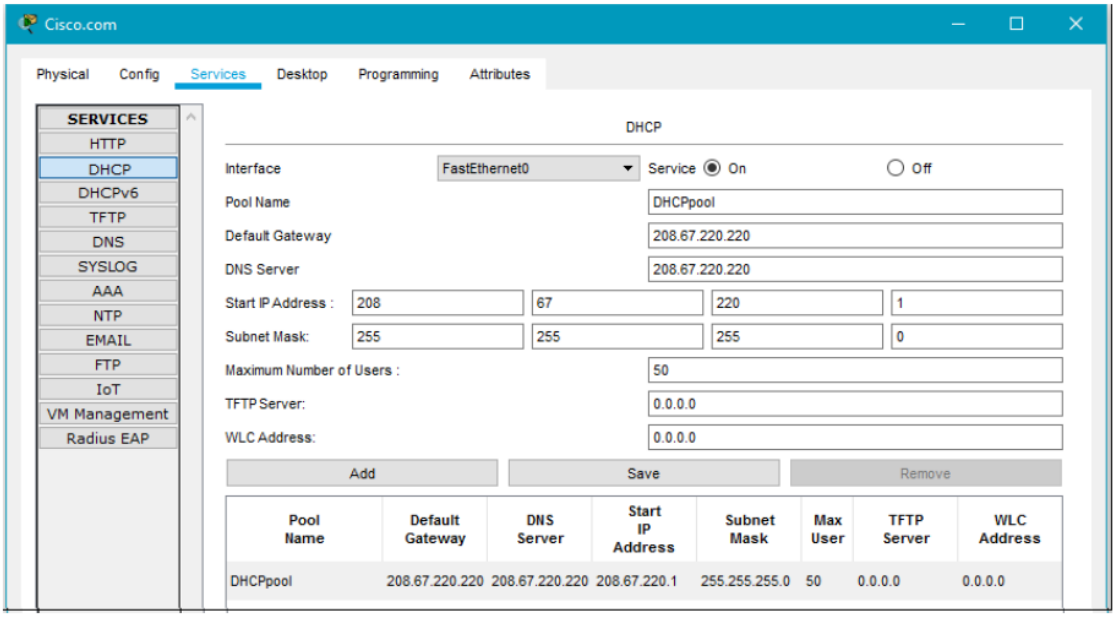
a. Configure the Cisco.com server as a DHCP server

Click on the Cisco.com server icon on the Packet Tracer **Logical** workspace and select the Services tab. Select **DHCP** from the **SERVICES** list in the left pane.

In the DHCP configuration window, configure a DHCP as shown in the figure with the following settings.

- Click **On** to turn the DCHP service on
- Pool name: DHCPpool
- Default Gateway: 208.67.220.220
- DNS Server: 208.67.220.220
- Starting IP Address: 208.67.220.1
- Subnet Mask 255.255.255.0
- Maximum number of Users: 50

Click **Add** to add the pool

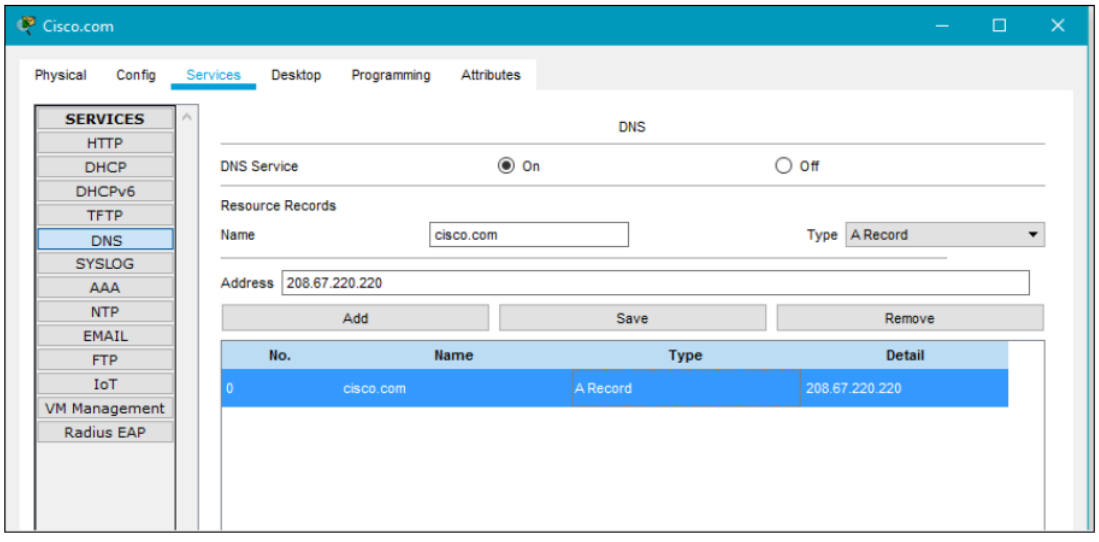



b. Configure the Cisco.com server as a **DNS** server to provide domain name to IPv4 address resolution. While still in the **Services** tab, select **DNS** from the **SERVICES** listed in the left pane.

Configure the DNS service using the following settings as shown in the figure.

- Click On to turn the DNS service on
- Name: Cisco.com
- Type: A Record
- Address: 208.67.220.220

Click **Add** to add the DNS service settings



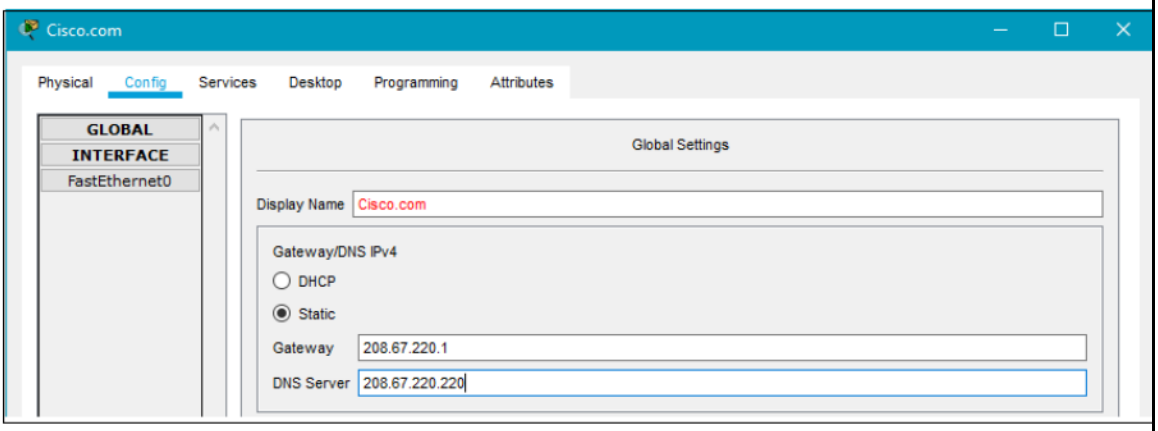

c. Configure the Cisco.com server Global settings.

Select the **Config** tab.

Click on Settings in left pane.

Configure the Global **settings** of the server as follows:

- Select **Static**
- Gateway: 208.67.220.1
- DNS Server: 208.67.220.220

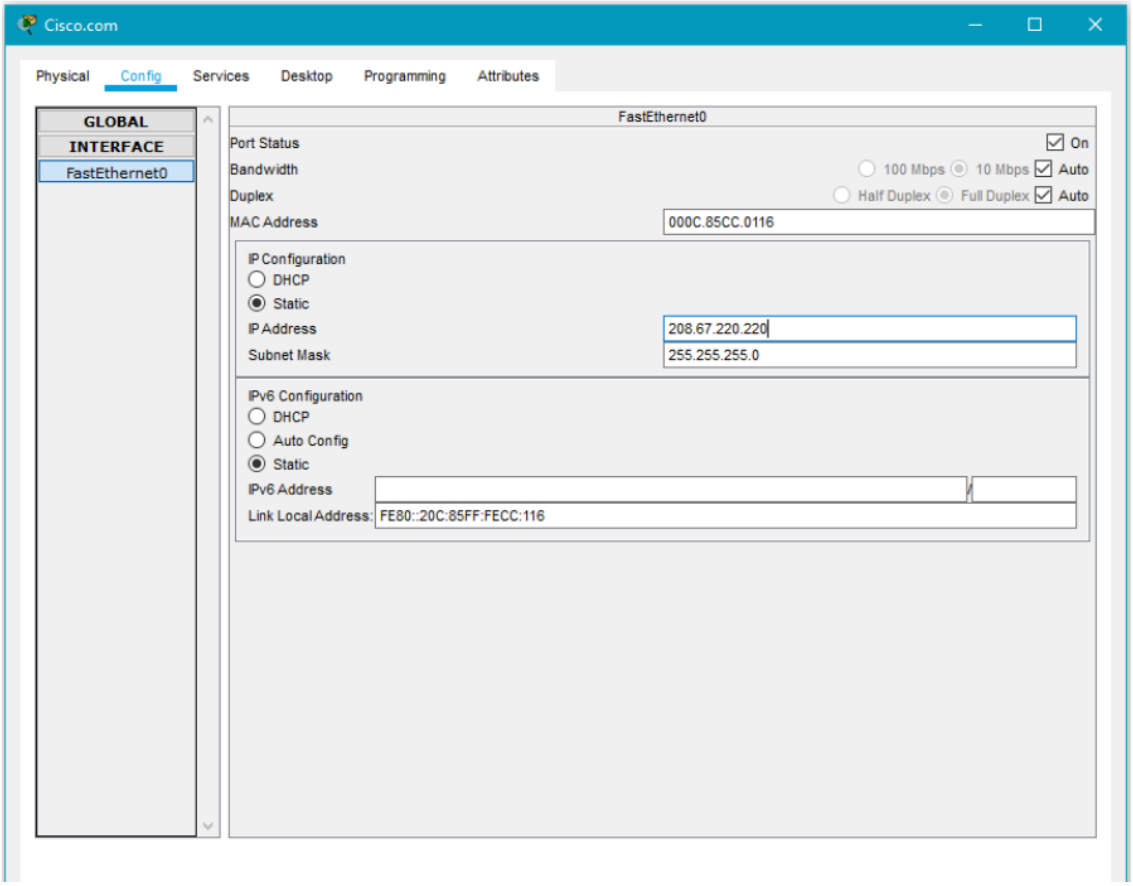



d. Configure the Cisco.com server FastEthernet0 Interface settings.

Click on **FastEthernet** in left pane of the **Config** tab

Configure the FastEthernet Interface settings of the server as follows:

- Select Static under IP Configuration
- IP Address: 208.67.220.220
- Subnet Mask: 255.255.255.0




- **Part 3: Verify Connectivity**

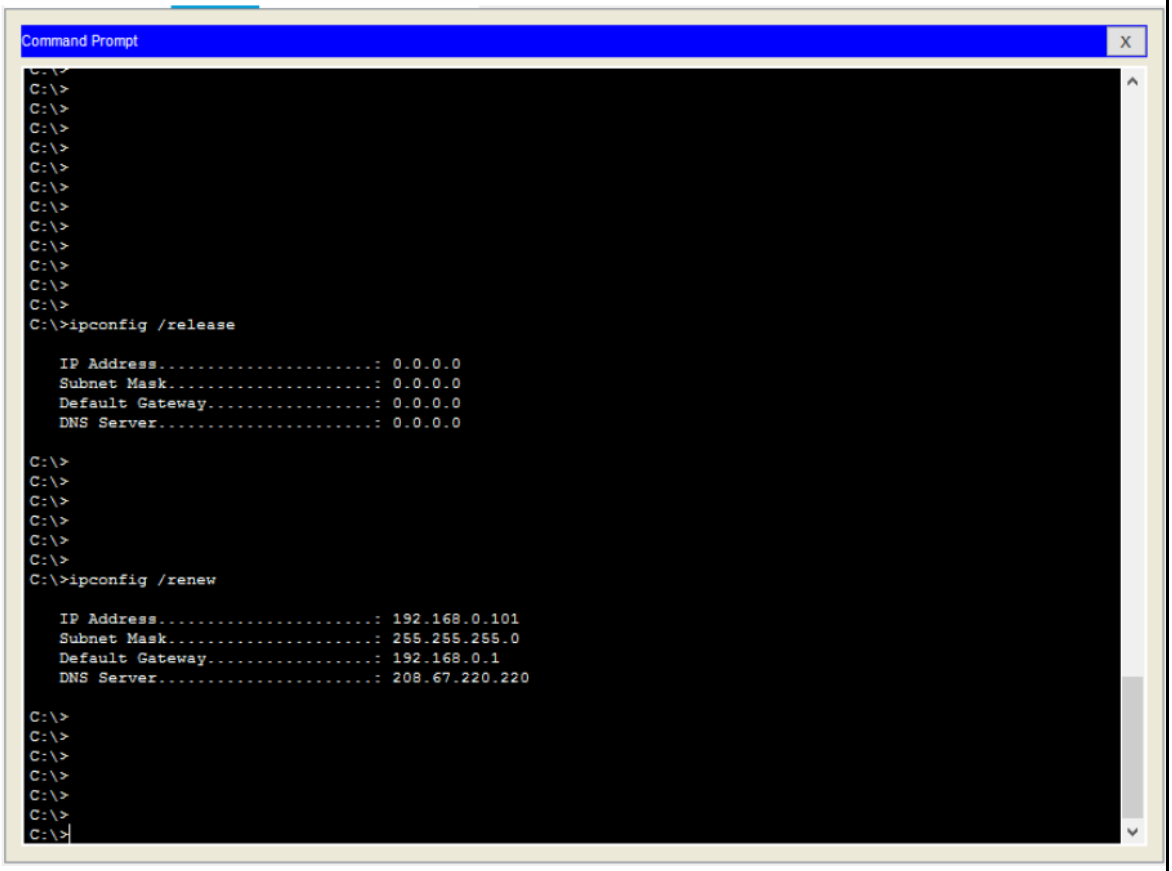
**Step 1: Refresh the IPv4 settings on the PC**

a) Verify that the PC is receiving IPv4 configuration information from DHCP.

Click on the PC on the Packet Tracer Logical workspace and then the select the Desktop tab of the PC configuration window.

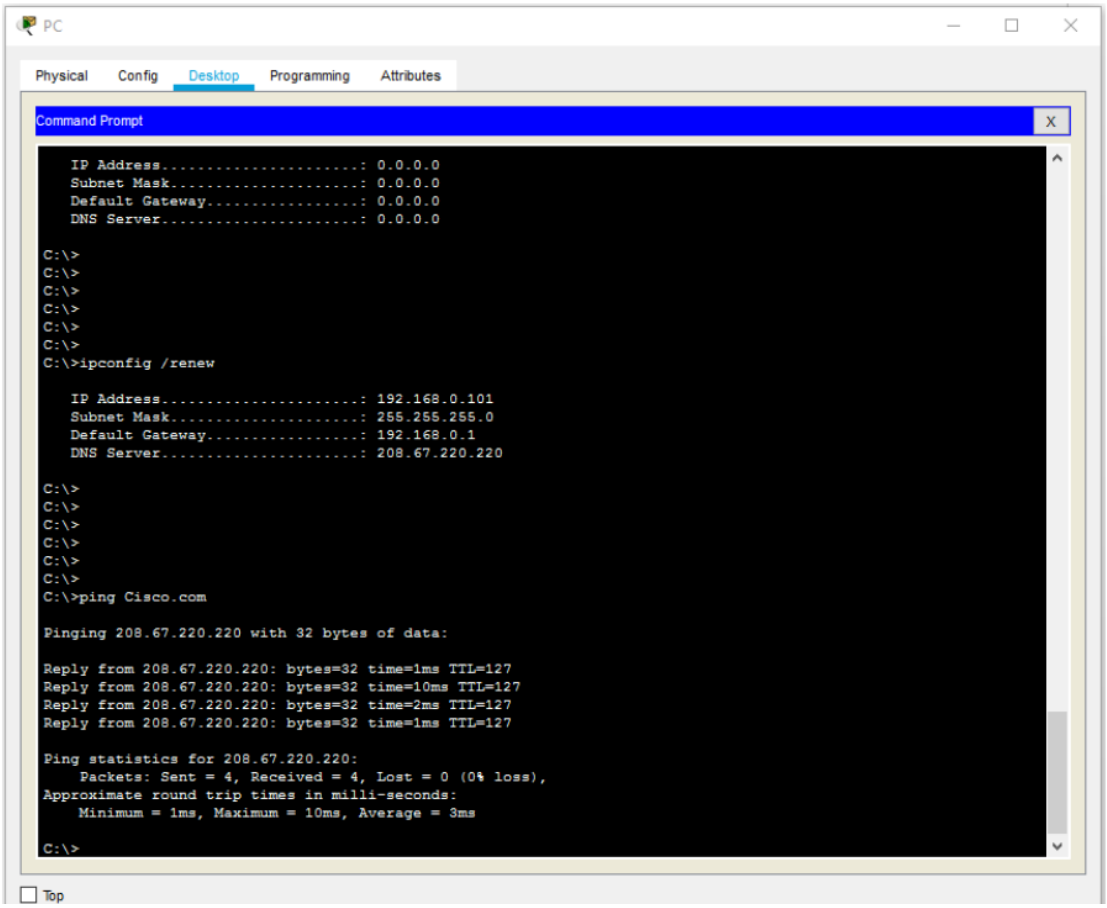
Click on the **Command Prompt** icon

In the command prompt refresh the IP settings by issuing the commands ipconfig /release and then ipconfig /renew. The output should show that the PC has an IP address in the 192.168.0.x range, a subnet mask, a default gateway, and DNS server address as shown in the figure.




b) Test connectivity to the Cisco.com server from the PC

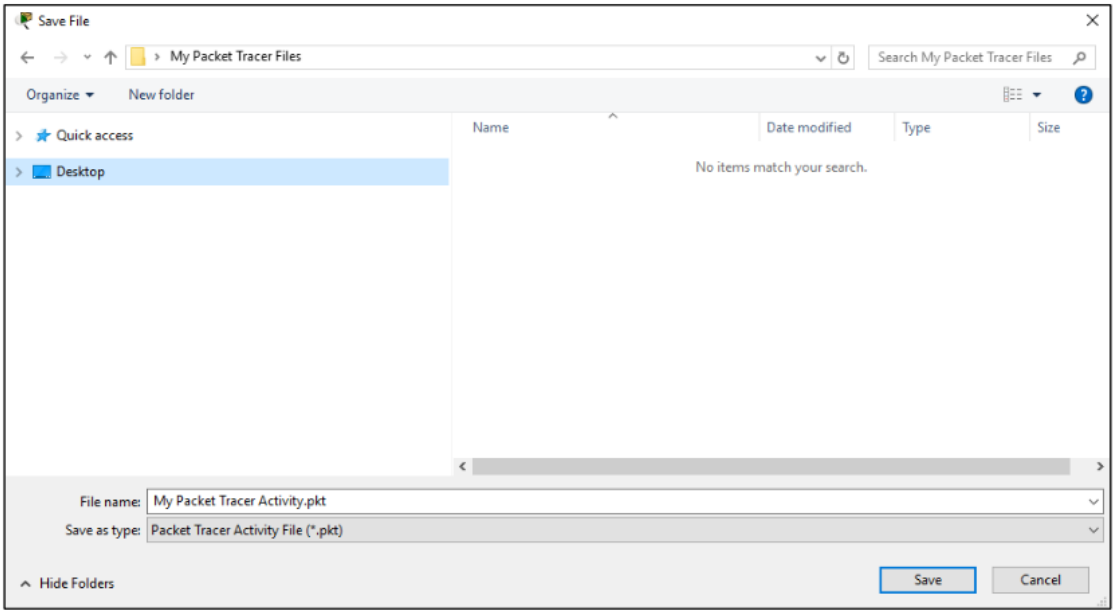
From the command prompt, issue the command ping Cisco.com. It may take a few seconds for the ping to return. Four replies should be received as shown in the figure




**Part 4: Save the File and Close Packet Tracer**

**Step 1: Save the File as a Packet Tracer Activity File (*.pkt)**

To save the completed network, click on **File** in the Packet Tracer menu bar and then select Save A from the dropdown menu. In the the **Save File** window choose a directory to save the file to and give the file an appropriate file name. The Save as type defaults to Packet Tracer Activity File (*.pkt). Click **Save** to save the file




**Step 2: Close Packet Tracer**

To close Packet Tracer you can either click the **"X"** in the top right corner of the Packet Tracer window, or click on **Exit** in the file drop down menu.

**Theoretical Activity 2.2.4: Description of GNS3 as designing tool**

**Tasks:**

1. In small groups, discuss on the following question related to Wireless Network Designing Tools.

    1. Describe GNS3 as designing tool

2. Discuss with your members, sharing ideas from different expert groups about the given tasks.
3. Listen to the expert view from trainer, take short note and ask clarification if necessary.
4. In addition, read the key readings2. 2.2 In this manual.



**Key readings 2.2.4 Description of GNS3 as designing tool**

- **GNS3 (Graphical Network Simulator-3)** is a network emulator that allows you to run real network operating systems (NOS) such as Cisco IOS, Juniper JunOS, and others on your computer.
- It uses a combination of Dynamips (a Cisco emulator) and QEMU (a generic emulator) to run NOS images. GNS3 has been around for over a decade and has a large and active user community.
- Some of the key features of GNS3 are:
  - ✓ Support for a wide range of NOS images from various vendors
  - ✓ Integration with cloud platforms such as Amazon Web Services (AWS) and Google Cloud Platform (GCP)
  - ✓ Support for network automation and programmability through Python scripting
  - ✓ A range of network topologies and devices to choose from
- **Pros:**
  - ✓ Supports a wide range of network devices and software.
  - ✓ Provides a high degree of customization and flexibility.
  - ✓ Allows the user to run real-world device images.
  - ✓ It is free to use
- **Cons:**
  - ✓ It may require significant hardware resources to run multiple network topologies simultaneously.
  - ✓ It can be challenging to set up and configure for beginners.
  - ✓ It may not support all network devices and configurations.
  - ✓ It may require additional software and plugins to work correctly.
- GNS3 (Graphical Network Simulator-3) is an open-source network software emulator that allows users to simulate complex networks. It enables the combination of virtual and real devices to create a comprehensive learning environment for network engineers. GNS3 supports multiple router software including Cisco IOS, Juniper's JunOS, and even open-source solutions like VyOS, allowing for versatile training and testing scenarios.

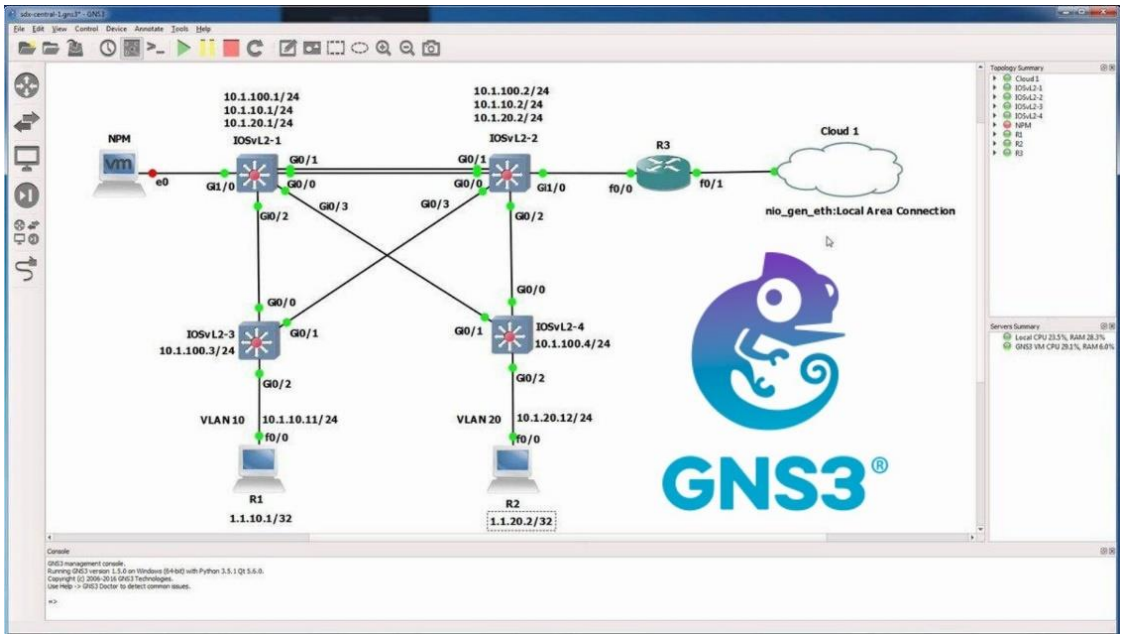
**Key Features and Functionalities**

- **Real-world environment simulation**: GNS3 excels in providing an environment that mirrors real network setups, which is essential for advanced learning and preparation for professional certifications
- **Integration with real hardware:** It allows the integration of actual devices into the virtual network, offering a hybrid setup that is invaluable for troubleshooting and testing.

- **Support for multiple network appliances**: Routers, switches, and firewalls from various vendors can be simulated, providing a flexible and scalable learning tool.

**Typical Use Cases in Network Engineering**

- **Professional certification preparation**: Ideal for preparing for higher-level certifications where a deep understanding of network technologies and troubleshooting is required.
- **Complex network modeling and testing:** Useful for simulating large networks with mixed equipment, helping professionals understand interactions in a multi-vendor environment.




**Comparing GNS3 and Packet Tracer**

- ✓ Both GNS3 and Packet Tracer offer unique user interfaces designed to cater to different levels of expertise in network engineering.
- ✓ GNS3 boasts a more complex interface that provides detailed control over network configurations. This complexity, while potentially daunting to beginners, is highly valued by advanced users who require a deeper level of simulation:
  - Detailed network topology creation: Allows for intricate network designs, closely mimicking real-world configurations.
  - High customization: Users can adjust and fine-tune each element of the network, from routers to individual interface settings.

**In contrast, Packet Tracer focuses** on simplicity and ease of use, making it accessible for newcomers:

- ✓ Intuitive drag-and-drop interface: Simplifies the process of network creation, making it easy for those with limited technical knowledge to start learning.

- ✓ Guided simulations and tutorials: These help users understand the flow of data through networks, offering a more guided learning experience.

**Simulation Capabilities**

- ✓ GNS3 excels in simulating more complex, real-world network scenarios that might involve multiple vendors and legacy systems. Its capability to integrate with actual hardware and virtual machines allows for a more thorough testing environment:
  - Advanced scenario simulation: Ideal for testing network setups under various conditions and stress tests.

**Packet Tracer,** while effective for educational purposes, has limitations in the depth of simulation it can provide:

- Basic to intermediate scenarios: Best suited for classrooms and individuals new to networking, focusing on learning rather than full-scale deployment testing.

**Educational Value**

- ✓ GNS3 is highly beneficial for those preparing for professional roles or advanced certifications, providing an environment that closely resembles actual networks, as demonstrated in courses like our Cyber Security course.
- ✓ Packet Tracer, with its straightforward setup and educational tools, serves as an excellent introduction to networking, especially for those just starting their educational journey in network engineering.

**Resource Requirements**

- Understanding the system requirements for both GNS3 and Packet Tracer is crucial for ensuring optimal performance and usability. Each tool has distinct requirements based on its capabilities and intended use.
- GNS3 is a more resource-intensive application due to its ability to run real network operating systems and connect with live networks:
- Higher CPU and RAM requirements: To simulate complex networks effectively, GNS3 requires a robust computer system, typically with a multi-core processor and substantial RAM.
- Storage space: Depending on the network devices and operating systems emulated, GNS3 may require significant storage space for IOS images and project files.

**Packet Tracer**

Packet Tracer, designed primarily for educational simulations, is less demanding in terms of hardware resources:

- Moderate CPU and memory usage: It runs comfortably on standard desktops and laptops, making it accessible to students and educators without the need for high-end hardware.
- Low storage requirements: As a simulation tool that uses pre-built commands and devices, it requires much less storage space compared to GNS3.

**When to Use GNS3 vs. Packet Tracer**

- Choosing between GNS3 and Packet Tracer depends largely on the user's specific needs and the complexity of the network scenarios they wish to explore.
  - ✓ Use GNS3 if you are a network professional looking to gain hands-on experience with complex network environments or preparing for advanced certifications. GNS3's ability to simulate real-world scenarios makes it invaluable for in-depth learning and professional growth.
  - ✓ Use Packet Tracer if you are a beginner or involved in academic settings where learning fundamental networking concepts is the primary goal. Its user-friendly interface and educational content make it ideal for those starting their journey in network engineering.

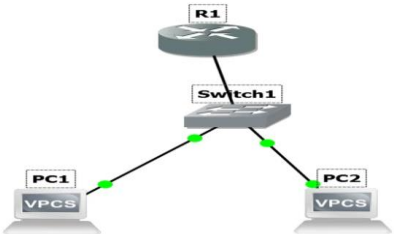


**Practical Activity 2.2.5 Design Wireless network using GNS3**




**Task:**

1. Go in the computer lab and design a given image in GNS3 from scratch and then save the network.




2. Listen to the guidelines and explanation on how the task should be performed
3. Follow the Demonstration on how to design wireless networks in GNS3 including the wireless router, wireless access points, a wired switch, and client devices through the following steps:
   1. Go to network component box and select the component to use

2. Drag and drop it into the designing area(workspace)
3. Repeat step 2 till the all the components are available in the workplace
4. Interconnect the components accordingly
5. Save the design for the future modification.
6. Export wireless network design document

4. Based on the similar task given, repeat the procedure and steps as demonstrated by trainer to perform the task.
5. After that, ensure that all process is followed based on founded output.
6. Read key readings 2.2.5
7. Perform the task provided in application of learning 2.2



**Key readings 2.2.5: Design Wireless network using GNS3**

**Design a Simple Network using GNS3**

**1.Install GNS3:**

- Download and install GNS3 on your computer. You can find the installer on the GNS3 website (https://www.gns3.com/).
- Follow the installation instructions for your specific operating system. (Linda, 2023)

**2.Install Required Software:**

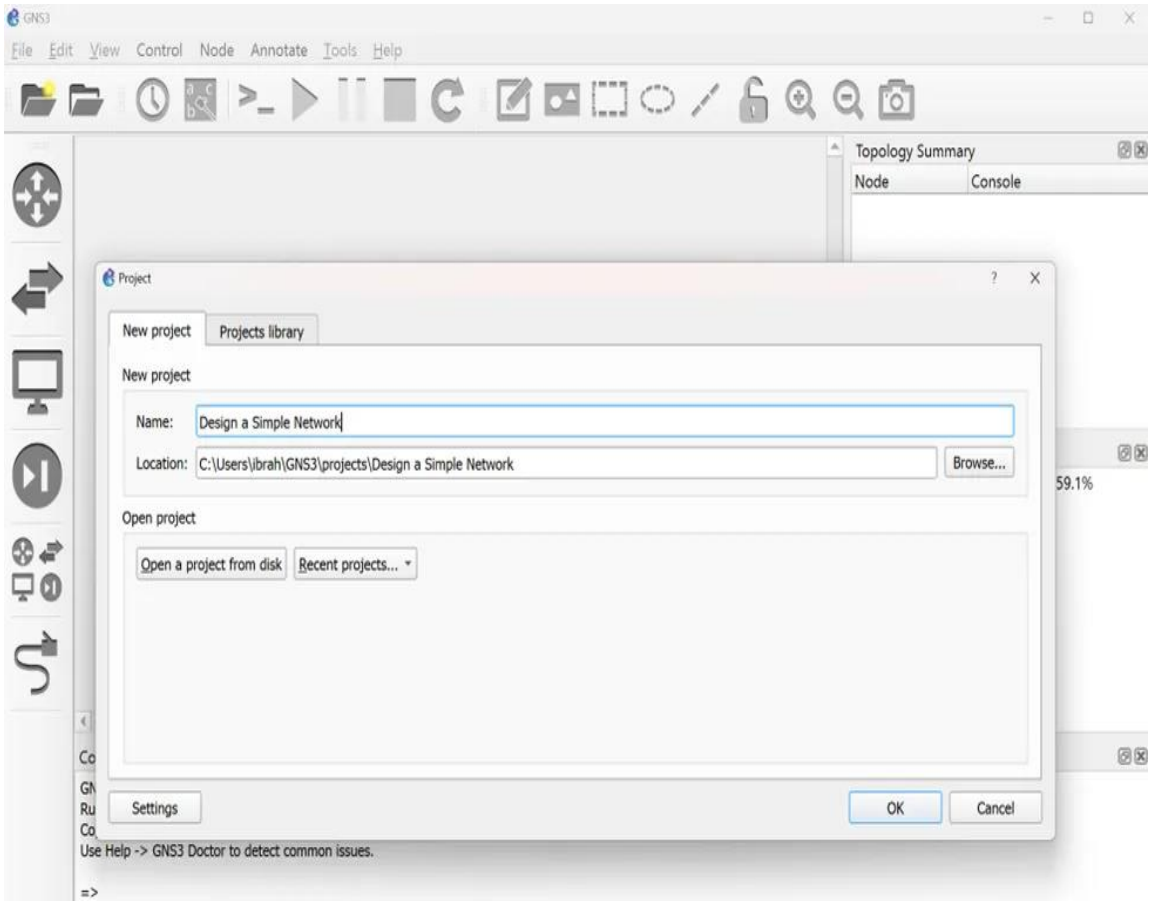
GNS3 relies on external software for virtualization. You'll need to install VirtualBox or VMware Workstation to run the virtual machines in GNS3.

**3.Import Router and PC Images:**

To use routers and PCs in GNS3, you'll need to import their images. You can find these images on the GNS3 website or other sources. Once you've obtained the images, import them into GNS3.

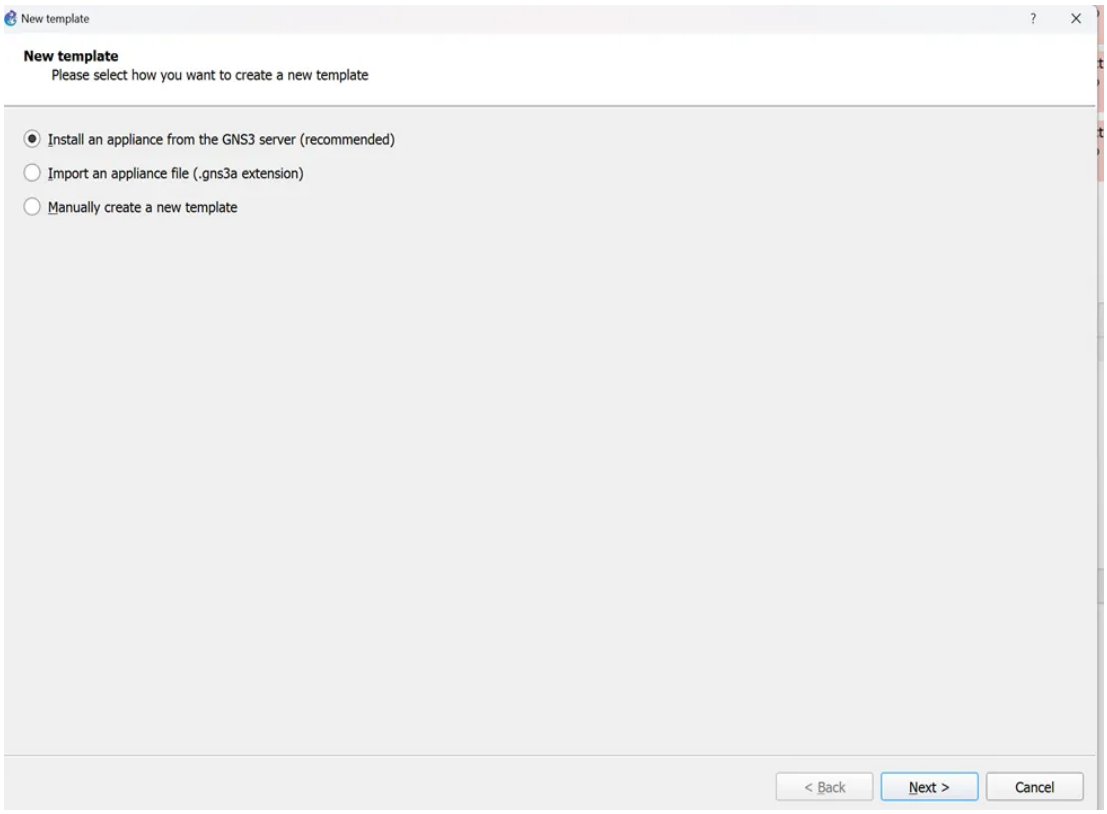
**4.Create a New Project:**

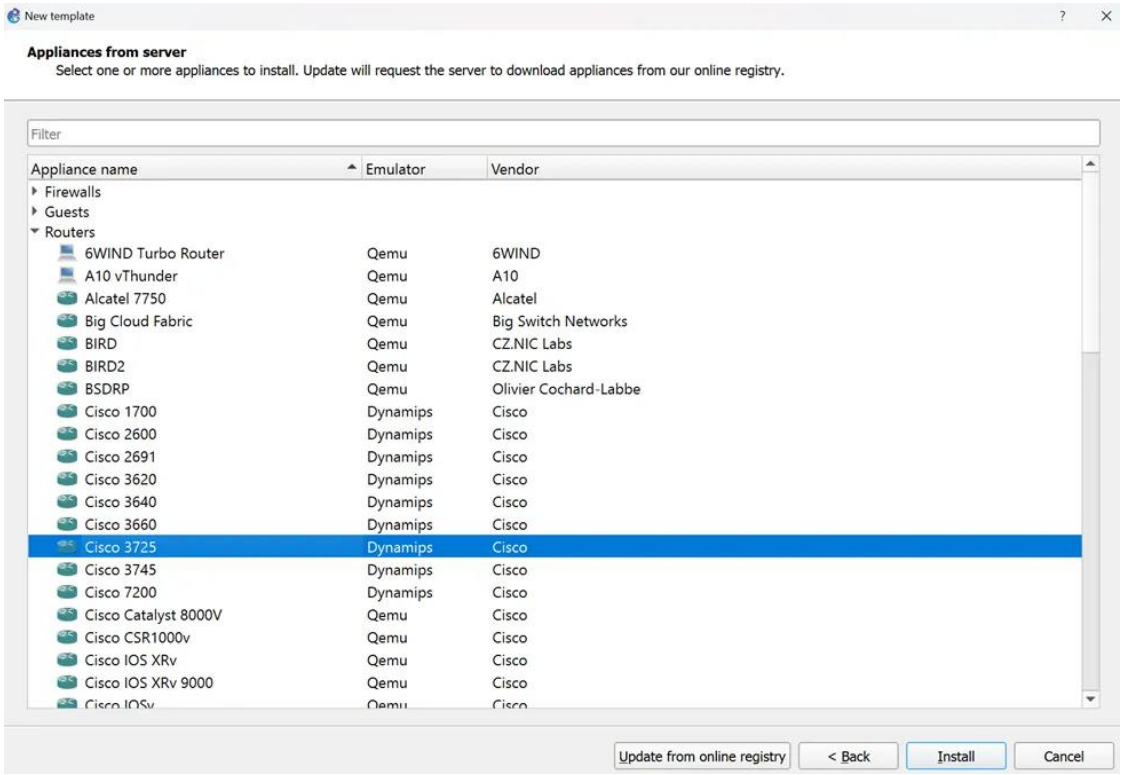
Open GNS3, create a new project, and give it a meaningful name.




**5.Add a Router:**

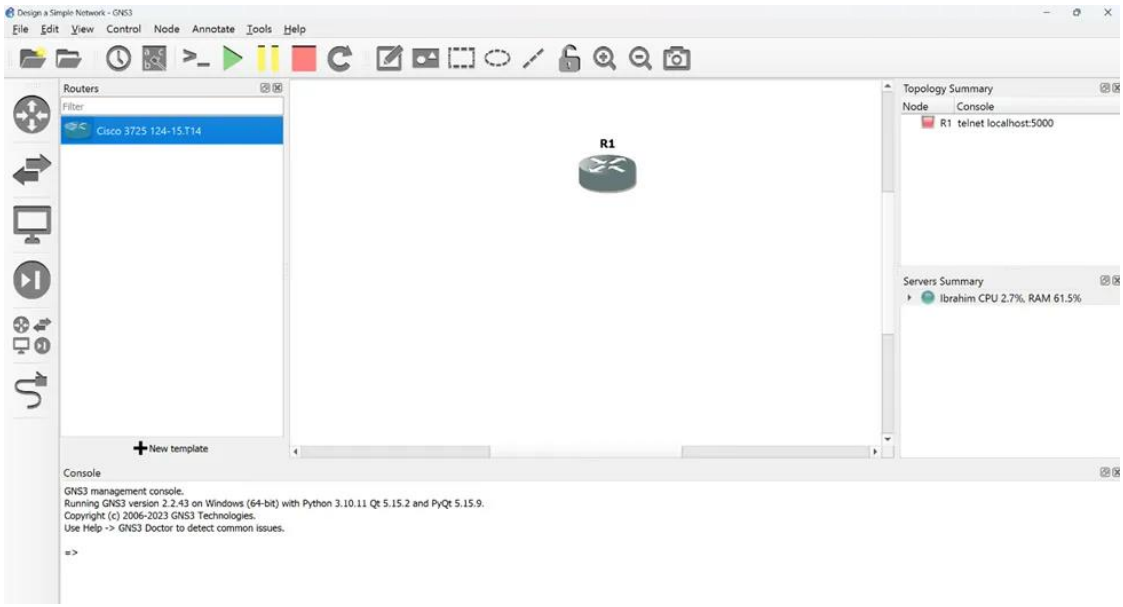
Drag and drop a router model (e.g., Cisco IOS, Juniper, etc.) into the GNS3 workspace. Connect the router to the cloud symbol (Internet) for external connectivity.



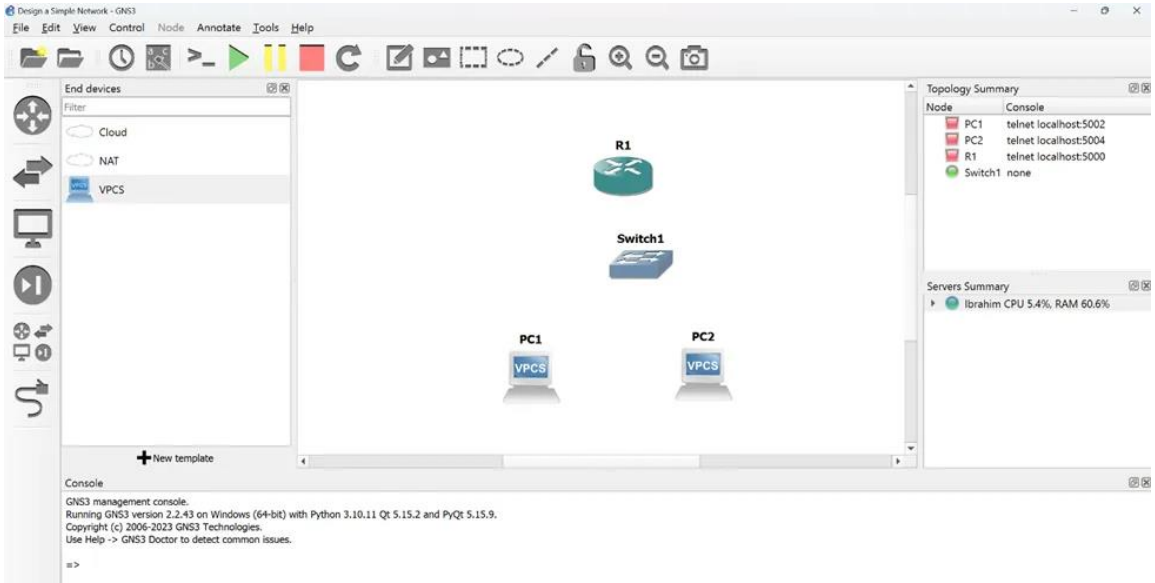

## 6.Configure the Router:

Right-click on the router and select “Configure.” Set basic parameters like the hostname and interface IP addresses. This can vary depending on the router image you’re using.




**7.Add PCs:**

Drag and drop PC nodes (usually called “Ethernet” or “VPCS”) onto the workspace. You can add as many PCs as you like, but for this example, add at least two.

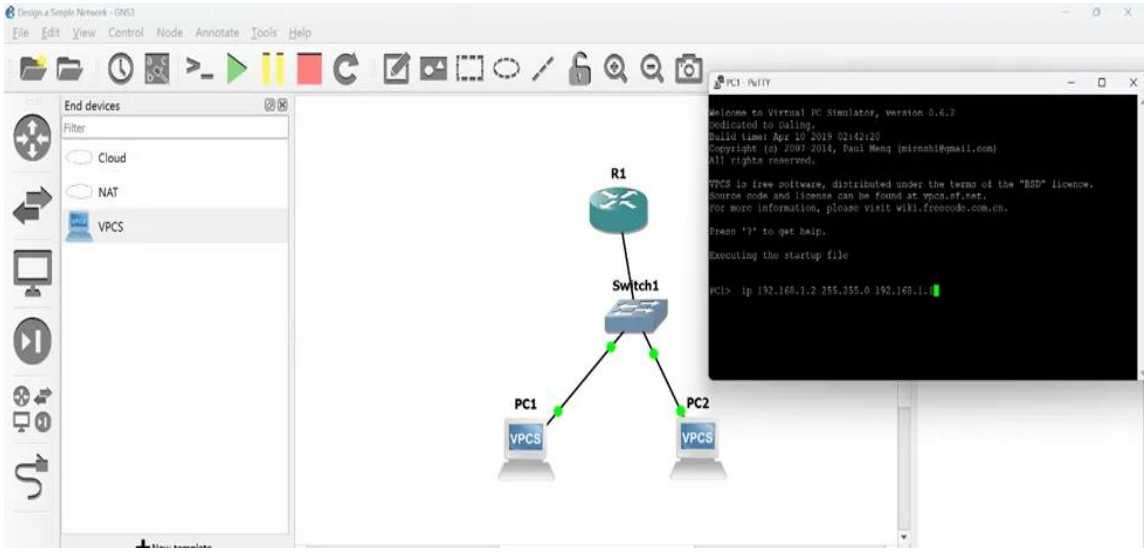



**8.Connect PCs to the Router:**

To connect PCs to the router, select a network cable from the toolbar and connect the Ethernet interfaces of the PCs to the router’s interfaces.

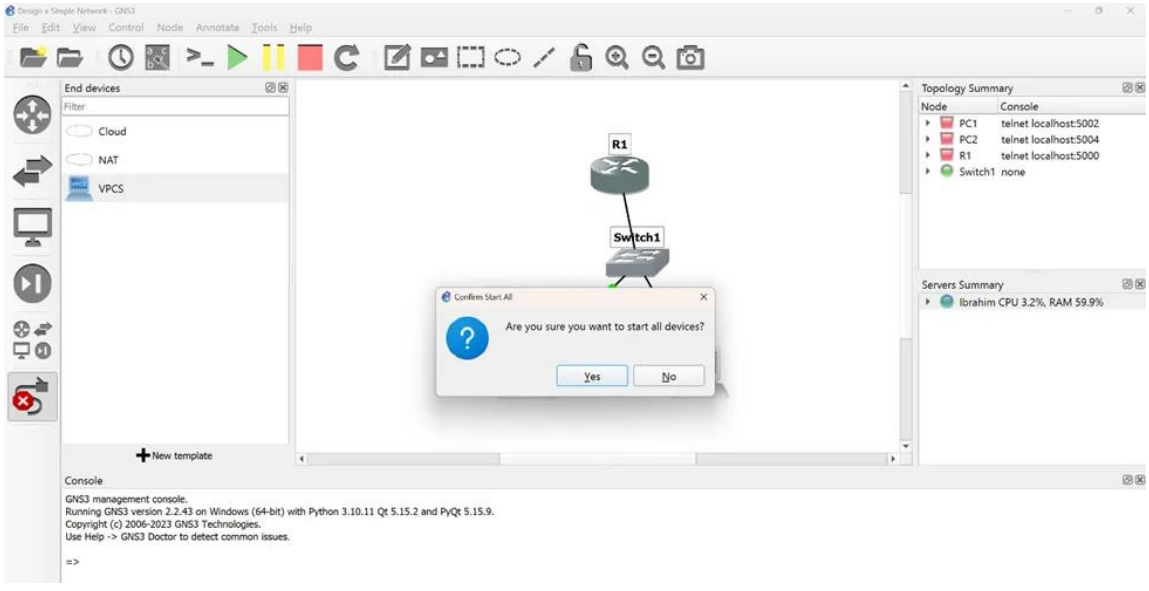
**9.Configure PCs:**

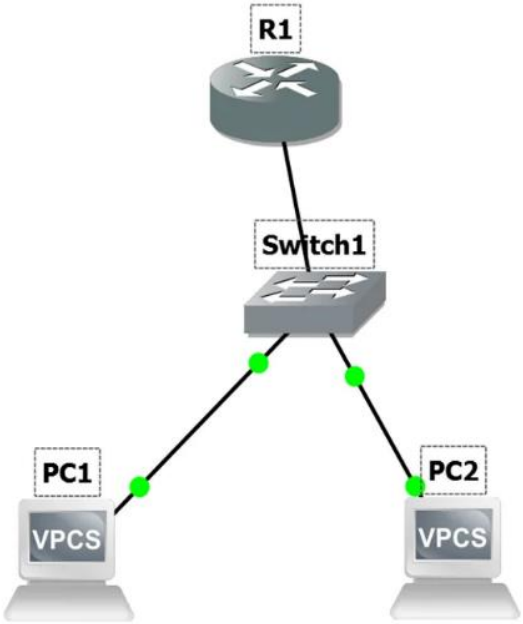
Right-click on each PC and select “Start.” Then, right-click again and choose “Console” to access the command line of the PC. Configure IP addresses and other network settings on the PCs as needed.




**10.Interconnect the PCs (Optional):**

If you want to make the network more interesting, you can connect the PCs to each other. To do this, add a hub or switch from the GNS3 toolbar and connect the PCs to it.



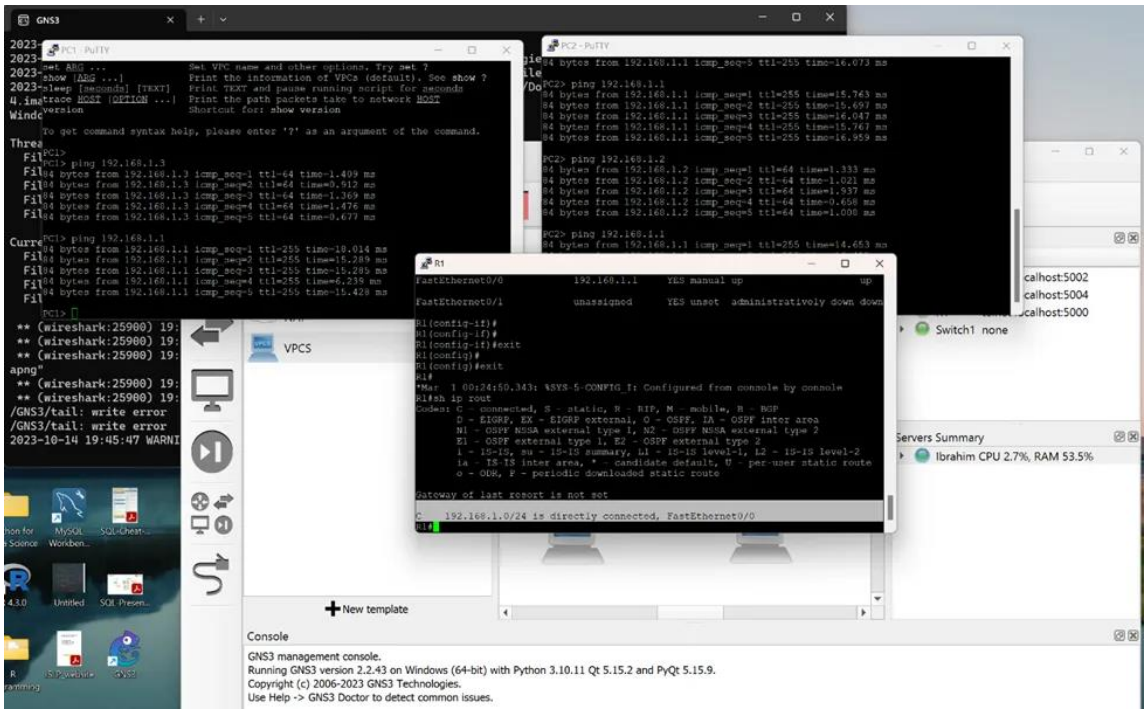


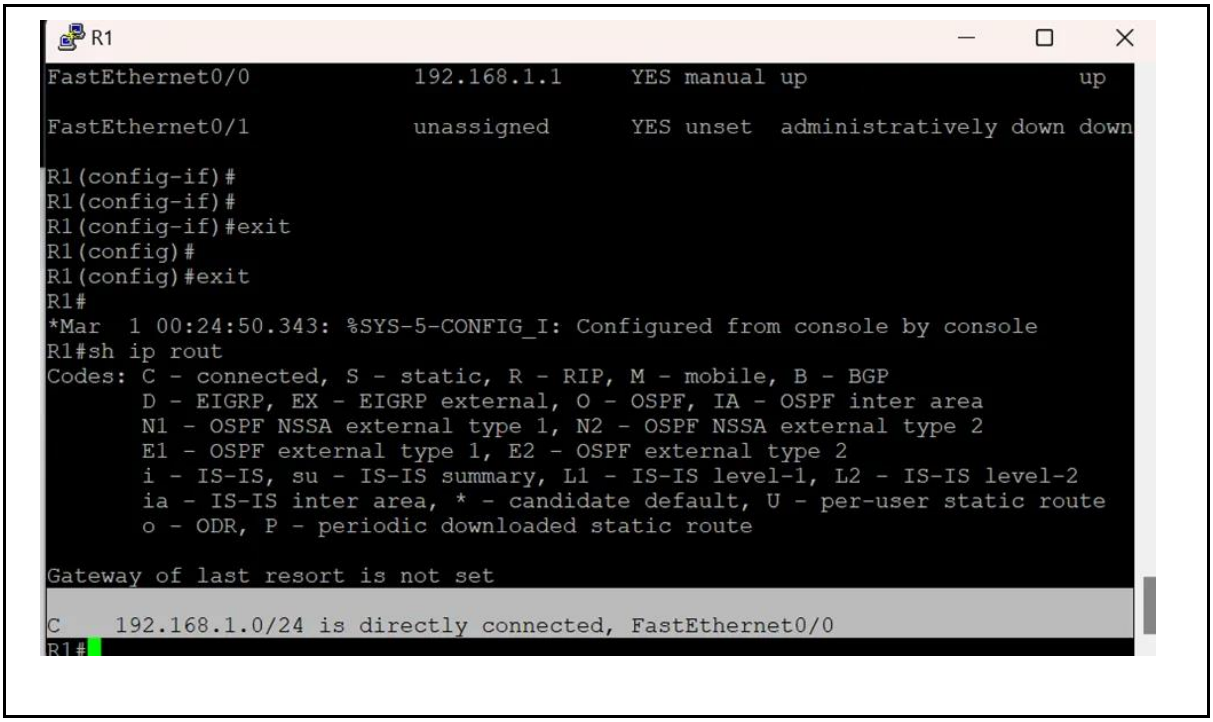

**11.Start the Network:**

Save your project and click the “Start” button in the GNS3 toolbar. This will start the virtual router and PCs.

**12.Test Your Network:**

Open a console to each PC and test the connectivity between them and the router by pinging IP addresses.






**Theoretical Activity 2.2.6: Description of E-draw max as designing tool**

**Tasks:**

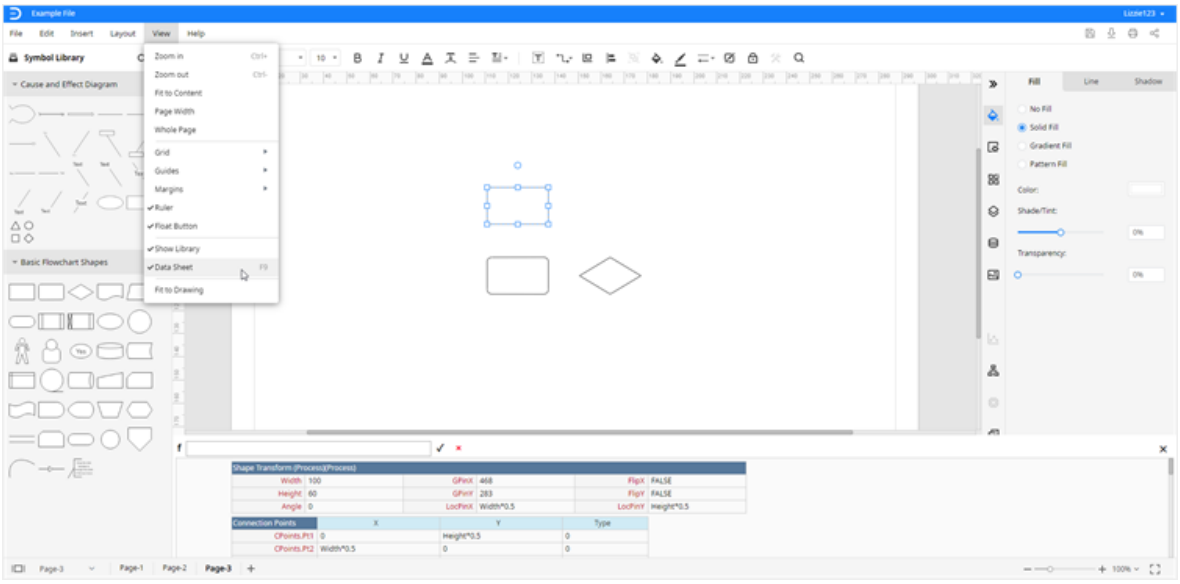
1. Observe the image showed by trainer and listen to his/her clarifications
2. In small groups, discuss on the following question related to Wireless Network Designing Tools.

    1. Describe Cisco packet tracer as designing tool

3. Discuss with your members, sharing ideas from different expert groups about the given tasks.
4. Listen to the expert view from trainer, take short note and ask clarification if necessary.
5. In addition, read the key readings2. 2.2 In this manual.

**Key readings 2.2.6: Description of E-draw max as designing tool**

- **E-Draw Max:**
  - **Description:** E-Draw Max is a diagramming tool used for creating detailed and visually appealing network diagrams and schematics. It provides a variety of shapes and symbols for representing network devices, connections, and layouts.
  - **Functions:** E-Draw Max helps in designing network blueprints, planning network expansions, and documenting network infrastructure. Its intuitive interface and extensive library of network symbols facilitate the creation of clear and informative network diagrams.



**Practical Activity 2.2.7 Design Wireless network using E-draw max**

**Task:**

1. **Read carefully this task:**
   **In your computer lab, open a computer, draw a Network Topology Diagram in EdrawMax** that show a central router with two switches connected to it. The switches will have computers, a server, and a network printer connected. The router will also be connected to a wireless access point, with 3 wireless devices linked to it.

2. Listen to the guidelines and explanation on how the task should be performed referring to the wireless network indoor site survey document.
3. Follow the Demonstration on how to design wireless networks in **EdrawMax** including the wireless router, wireless access points, a wired switch, and client devices.
4. Based on the similar task given, repeat the procedure and steps as demonstrated by trainer to perform the task.
5. After that, ensure that all process is followed based on founded output.
6. Read key readings 2.2.2
7. Perform the task provided in application of learning 2.2



**Key readings 2.2.7: Design Wireless network using E-draw max**

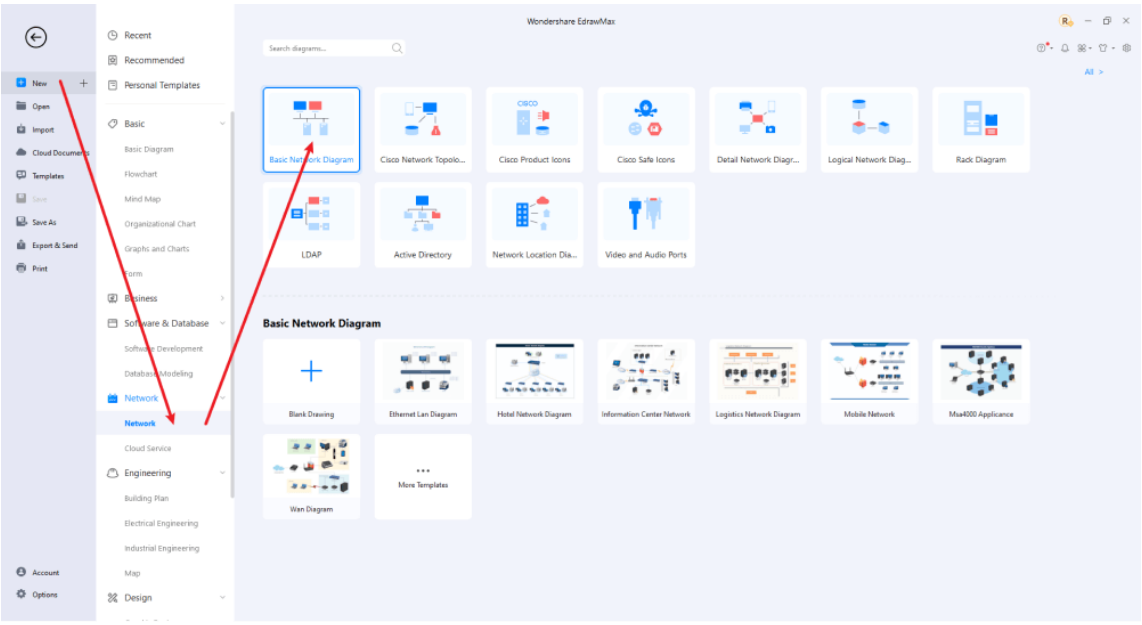
**1. How to Draw a Network Diagram in EdrawMax**

Having decided that you need a network diagram for your project activities, you must be wondering how to create one? There are two ways that you can do it. There are two methods. One is drawing a network diagram layout from scratch using EdrawMax, and the other method is when you choose a template provided by EdrawMax. If you want to design a network layout, you must consider what kind of network you are trying to build. Let's walk you through the steps:

**Draw Network Diagrams from Scratch**

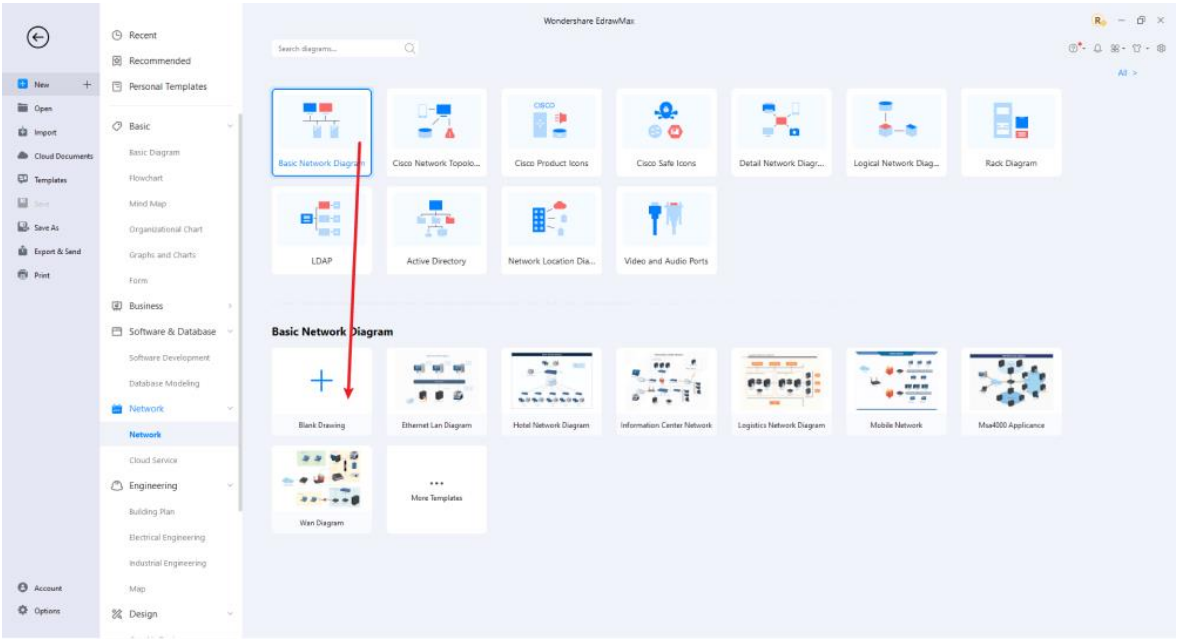
**Step1: Open EdrawMax & Login**

If you are using the offline version of EdrawMax, then open the tool in your system. If you wish to have remote collaboration, head to **EdrawMax Download** and log in using your registered email address. If this is your first time using the tool, you can register yourself using your personal or professional email address. Go to **'Network'** and click on **'Basic Network Diagram'**.




**Step2: Open a New Canvas**

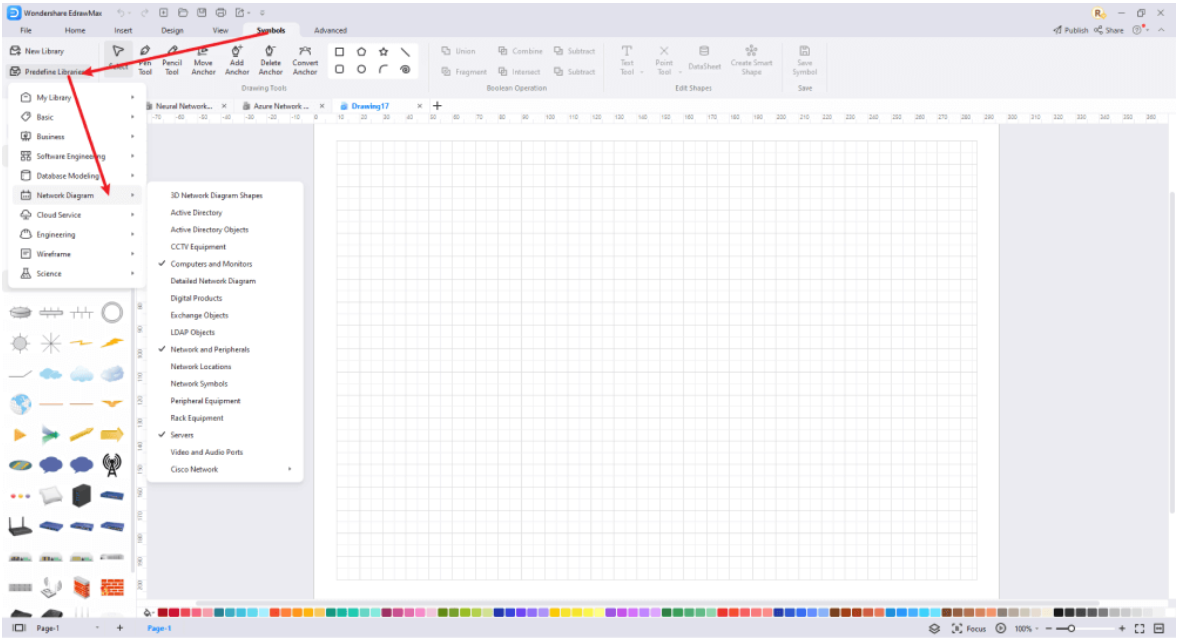
When you're building your network diagram from scratch, just hit the + button on the left, and it will take you to the EdrawMax user-friendly canvas, where you'll find features like importing images, adding text, drawing shapes, and changing shapes, and more. Create the basic sketch and then add shapes (nodes).




**Step3: Select Symbols**

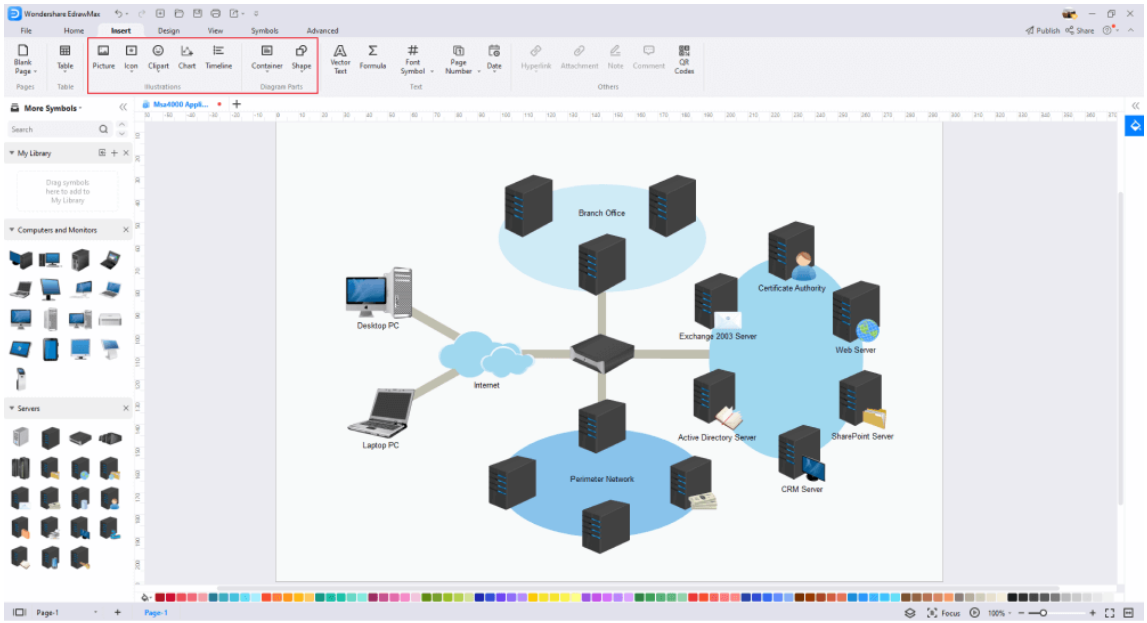
Create the basic outline of your network diagram. Go to Symbols Tab and click on **Predefine Libraries > Network Diagram** to get more symbols. The Network Diagram

window is your best friend! It will show you a lot of pre-built and valuable icons, such as digital products, network symbols, etc. Use the symbol of your choice and import them to your library.



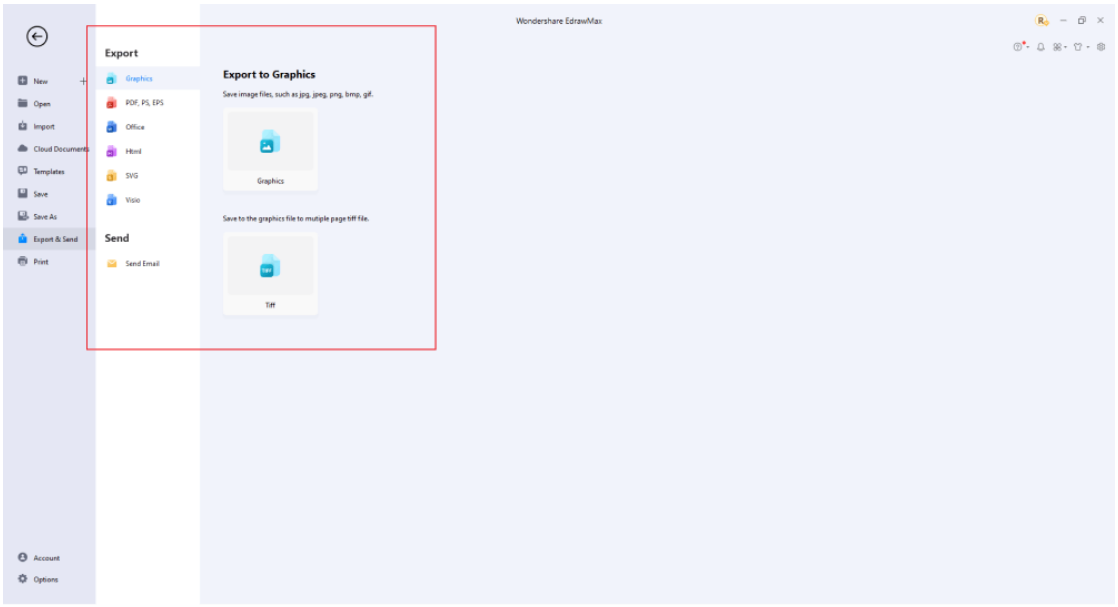
**Step4: Add Elements**

Once you have imported all the symbols, you can start placing them in the network drawing created in Step #2. When planning the landscaping design, you'll also need to add other landscaping items from the element section. Use the toolbar on the right to change the font, color, theme, size, and more.



**Step5: Save & Export**

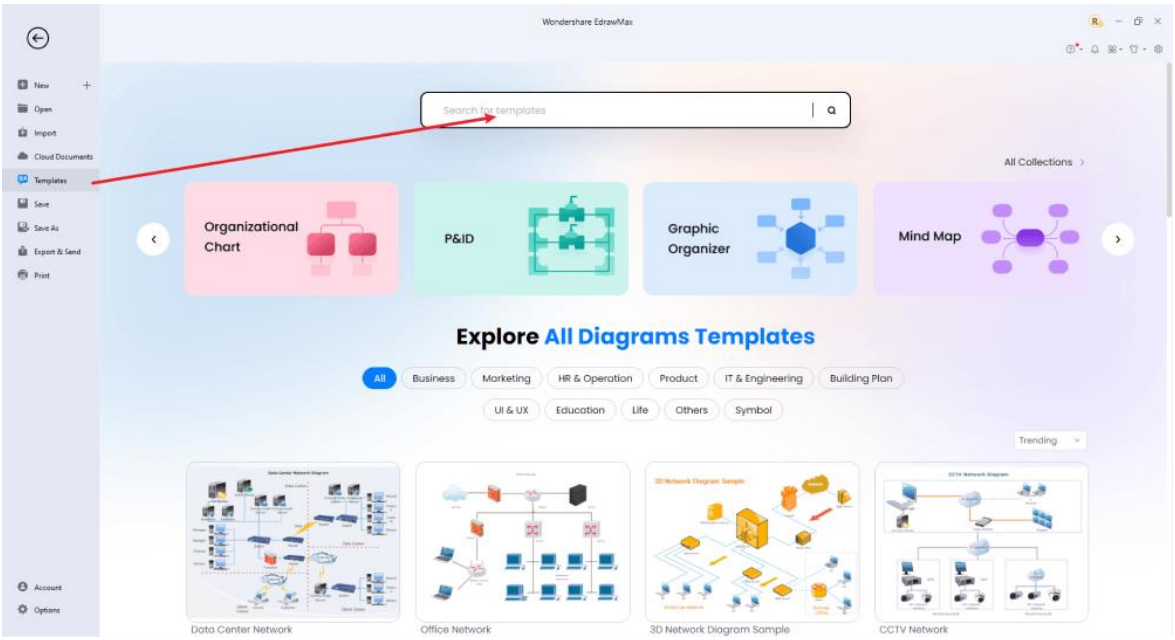
After designing the network Diagram, you can export it into multiple formats, like JPEG, JPG, PNG, PDF, and other Microsoft formats, like MS Word, MS Excel, PPTX, etc. You can also share the network layout design on different social media platforms, like Facebook, Twitter, LinkedIn, etc.




### 4.2 Draw a Network Diagram from Built-in Templates

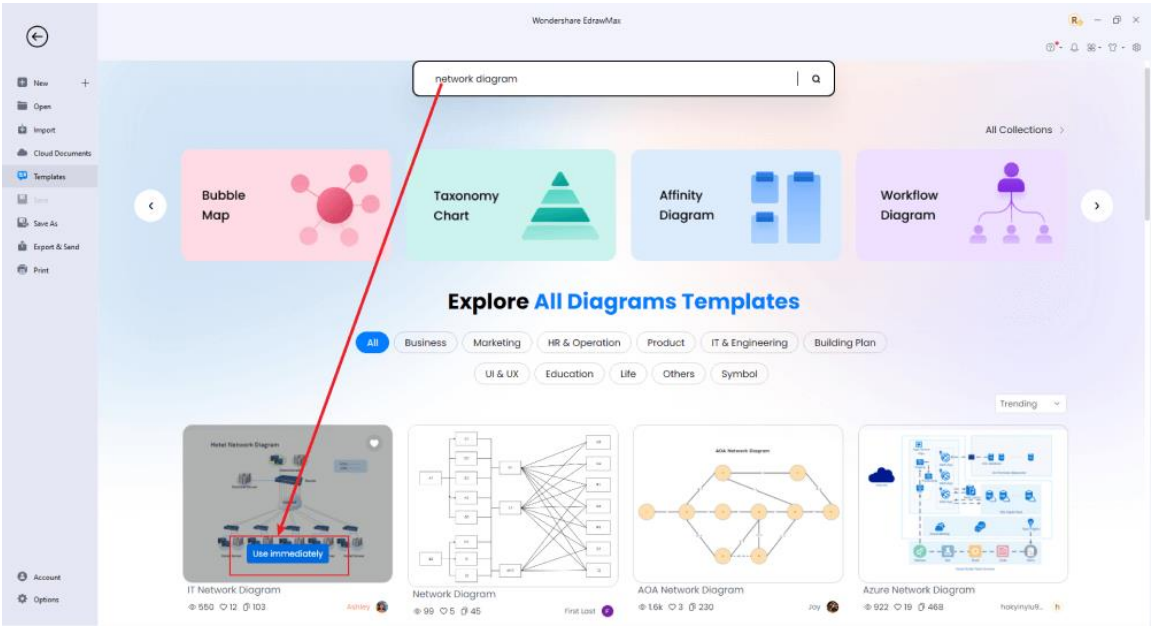
**Step1: Login to EdrawMax**

If you are using the offline version of EdrawMax, then open **t**he tool in your system. If you wish to have remote collaboration, head **to EdrawMax Download** and log in using your registered email address. If this is your first time using the tool, you can register yourself using your personal or professional email address. Go to the '**Templates**' section on the left and type '**Network Diagram**.'




**Step2: Choose a Template**

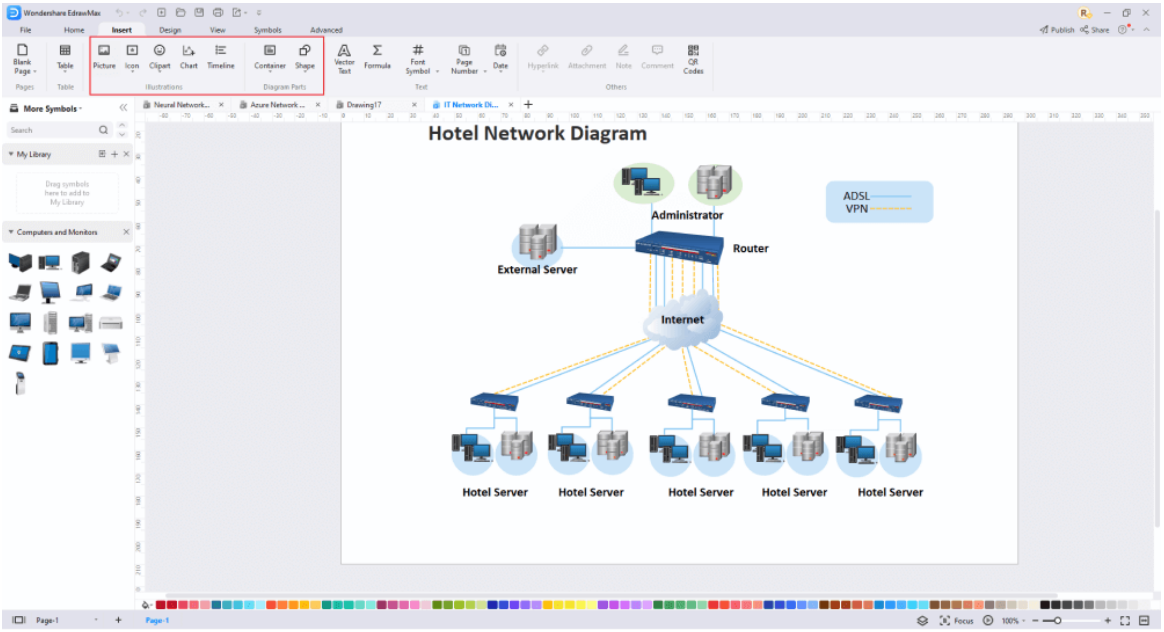
In the **template community,** you will find hundreds of built-in templates readily available to use. Choose any templates that closely resemble your ideal network diagram and duplicate it to '**Use Immediately**.'




**Step3: Change Details**

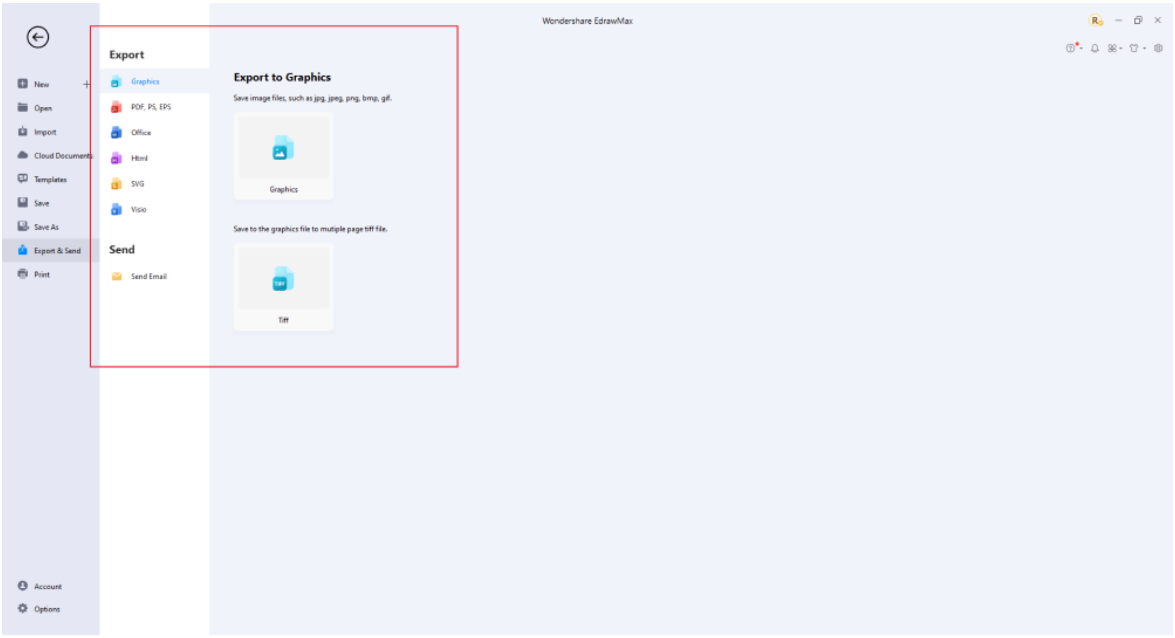
In addition to importing the template to the EdrawMax canvas, you can change some details as required. You can use images, create your layouts, choose your colors, and

add extra features. You can use these server symbols in your network drawing based on your vision.




**Step4: Export or Publish**

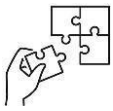
Once your network diagram is complete, you can save it to multiple file formats as described in the instructions or post it to the template community for the other 25 million registered users to see and get inspired by your neural network drawing. Click the "**Publish**" button on the top right area of the canvas, and fill in your template details, like template title, description, and tag to share your drawing with others.



**Points to Remember**

- **Here are the steps to design a network topology using Edraw Max:**

1. Install and Open Edraw Max
2. Select a Network Diagram Template
3. Define Network Requirements
4. Drag and Drop Network Symbols
5. Arrange and Connect Devices
6. Label Components and Connections
7. Customize Design Elements
8. Add Notes and Annotations
9. Review and Adjust the Diagram
10. Export and Share the Diagram



**Application of learning 2.2.**

Oakwood Elementary School has recently conducted a site survey to assess its current wireless network infrastructure and identify areas for improvement. The site survey report indicates that the existing network setup is inadequate, with several areas suffering from weak or no Wi-Fi coverage. The school administration wants to enhance its wireless network to support digital learning and administrative tasks more effectively.

Based on the site survey, the following network components are planned for the new setup:

1. **1 Wireless Router:** The central device that will connect to the internet and manage network traffic.
2. **2 Access Points (APs):** To be placed strategically to extend wireless coverage across the school.
3. **4 PCs:** To be connected wirelessly to the network via the access points.

**As a network designer specialist, y**our task is to design a wireless network topology for Oakwood Elementary School using the information provided in the site survey documents.

## Indicative content 2.3: Estimation of Wireless Network Indoor Installation Cost.

**Duration: 2**

### Theoretical Activity 2.3.1: Identify the Item specifications and costs

**Tasks:**

1. In small groups, you are requested answer the following questions
   i. What do you understand by a wireless network indoor cost estimation?
   ii. What does a wireless network indoor cost estimation document contain?
   iii. Why do we need to estimate the cost of any wireless network indoor before its implementation?
   iv. What are the key components that are typically included in a wireless network indoor cost estimation document?
2. Write the answers on provided space
3. Present your findings in front of a whole class
4. Listen to the expert view of trainer, note the key points and ask questions when necessary.
5. Read the key reading 2.3.1 in this manual

### Key readings 2.3.1.: Identify the Item specifications and costs

- Wireless network indoor cost estimation refers to the process of determining the expenses associated with planning, designing, implementing, and maintaining a wireless network within an indoor environment. Estimating the cost of an indoor wireless network involves various components, from hardware and cabling to labour and ongoing operational expenses.
- An indoor wireless network cost estimation document typically contains detailed information about the various expenses associated with planning, designing, implementing, and maintaining a wireless network within an indoor environment. This document is an essential part of the project planning process, providing a clear breakdown of costs and helping stakeholders make informed decisions.
- Estimating the cost of a wireless network indoor installation before implementation is crucial for several reasons:
  - ✓ Budget Planning
    - Resource Allocation: Knowing the cost upfront helps allocate the necessary financial resources. It ensures that the project can be funded adequately without unexpected financial strain.

- Preventing Overspending: A detailed cost estimate helps prevent overspending by setting clear financial boundaries for the project.

✓ Project Feasibility
- Decision Making: Estimating costs allows decision-makers to assess whether the project is financially viable and aligns with the organization's budgetary constraints.
- Comparing Alternatives: It enables the comparison of different network designs, technologies, and equipment to find the most cost-effective solution that meets the requirements.

✓ Scope Definition
- Clear Understanding of Requirements: Cost estimation helps define the scope of the project by providing a clear understanding of what is required in terms of equipment, labor, and other resources.
- Avoiding Scope Creep: By having a clear cost estimate, you can avoid scope creep, which occurs when additional features or changes are added to the project without corresponding adjustments to the budget.

✓ Risk Management
- Identifying Potential Risks: Cost estimation helps identify potential financial risks early in the planning process. This includes risks related to fluctuating prices of materials, unexpected labor costs, or the need for additional equipment.
- Contingency Planning: It allows for the inclusion of a contingency budget to cover unexpected expenses, reducing the risk of project delays or failures.

✓ Vendor and Supplier Negotiation
- Better Negotiation Power: With a clear cost estimate, you can negotiate better terms with vendors and suppliers. It ensures that you are not overpaying for equipment or services.
- Competitive Bidding: Cost estimates enable you to solicit competitive bids from multiple vendors, ensuring that you get the best value for your money.

✓ Resource Management
- Efficient Use of Resources: A detailed cost estimate ensures that resources, including time, money, and labor, are used efficiently, avoiding waste and optimizing the project's overall effectiveness.

✓ Timeline and Scheduling
- Accurate Project Timeline: Estimating costs helps in creating a more accurate project timeline, as financial resources are closely tied to project milestones and deadlines.

- Timely Procurement: Ensures that necessary funds are available for timely procurement of equipment and services, preventing delays in the project.

- The key components that are typically included in a wireless network indoor cost estimation document:
  - ✓ Project Overview
  - ✓ Scope of Work: An outline of the tasks and activities involved in the project, specifying what the cost estimation covers
  - ✓ Hardware and software costs
  - ✓ Installation Costs: A breakdown of expenses for deploying and installing the network infrastructure, including labour charges, electrical work, and construction modifications.
  - ✓ Site Survey Costs: Costs associated with conducting site surveys, including equipment and personnel expenses.
  - ✓ Other costs: any other costs for the needed activities like cabling, testing and validation …
  - ✓ Total Estimated Cost: The sum of all costs listed in the document, providing the overall budget for the indoor wireless network project



**Practical Activity 2.1.2: Estimate wireless network indoor installation**




**Task:**

1. **Read carefully the following task:**
   Your school is finishing its new office space designed for teachers and staff teams. The space spans 3,000 square feet on a single floor and needs a reliable wireless network to support up to 50 simultaneous users. The network must provide high-speed internet access for activities such as video conferencing, large file transfers, and cloud computing. The co-working space also requires a secure guest network for visitors.
   Your task as a network consultant is to estimate the total cost of setting up this wireless network, including hardware, labour, software, and other necessary expenses.
2. Receive the table of specification for all required items given by trainer
3. Follow the demonstration of the process of estimating cost for each item and calculation of total cost
4. Repeat the previous step by yourself, estimate the cost of each item and calculate the total cost
5. Prepare a wireless network indoor cost estimation report

6. Read the key readings 2.3.2
7. Perform the task provided in application of learning 2.3



**Key readings 2.3.2.: Estimate wireless network indoor installation**

- Estimating the cost and calculating the total cost of a wireless network installation involves several key steps. This process ensures that all necessary components and labour are accounted for and that the project stays within budget
- **Requirement Analysis**
  - ✓ **Identify Network Needs**: Determine the scope of the project, including the size of the area to be covered, the number of users, and the types of devices that will be connected to the network.
  - ✓ Technical Specifications: Identify the technical requirements, such as wireless standards (e.g., Wi-Fi 6), security protocols, and bandwidth needs.
- **Conduct a Site Survey**
  - ✓ **Physical Layout:** Assess the physical layout of the installation site, noting the size, structure, materials, and any potential obstacles that could affect signal propagation.
  - ✓ **Interference Sources:** Identify potential sources of interference, such as other wireless networks, electronic devices, or physical barriers.
  - ✓ **Access Point Locations:** Determine the optimal placement of access points (APs) to ensure full coverage and strong signal strength.

**Network Design**

- ✓ Number of Access Points: Based on the site survey, calculate the number of access points needed to cover the area adequately.
- ✓ Network Components: List all the necessary components, including routers, switches, cabling, mounting hardware, and network management tools.
- ✓ Redundancy and Scalability: Consider adding extra components or higher-capacity equipment to allow for future expansion or redundancy.

**Component Cost Estimation**

- ✓ Research Prices: Use vendor catalogues, online stores, or supplier quotes to find the prices of the required components.
- ✓ Budgeting for Each Item

**Labour Cost Estimation**

- ✓ Calculate Labour Costs: Multiply the estimated time by the hourly rate to calculate the total labour cost

**Additional Costs**

- ✓ Contingency Budget: Set aside a contingency budget (typically 10-15% of the total estimated costs) to cover unexpected expenses or changes in project scope.
- ✓ Miscellaneous Expenses: Account for additional costs such as shipping fees, taxes, and any special equipment needs

**Calculate the Total Cost**

- ✓ Sum of All Costs: Add together all the estimated costs for components, labor, software, and additional expenses.

- **Estimating the cost of a wireless network indoor installation before implementation is crucial for several reasons:**
  - ✓ **Budget Planning**
    - Resource Allocation: Knowing the cost upfront helps allocate the necessary financial resources. It ensures that the project can be funded adequately without unexpected financial strain.
    - Preventing Overspending: A detailed cost estimate helps prevent overspending by setting clear financial boundaries for the project.
  - ✓ **Project Feasibility**
    - Decision Making: Estimating costs allows decision-makers to assess whether the project is financially viable and aligns with the organization's budgetary constraints.
    - Comparing Alternatives: It enables the comparison of different network designs, technologies, and equipment to find the most cost-effective solution that meets the requirements.
  - ✓ **Scope Definition**
    - Clear Understanding of Requirements: Cost estimation helps define the scope of the project by providing a clear understanding of what is required in terms of equipment, labor, and other resources.
    - Avoiding Scope Creep: By having a clear cost estimate, you can avoid scope creep, which occurs when additional features or changes are added to the project without corresponding adjustments to the budget.
  - ✓ **Risk Management**
    - Identifying Potential Risks: Cost estimation helps identify potential financial risks early in the planning process. This includes risks related to fluctuating prices of materials, unexpected labor costs, or the need for additional equipment.
    - Contingency Planning: It allows for the inclusion of a contingency budget to cover unexpected expenses, reducing the risk of project delays or failures.
  - ✓ **Vendor and Supplier Negotiation**

- Better Negotiation Power: With a clear cost estimate, you can negotiate better terms with vendors and suppliers. It ensures that you are not overpaying for equipment or services.
- Competitive Bidding: Cost estimates enable you to solicit competitive bids from multiple vendors, ensuring that you get the best value for your money.

✓ **Resource Management**

- Efficient Use of Resources: A detailed cost estimate ensures that resources, including time, money, and labor, are used efficiently, avoiding waste and optimizing the project's overall effectiveness.

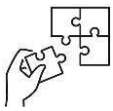
✓ **Timeline and Scheduling**

- Accurate Project Timeline: Estimating costs helps in creating a more accurate project timeline, as financial resources are closely tied to project milestones and deadlines.
- Timely Procurement: Ensures that necessary funds are available for timely procurement of equipment and services, preventing delays in the project.



**Points to Remember**

- Thorough cost estimation is essential to ensure that all components and labor for a wireless network installation are accounted for and within budget.
- Conducting a site survey and analyzing technical requirements helps determine the necessary equipment and optimal placement of access points.
- Including a contingency budget and estimating labor costs are critical steps to manage unexpected expenses and ensure project feasibility.
- Accurate cost estimation supports better resource management, vendor negotiation, and the creation of a realistic project timeline.



**Application of learning 2.3.**

KENZ School has recently conducted a site survey to assess its current wireless network infrastructure and identify areas for improvement. The site survey report indicates that the existing network setup is inadequate, with several areas suffering from weak or no Wi-Fi coverage. The school administration wants to enhance its wireless network to support digital learning and administrative tasks more effectively.

Based on the site survey, the following network components are planned for the new setup:

1. **1 Wireless Router:** The central device that will connect to the internet and manage network traffic.
2. **2 Access Points (APs):** To be placed strategically to extend wireless coverage across the school.
3. **4 PCs:** To be connected wirelessly to the network via the access points.

**As a network designer specialist, y**our task is to estimate the total costs for the installation of Oakwood Elementary School wireless network using the information provided in the site survey documents.

## Learning outcome 2 end assessment

**Theoretical assessment**

1. Circle the correct answer:

1. The primary purpose of conducting a wireless network site survey?

A. To determine the best location for a coffee machine

B. To measure signal interference and optimize coverage

C. To assess the physical layout of an office for interior design

D. To calculate the electrical power requirements of the office

2. Which tool is commonly used during a wireless network site survey?

A. Multimeter

B. Signal analyser

C. Spectrum analyser

D. Thermal camera

2. Answer by true or false
   a. The physical topology of a network refers to the actual layout of the devices and cables
   b. The physical topology of a network has no impact on its performance or scalability.
   c. In a logical ring topology, data travels in a circular direction, passing through each device on the network.
3. Match the network designing tools with their features/unction

| answers | Network designing tools | Features/functions |
|---|---|---|
| 1…… | NetSpot | Monitors network performance and provides detailed reports |
| 2…… | Wireshark | Provides a user-friendly interface for basic Wi-Fi surveys and troubleshooting |
| 3…… | Solar Winds Network Performance Monitor | Captures and analyses network traffic to diagnose issues |

**Practical assessment**

SZZ SACCO is in the final stages of completing its new building, which consists of two floors. The first-floor houses four rooms: a customer waiting hall, a cashiers' room, a credit line room, and an operational room. The second floor comprises three rooms: the financial manager's office, the director manager's office, and an additional room yet to be assigned.

The organization plans to deploy a wireless network within this building to ensure that all employees working inside, as well as visitors in the surrounding compound, have reliable access to the network.

As a network technician, you have been hired to perform the following tasks:

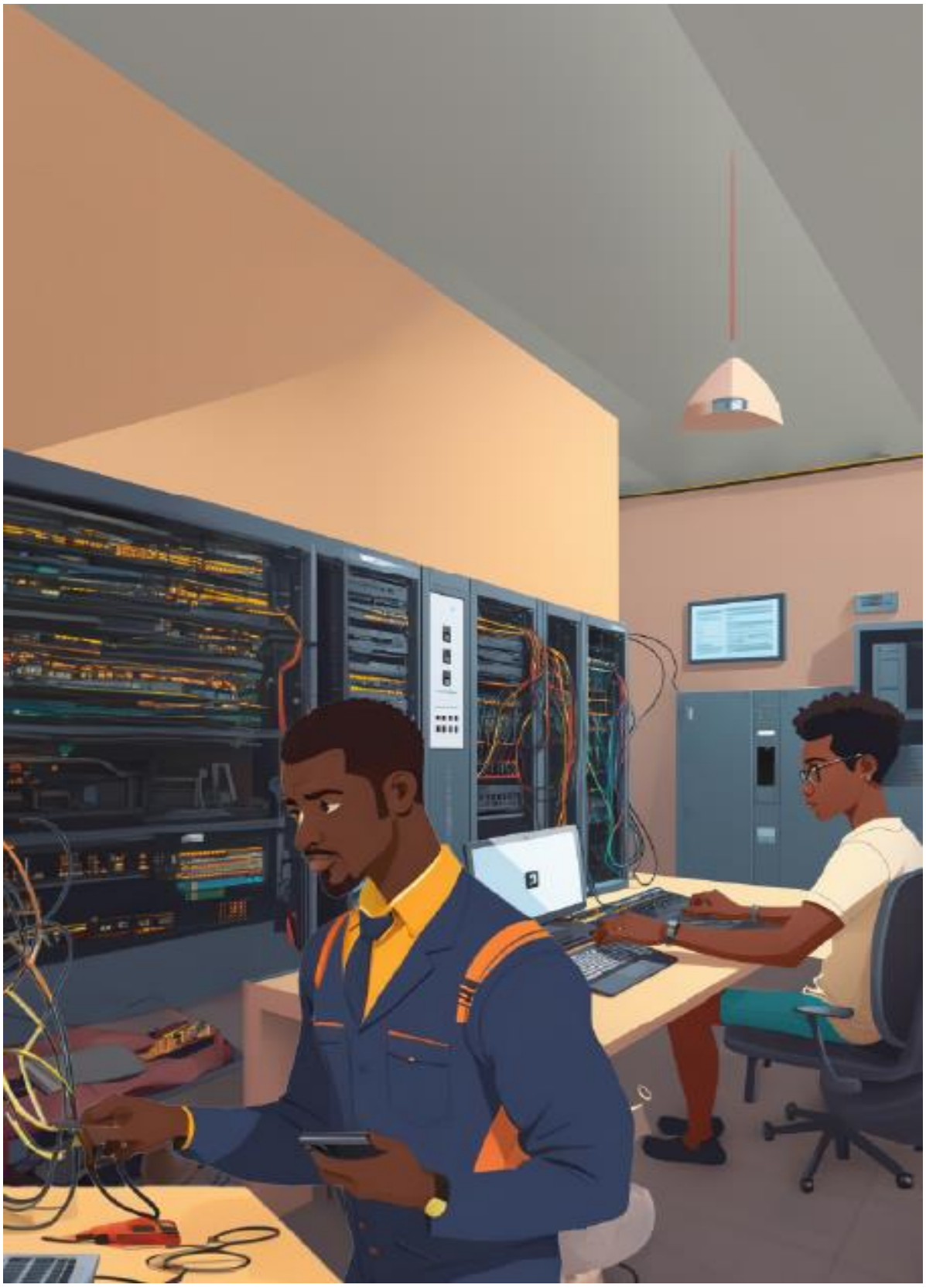
1. Conduct a site visit to assess the building's layout, materials, and any potential obstacles that might impact the wireless network deployment.
2. Identify optimal locations for wireless access points (APs) to ensure full coverage across both floors and the outdoor compound.
3. Determine the necessary equipment and configurations to support both employee and guest access, ensuring network security and performance.
4. Design a network topology of the new building
5. Develop a detailed deployment plan, including a timeline and cost estimate, to present to Umukiro SACCO's management.

**END**



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3. https://infosyte.com/network-simulators/
4. Balston, D. M., and Macario, R. C. V. Cellular Radio Systems. Norwood, MA: Artech House, 1993.
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7. Linda. (2023). *How to Install and Run A GNS3 VM [A Step-by-Step Guide]*. Retrieved from minitool.com: https://www.minitool.com/partition-disk/gns3-vm.html



**Indicative contents**

**3.1 Mounting Wireless Network equipment**

**3.2 Basic Configuration of wireless network devices**

**3.3 Application of Access restrictions parameters**

**3.4 Testing of deployed wireless network indoor**

**3.5 Documentation of deployed wireless network**

## Key Competencies for Learning Outcome 3: Deploy wireless network indoor

| Knowledge | Skills | Attitudes |
|---|---|---|
| • Identification of the cable colour coding for making a cable<br>• Identification of element of maintenance report<br>• Identification of common wireless network problems and their causes | • Mounting Wireless Network equipment<br>• Configure Basically wireless network devices<br>• Selecting tools, Materials and equipment<br>• Testing of deployed wireless network indoor<br>• Applying Access restrictions parameters | • Being a good observer<br>• Having Self confidence<br>• Being a Critical thinker |

**Duration: 15 hrs**

**Learning outcome 3 objectives**:

By the end of the learning outcome, the trainees will be able to:

1. Mount correctly Wireless Network equipment based on the installation environment.
2. Select correctly the tools, Materials and equipment to be used in accordance with network requirements.
3. Perform clearly the Basic Configuration of wireless network devices based on device specifications.
4. Apply correctly Access restrictions parameters based on user requirements.
5. Test effectively wireless network indoor deployed in accordance with network functionality.
6. Document correctly the deployed wireless network in accordance with network functionality and its maintenance.

**Resources**

| Equipment | Tools | Materials |
|---|---|---|
| • Router<br>• Switch<br>• Access point<br>• Wireless NIC<br>• Rack<br>• Computers<br>• wireless repeaters | • Drilling machine<br>• Multimeter<br>• Crimping tool<br>• Wi-Fi Analyzer | • Network cable<br>• internet bundles<br>• Connectors<br>• Cable ties |

## Indicative content 3.1: Mounting Wireless Network Equipment

**Duration: 3**

**Theoretical Activity 3.1.1: Description of key terms, materials, and tools used in mounting Wireless Network Equipment**

**Tasks:**

1. In small groups, you are requested answer the following questions

   i. Identify tools, Materials and equipment used in mounting Wireless Network
   ii. Explain these key terms used in Mounting Wireless Network:
   - laying cables
   - Fixing equipment
   - Connecting devices in Mounting Wireless Network
   - Cable color coding
   - Cable termination
   - Cable testing
   - Interconnection
2. After this discussion in expert group, return back to your home team and share the ideas you discussed with your members
3. Listen to the expert view from trainer and take notes, if necessary, then ask questions or clarifications
4. Read the key readings 3.1.1 and do the application of learning 3.1

**Key readings 3.1.2: Description of key terms, materials, and tools used in mounting Wireless Network Equipment**

Here's a detailed description of each phase involved in mounting wireless network equipment:

**1. Identifications of Tools, Materials, and Equipment**

**Tools:**

- **Cable Stripper**: Used to remove the insulation from network cables, exposing the internal wires without damaging them.



- **Crimping Tool**: Essential for attaching connectors (like RJ45) to the ends of Ethernet cables. Ensures a secure and reliable connection.



- **Cable Tester**: Tests cables for continuity, signal strength, and faults to ensure proper connectivity.



- **Drill**: Required for creating mounting holes in walls or ceilings for brackets and other fixtures.



- **Screwdrivers**: Used to secure equipment and mounting brackets to surfaces.



- **Fish Tape or Cable Puller**: Helps in threading cables through conduits or walls, especially in challenging installations.



- **Cable Management Tools**: Includes cable ties, clips, and raceways to organize and secure cables, preventing tangling and damage.



**Materials:**

- **Ethernet Cables**: Types such as Cat5e, Cat6, or Cat6a are used for network connections, chosen based on bandwidth and distance requirements.




- **Connectors**: RJ45 connectors are commonly used for Ethernet cables, ensuring a stable connection between devices.

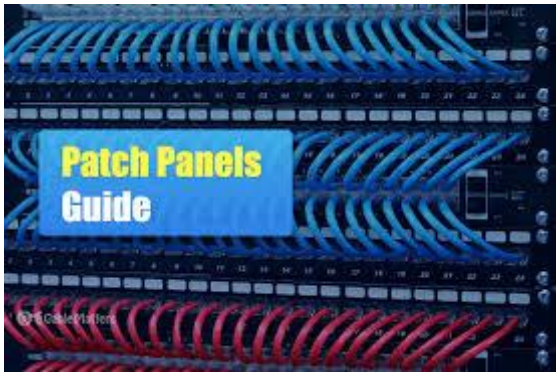





- **Cable Supports**: Cable ties, clips, or saddles to keep cables secured and prevent them from being pulled or damaged.



- **Patch Panels**: Used to organize and connect multiple network cables in a structured manner.




- **Mounting Brackets**: Hardware for securing network equipment like access points, switches, or routers to walls or ceilings.



**Equipment:**

- **Access Points**: Devices that provide wireless connectivity to the network, expanding coverage and improving connectivity.



- **Switches and Routers**: Essential network devices that manage traffic between devices and direct data flow within the network.



- **Patch Panels**: Centralized points for managing and organizing network cables, facilitating easier troubleshooting and management.
- **Rack or Cabinet**: Enclosures for housing network equipment, providing a clean and organized setup.



**2. Laying Cables**

- **Plan Cable Routes**: Determine the most efficient paths for cables to follow, considering aesthetic factors and performance needs. Avoid running cables alongside power lines to reduce interference.
- **Run Cables**: Install cables through walls, ceilings, or conduits, ensuring they are not kinked or pinched. Use appropriate tools to pull cables through tight spaces.
- **Secure Cables**: Use cable ties, clips, or raceways to keep cables organized and prevent them from sagging or becoming damaged.
- **Label Cables**: Clearly label both ends of each cable to facilitate easier identification, troubleshooting, and future upgrades.



**3. Fixing Equipment**

- **Mounting Brackets**: Attach brackets securely to walls, ceilings, or racks based on the equipment's specifications. Ensure the brackets are level and aligned.
- **Secure Equipment**: Use screws or other fasteners to attach network devices like access points, switches, or routers to the mounted brackets. Check that all devices are firmly attached.

- **Ensure Stability**: Verify that the equipment is stable and does not wobble. Make adjustments if necessary to ensure a secure installation.

**4. Connecting Devices**

- **Terminate Cables**: Attach connectors to the ends of cables using a crimping tool. Ensure that wires are properly aligned and securely connected.
- **Patch Panels**: Insert the cables into the patch panel, ensuring they are correctly connected to the appropriate ports.
- **Plug into Equipment**: Connect cables from the patch panel or directly from the wall to the network devices, such as routers or switches.
- **Test Connections**: Use a cable tester to check the integrity and performance of each cable connection. Ensure that all devices are receiving proper signals and functioning as expected.

**In the context of mounting wireless network equipment, here's a description of each term:**

**Cable Color Coding**

Cable color coding is a standardized system used to identify and organize the wires within Ethernet cables. This system assigns specific colors to each wire in the cable (e.g., blue, orange, green, brown) according to a recognized standard (such as T568A or T568B). Proper color coding helps ensure that each wire is connected correctly to the corresponding pin on connectors and patch panels, which simplifies installation, troubleshooting, and maintenance of the network.



**Cable Termination**

Cable termination refers to the process of attaching connectors to the ends of network cables so they can be plugged into network devices. This involves stripping the insulation from the cable, arranging the internal wires in the correct order based on the wiring standard, and then crimping a connector (like an RJ45 plug) onto the cable. Accurate

cable termination is crucial for creating reliable connections and ensuring proper network functionality.




**Cable Testing**

Cable testing is the process of using specialized tools to verify that network cables are properly installed and functioning. Testing checks for continuity, signal strength, and potential faults such as shorts or miswires. By using a cable tester, you can ensure that cables are correctly terminated and free of defects, which is essential for maintaining a stable and efficient network.



**Interconnection**

Interconnection involves connecting various network devices, such as routers, switches, and access points, to establish a functional network. This process includes plugging cables into the appropriate ports on each device and ensuring that the connections are secure and correctly configured. Proper interconnection is key to ensuring seamless communication between devices and achieving optimal network performance.








**Practical Activity 3.1.2. Selecting tools, Materials and equipment to use in mounting of Wireless Network equipment**

**Task:**

1. **Read carefully the following task:**

   You have been assigned to install a wireless network in a small business office with 10 workstations and two conference rooms. The office requires strong and reliable Wi-Fi coverage throughout the space, as well as wired connections for critical devices such as printers and desktop computers. Your task is to select the appropriate tools, materials, and equipment needed to mount and install the wireless network equipment effectively.
2. After reading the task, receive the network design document given by your trainer, analyse the design and then select tools that can be used based on the network design /blueprint.
3. Follow the demonstration on reading the network blueprint /network design and choosing the corresponding components to be used in deploying wireless network.
4. Refers to similar network design document, select all tools, materials and equipment needed to deploy the network based on network design document following all process as demonstrated by your trainer.
5. Ensure that you have selected all necessary components to use in deploying wireless network based on network design document.
6. Read the key readings 3.2.2 and do the application of learning 3.2



**Key readings 3.2.1: Selecting tools, Materials and equipment to use in mounting of Wireless Network equipment**

When selecting tools, materials, and equipment for mounting wireless network equipment, it's important to establish specific criteria to ensure the installation is efficient, reliable, and scalable. Here are the key criteria to consider:

**1. Compatibility**

- **Network Standards**: Ensure that all tools, materials, and equipment are compatible with the latest network standards (e.g., Wi-Fi 6, Cat6 cables).
- **Device Integration**: Choose equipment that integrates seamlessly with existing network devices and infrastructure.

**2. Quality and Durability**

- **Tool Reliability**: Select high-quality tools that can withstand regular use and provide precise results (e.g., durable crimping tools and cable testers).
- **Material Longevity**: Opt for materials (e.g., cables, connectors) that are designed for long-term use and can withstand environmental conditions like temperature changes and physical wear.

**3. Performance**

- **Bandwidth Support**: Choose cables (e.g., Cat6, Cat6a) and equipment (e.g., routers, switches) that can handle the required bandwidth and data transfer rates.
- **Signal Strength**: Ensure that access points and wireless devices provide strong and consistent Wi-Fi coverage across the intended area.

**4. Ease of Installation**

- **User-Friendly Tools**: Select tools that are easy to use, even for those with basic technical skills, to streamline the installation process.
- **Pre-Configured Options**: Consider equipment that comes with easy setup options or pre-configuration to reduce installation time.

**5. Scalability**

- **Future-Proofing**: Choose equipment that allows for future upgrades and expansions without the need for significant overhauls (e.g., modular switches, expandable access points).
- **Port Availability**: Ensure there are enough ports and capacity in routers and

switches to accommodate potential future devices.

**6. Cost-Effectiveness**

- **Budget Alignment**: Select tools, materials, and equipment that offer the best value within the budget, balancing cost with performance.
- **Long-Term Investment**: Consider the total cost of ownership, including maintenance, support, and potential future upgrades.

**7. Safety and Compliance**

- **Regulatory Compliance**: Ensure that all selected materials and equipment meet local and international safety standards and regulations.
- **Electrical Safety**: Choose equipment that is safe to use, particularly when it involves electrical installations (e.g., power over Ethernet).

**8. Availability and Support**

- **Readily Available**: Opt for tools, materials, and equipment that are readily available to avoid delays in the installation process.
- **Technical Support**: Consider the availability of technical support and warranties for the equipment, in case issues arise during or after installation.

**9. Environmental Considerations**

- **Energy Efficiency**: Select equipment that is energy-efficient to reduce operational costs and environmental impact.
- **Sustainable Materials**: Consider the use of eco-friendly materials and tools where possible.

**10. Aesthetics and Cable Management**

- **Cable Organization**: Choose materials and tools that facilitate neat and organized cable management, reducing clutter and making future maintenance easier.
- **Aesthetic Impact**: Consider the visual impact of the installed equipment, particularly in customer-facing or high-visibility areas.






**Practical Activity 3.1.3. Laying cables in mounting of Wireless Network equipment**

**Task:**

**1.** Read carefully this task:

You have been tasked with installing a wireless network in a newly renovated office space designed for a small tech start-up. The office includes an open workspace, a meeting room, and a server room. The company requires a robust and reliable network to support 20 workstations, several printers, and video conferencing equipment. Your primary task is to lay the network cables that will connect all these devices, ensuring that both wired and wireless connections are stable and efficient.

**2.** Receive the wireless network design document and make analysis on the cables that are needed to connect all these devices, ensuring that both wired and wireless connections are stable and efficient depending on the wireless network design /blueprint

**3.** Follow the demonstration step by step for the process of laying the cables on one cable taken as a sample and write the whole process.

**4.** Repeat what you have seen to lay all needed cables by following the written steps as demonstrated on previous step

**5.** Ensure that you are able to lay all needed cables for connecting all these devices as shown on the network blueprint

**6.** Read the key readings 3.2.2 and do the application of learning 3.2

**Key readings 3.2.1: Laying cables in mounting of Wireless Network equipment**

When tasked with laying network cables for a wireless network installation in a newly renovated office space, follow these steps to ensure a successful and efficient setup:

**Step 1: Assess the Office Layout**

**Review the Floor Plan:** Analyze the office layout, including the open workspace, meeting room, and server room, to understand where each device (workstations, printers, access points) will be located.

**Identify Key Locations:** Determine the optimal locations for wireless access points and wired connections, considering coverage needs, device placement, and proximity to the server room.

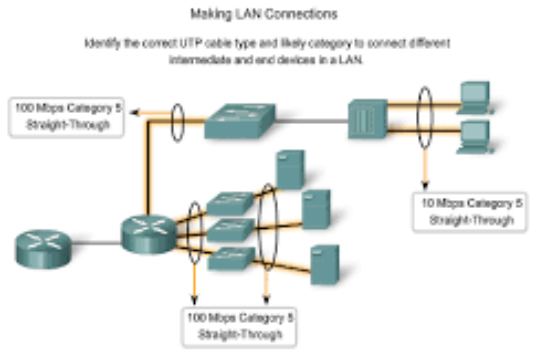


**Step 2: Plan Cable Routes**

**Determine Cable Paths:** Map out the most efficient cable routes from the server room to each workstation, printer, and access point. Consider factors such as the shortest distance, minimal interference, and accessibility for future maintenance.

**Avoid Interference:** Plan routes that avoid electrical wiring, HVAC systems, and other sources of potential interference. This helps to maintain signal integrity and network performance.

**Select Conduits and Pathways:** Decide on the use of conduits, raceways, or cable trays to organize and protect the cables along their paths.




**Step 3: Gather Tools and Materials**

**Select Tools:** Prepare essential tools such as cable strippers, crimping tools, fish tape, drills, screwdrivers, and cable testers.

**Choose Materials:** Gather the necessary materials, including Cat6 or Cat6a Ethernet cables, RJ45 connectors, cable ties, clips, raceways, and labeling supplies.

**Ensure Equipment Compatibility:** Verify that the chosen materials and tools are compatible with the network equipment being installed, such as access points, switches, and patch panels.

**Step 4: Install Conduits and Pathways**

**Prepare the Infrastructure:** Install conduits, raceways, or cable trays along the planned routes to organize and protect the cables. Ensure that these pathways are securely fastened to walls, ceilings, or floors.

**Create Access Points:** Drill holes or create openings in walls or ceilings as needed to pass cables through, ensuring that they align with the planned cable routes.



**Step 5: Run the Network Cables**

**Pull Cables:** Begin pulling Ethernet cables from the server room to each designated endpoint (workstations, printers, access points). Use fish tape or cable pullers to

navigate tight or challenging spaces.

**Avoid Damage:** Ensure that cables are not kinked, twisted, or excessively bent during installation. Lay cables smoothly to prevent signal degradation and physical damage.

**Organize and Group Cables:** Where multiple cables run together, group them using cable ties or clips to maintain organization and prevent tangling.



**Step 6: Secure and Label Cables**

**Secure Cables:** Fasten cables along their routes using cable ties, clips, or within raceways. Ensure that cables are secured at regular intervals to prevent sagging or movement.

**Label Each Cable:** Clearly label both ends of each cable with identifiers that correspond to their destination (e.g., "Workstation 1," "Conference Room AP"). This labeling will aid in troubleshooting and future maintenance.



**Step 7: Terminate Cables**

**Attach Connectors:** Use a crimping tool to terminate the cables with RJ45 connectors, ensuring that the wires are properly aligned and securely fastened.

**Connect to Patch Panels:** If using a structured cabling system, terminate cables at a patch panel in the server room. Connect the patch panel to network switches and other devices.



**Step 8: Test the Network Cables**

**Conduct Cable Testing:** Use a cable tester to verify the integrity, continuity, and signal strength of each cable. Ensure that all connections are functioning correctly and that there are no faults.

**Document Results:** Record the test results for each cable, noting any issues that require correction. Retest if necessary to ensure reliability.



**Step 9: Connect Devices**

**Plug into Devices:** Connect the labeled cables to the appropriate network devices (workstations, printers, access points). Ensure that each device is properly connected and powered on.

**Verify Connectivity:** Check that all devices have network connectivity, both wired and wireless. Perform basic network tests to confirm that data is flowing correctly.



**Step 10: Finalize and Review Installation**

**Inspect the Installation:** Conduct a final inspection of the cable layout, ensuring that all cables are securely fastened, properly labeled, and neatly organized.

**Cleanup:** Tidy up any remaining installation materials and tools. Make sure that the office space is left clean and free of any debris or leftover materials.

**Document the Setup:** Create a final report or diagram of the network setup, including cable routes, device connections, and test results. This documentation will be useful for future maintenance or troubleshooting.



**Selection of tools, Materials and equipment**



**Points to Remember**

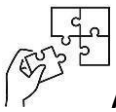
**Factors that should be taken into consideration:**

- Size of Physical Location
- Intended Use of the Wireless Network
- Number of Wireless Devices
- Wireless Client Device Capabilities
- The Environment
- Performance Expectations
- Bring Your Own Device (BYOD) Acceptance
- Building Age and Construction Materials
- Network Infrastructure Devices

**Laying cables**

**Factors that should be taken into consideration:**

- **Topology:** The first thing that you need to think about when designing your networking infrastructure is the type of topology. It will help you in determining what hardware components and the type of cable you'll be needing to build the networking infrastructure.
- **Bandwidth:** The amount of data that may be transferred at one time is referred to as bandwidth. The bigger the bandwidth of the cable, the faster the information is transported across it.
- **Signal Attenuation:** A cable with a higher signal attenuation allows information to be sent without the use of any equipment to increase the electric signal.
- **Electromagnetic Interface:** Transmission over networks may be interrupted by EMI from equipment such as fluorescent lights, photocopiers, and other sources. In such a case, EMI-resistant cables can be useful.
- **Expansion Capabilities:** Depending on how easy cable installation and the requirement for extra equipment are in the process of expanding every network.



**Application of learning 3.1.**

XNC photo editing company located at mulinga city, need a qualified technician who will setting up a wireless network in a new office building. As a network technician, you need to select the appropriate tools for different stages of the network installation process. The manager of the company has listed all tasks to be performed and all tools that are in a store and he needs you select the appropriate tools among provided one to accomplish each task.

Tasks:

1. Conducting a Site Survey: Determine the optimal locations for access points and identify potential sources of interference.
2. Mounting Access Points: Securely install access points on walls or ceilings at designated locations.
3. Running Ethernet Cables: Lay out and secure Ethernet cables from the network switches to the access points.
4. Configuring Network Settings: Set up the access points, router, and other network devices with the necessary configurations.
5. Testing Network Performance: Evaluate the network performance to ensure proper coverage and connectivity.

Tools:

1. Drill and Mounting Hardware
2. Ethernet Cable Tester
3. Wi-Fi Analyzer (e.g., Ekahau, NetSpot, AirMagnet Survey
4. Screwdriver Set
5. Laptop with Network Configuration Software
6. Cable Management Clips
7. Wireshark

## Indicative content 3.2: Basic Configuration of Wireless Network Devices

**Duration: 4**

**Theoretical Activity 3.2.1: Describing the basic configuration of wireless network devices**

**Tasks:**

1. Observe the visual aid and express your thoughts about it.
2. Receive task sheet given by trainer that include the description of the basic configuration of wireless network devices.
   - ✓ Administration parameters
   - ✓ Access settings
   - ✓ Security mode settings
3. Make summary of what you read and explain them to your colleague and then listen to them while they are explaining what they summarise.
4. Be attentive while trainer is giving expert view and clarifications for what you have discussed.
5. Ask questions and clarifications if necessary.
6. Read the key readings 3.2.1 in this manual

**Key readings 3.2.1.: : Describing the basic configuration of wireless network devices**

- **Administration parameters setting:** It refers to the options and settings that control the overall management and configuration of the router itself. These settings are typically accessible through a web-based interface, where users can log in to the router's management page to configure various aspects of the device. The specific options may vary depending on the router model and manufacturer, but common administration parameters include login credentials (username and password), gateway IP address
- **Accessibility settings:** Accessibility settings in a wireless router configuration typically refer to options that control how users and devices can connect to and interact with the router. While specific options may vary based on the router model and manufacturer, common accessibility settings include:
  - ✓ Wireless Network Name (SSID):

- ✓ SSID Broadcast: This setting determines whether the router broadcasts its SSID. Disabling SSID broadcast can make the network less visible to casual users, but it does not provide robust security.

- **Wireless Password (Pre-shared Key):**
  - ✓ Password Configuration: Admins can set and update the pre-shared key (password) required for devices to connect to the wireless network.
- **Security mode setting:**
  - ✓ Security Mode: Users can choose from various security modes such as WEP, WPA, WPA2, or WPA3. WPA2 and WPA3 are more secure options and are recommended over older standards like WEP.
  - ✓ Encryption Type: Within the selected security mode, users may need to choose the encryption algorithm (e.g., TKIP or AES).



**Practical Activity 3.2.1: Basic Configuration of wireless network devices**




**Task:**

**1.** Read carefully this task:

You have been assigned to deploy a wireless network within an indoor office environment for a growing startup. The office space is spread across two floors, with a mix of open workspaces, private offices, and conference rooms. The company needs seamless wireless connectivity throughout the entire office to support 30 employees, including those using laptops, smartphones, and tablets.

Your task is to perform the basic configuration of the wireless network devices, ensuring secure and efficient network management. This includes setting the administration parameters, configuring the gateway IP address, and establishing a secure username and password for accessing the network devices.

**2.** Follow the demonstration on configuring the access point and write each step.

**3.** repeat what you have seen by following written steps as per your respective task sheet instructions where you generally have to:
   - ✓ Connect access point.
   - ✓ Set gateway IP address, username, and password.
   - ✓ Configure accessibility settings by selecting radio mode, channel, SSID, and password.
   - ✓ Configure security mode settings.

**4.** Ensure that you are able to configure access point on your own, ask questions if necessary.

**5.** Read the key readings 3.2.2 and do the application of learning 3.2

**Key readings 3.2.2: Basic Configuration of wireless network devices**

To deploy a wireless network within the indoor office environment for the growing startup, Follow these step-by-step instructions to ensure secure and efficient network management:

**Step 1: Prepare for Configuration**

- **Review Office Layout**: Understand the layout of the office, including the open workspaces, private offices, and conference rooms. Identify the best locations for placing wireless access points to ensure full coverage across both floors.
- **Gather Equipment**: Ensure that all wireless network devices (routers, access points) are ready for configuration, and that you have the necessary tools (e.g., laptop, Ethernet cables) to access the devices.

**Step 2: Access the Wireless Router's Administration Interface**

- **Connect to the Router**: Use an Ethernet cable to connect your laptop to one of the LAN ports on the wireless router.
- **Login to the Router**: Open a web browser and enter the router's default IP address (usually 192.168.1.1 or 192.168.0.1) into the address bar. Enter the default username and password to log in (typically "admin" for both).
- **Update Firmware**: Check if there is a firmware update available for the router. If so, update the firmware to ensure you have the latest security features and performance improvements.



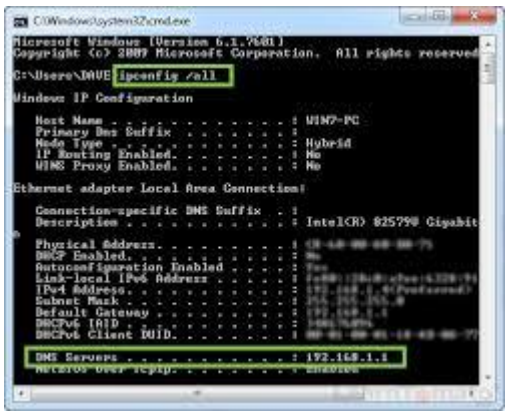
**Step 3: Set Administration Parameters**

- **Configure SSID (Network Name)**: Navigate to the wireless settings section and set the SSID (network name) that will be visible to users. Ensure it is easily identifiable by the company employees.
- **Select Wireless Mode**: Choose the appropriate wireless mode (e.g., 802.11ac or 802.11ax) based on the devices used in the office.
- **Set Wireless Channel and Bandwidth**: Choose an optimal wireless channel and bandwidth (e.g., 20/40/80 MHz) to minimize interference from neighboring networks and optimize performance.




**Step 4: Configure the Gateway IP Address**

- **Set the Gateway IP Address**: In the LAN settings, configure the router's IP address (e.g., 192.168.1.1). Ensure it does not conflict with any existing network devices.
- **Configure DHCP Settings**: Enable the DHCP server on the router to automatically assign IP addresses to connected devices within a specified range (e.g., 192.168.1.100 to 192.168.1.200).
- **Save Settings**: Save the configuration and reboot the router to apply the new gateway IP address.



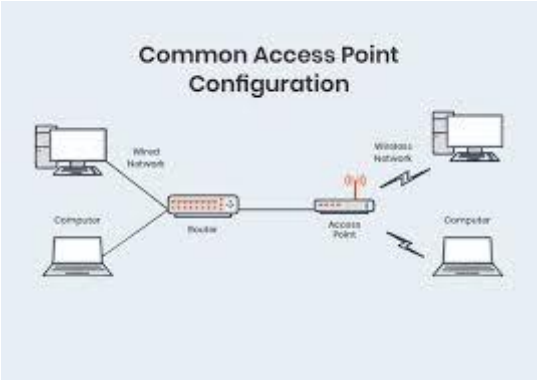

**Step 5: Set Up Secure Username and Password**

- **Change Default Username and Password**: In the administration settings, change the default username and password to something secure. Use a strong password with a mix of letters, numbers, and special characters.
- **Set WPA2/WPA3 Security**: Configure the wireless network security settings to use WPA2 or WPA3 encryption. Set a strong passphrase for users to connect to the wireless network.
- **Document Credentials**: Securely document the new admin credentials and wireless passphrase for future use by network administrators.




**Step 6: Deploy Wireless Access Points**

- **Position Access Points**: Place the wireless access points in the identified locations throughout the office to ensure optimal coverage. Connect them to the network using Ethernet cables.
- **Configure Access Points**: Access the administration interface of each access point and configure them with the same SSID and security settings as the main router.
- **Test Coverage**: Walk through the office with a Wi-Fi-enabled device to test signal strength and coverage in all areas, adjusting access point placement as needed.

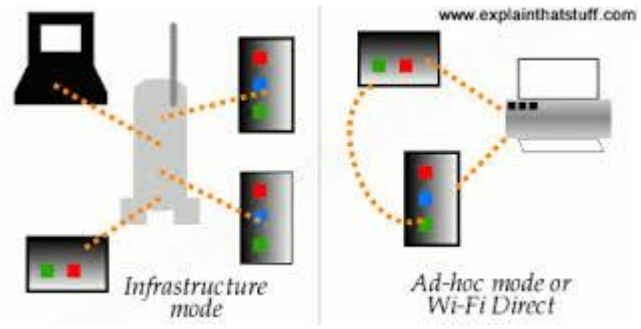



**Step 7: Test and Validate the Network**

- **Test Connectivity**: Connect multiple devices (laptops, smartphones, and tablets)

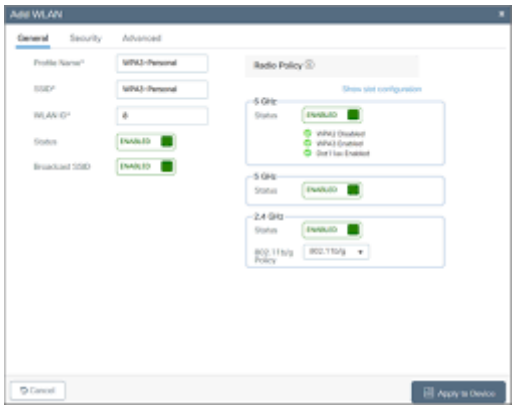
to the wireless network. Ensure that all devices can access the internet and communicate with each other.

- **Verify Admin Access**: Log back into the router's administration interface using the new credentials to confirm that they work and that unauthorized access is blocked.
- **Perform Speed Tests**: Use online tools to perform speed tests in different areas of the office to ensure that the network provides the required performance.




**Step 8: Finalize and Document**

- **Document Network Configuration**: Create a detailed document that includes the SSID, encryption type, admin credentials, and any other relevant network configuration details.
- **Educate Users**: Provide the employees with the SSID and wireless passphrase, along with basic instructions on how to connect to the network.
- **Cleanup and Review**: Ensure that all cables are neatly organized, and that the workspace is left clean and tidy. Review the installation to ensure that everything is functioning as expected.



**OTHER WAY**

**Steps to Configure WLAN in Cisco Packet Tracer:**

**Step1:** we need these devices to set up the network topology as shown in the table

below:

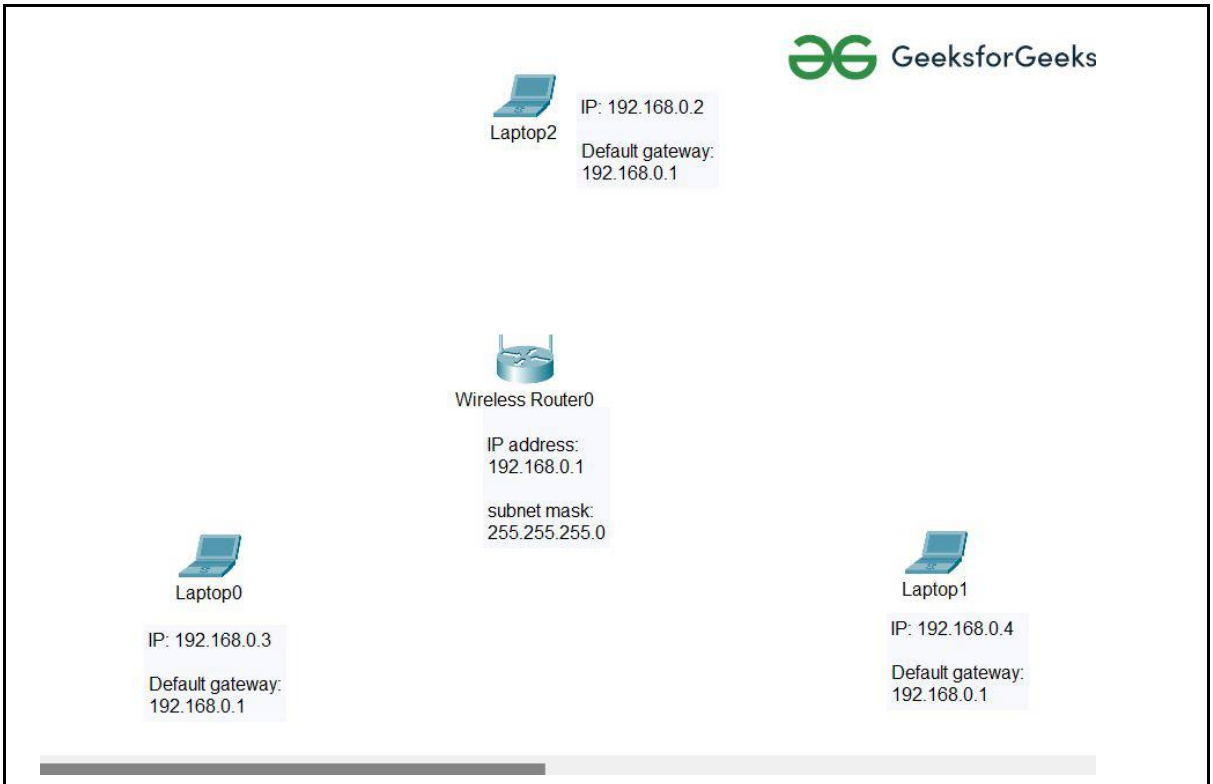
| S. No | Device | Model Name(as given in cisco packet tracer) | quantity |
|---|---|---|---|
| 1. | Router | WRT300N | 1 |
| 2. | Laptop | Laptop | 3 |

By using these devices we'll have to create a network like shown in the representation:

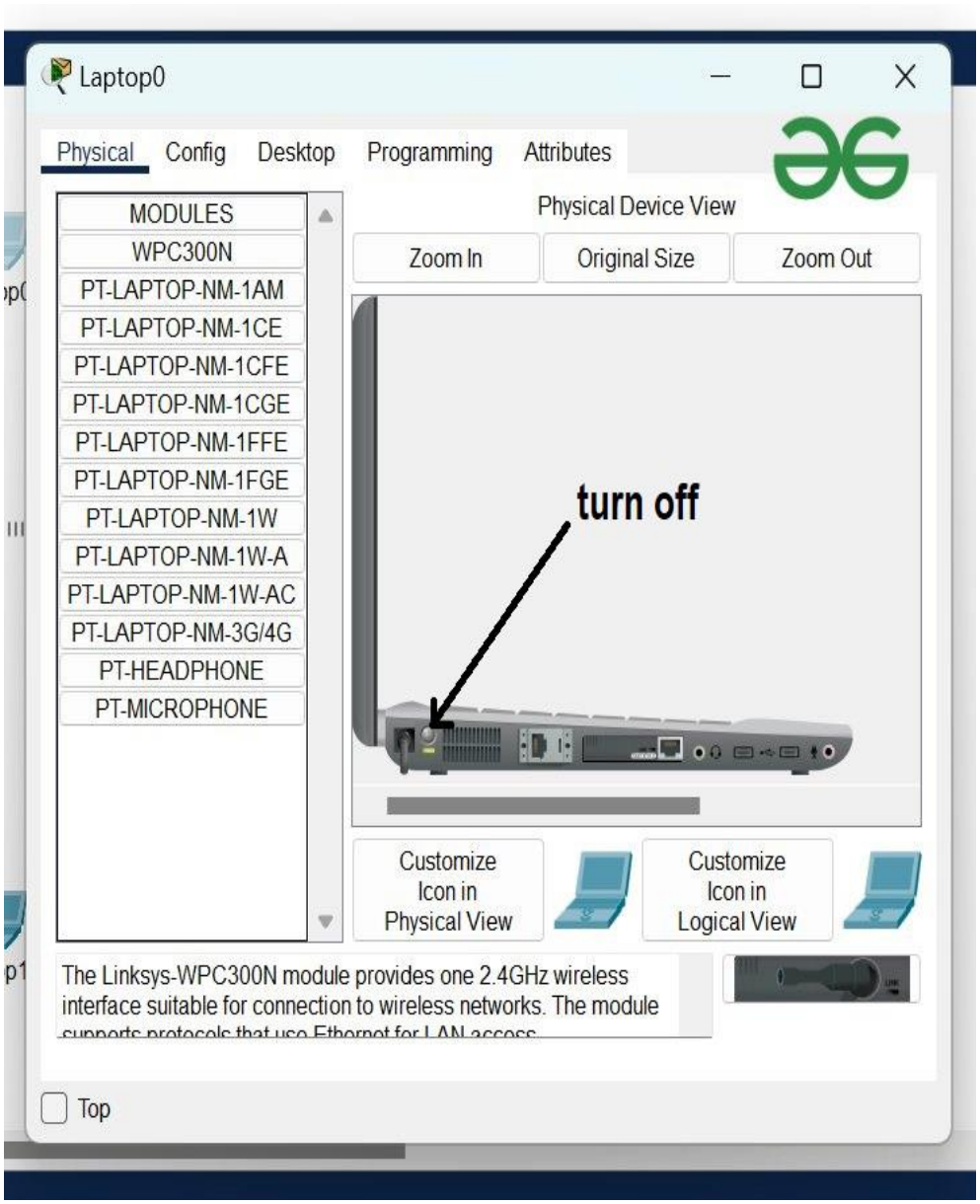
e below:

| S. No | Device | Model Name(as given in cisco packet tracer) | quantity |
|---|---|---|---|
| 1. | Router | WRT300N | 1 |
| 2. | Laptop | Laptop | 3 |

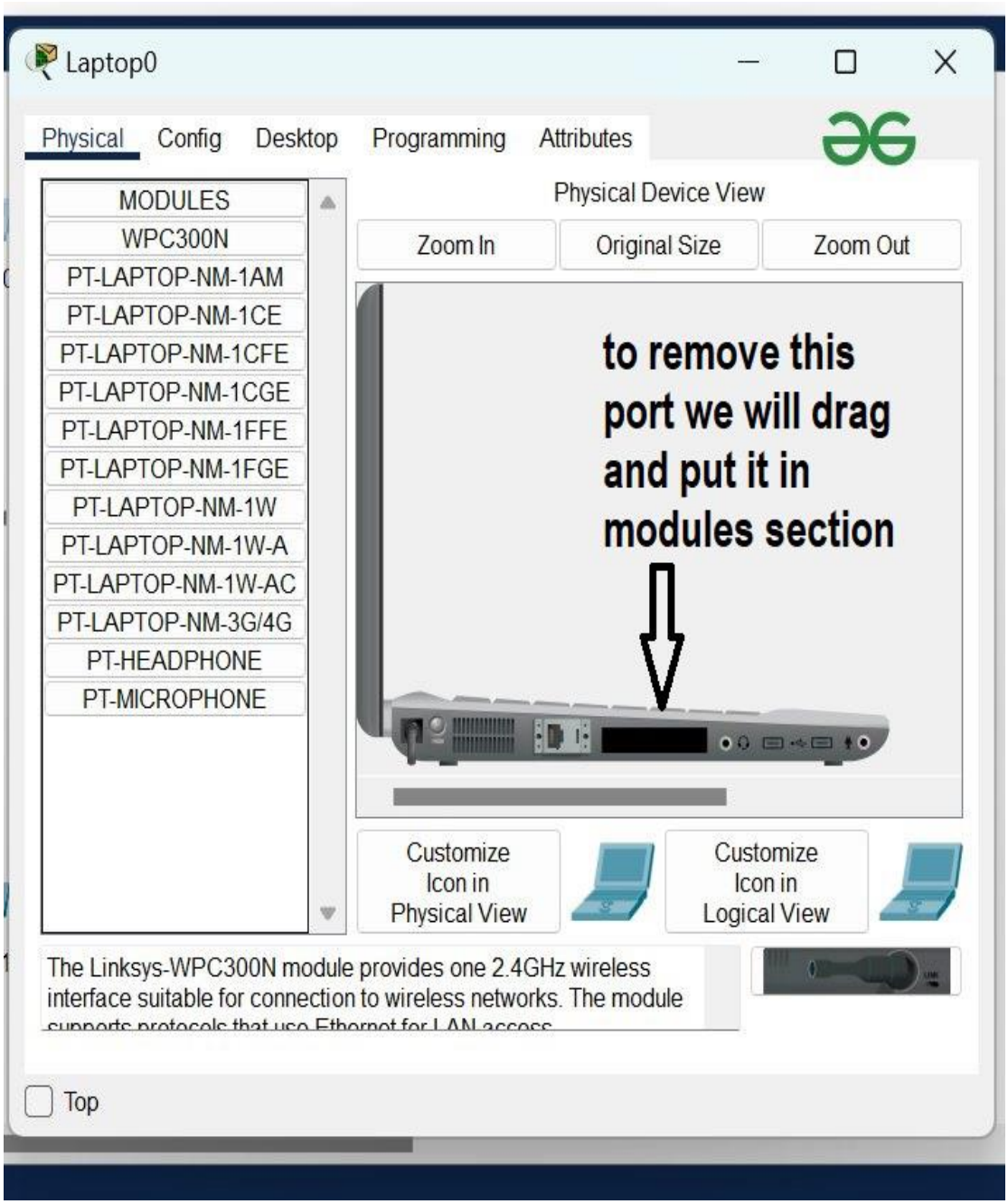
By using these devices we'll have to create a network like shown in the representation:




**Step 2:** Configuring laptops to make them wireless. First, click on the laptop0 and turn off its power to change the ports basically we are going to replace the wired port with the wireless port which is **WPC300N**.


Laptop0
Physical
Config
Desktop
Programming
Attributes
MODULES
WPC300N
PT-LAPTOP-NM-1AM
PT-LAPTOP-NM-1CE
PT-LAPTOP-NM-1CFE
PT-LAPTOP-NM-1CGE
PT-LAPTOP-NM-1FFE
PT-LAPTOP-NM-1FGE
PT-LAPTOP-NM-1W
PT-LAPTOP-NM-1W-A
PT-LAPTOP-NM-1W-AC
PT-LAPTOP-NM-3G/4G
PT-HEADPHONE
PT-MICROPHONE
Physical Device View
Zoom In
Original Size
Zoom Out
turn off
Customize Icon in Physical View
Customize Icon in Logical View
The Linksys-WPC300N module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module
Top

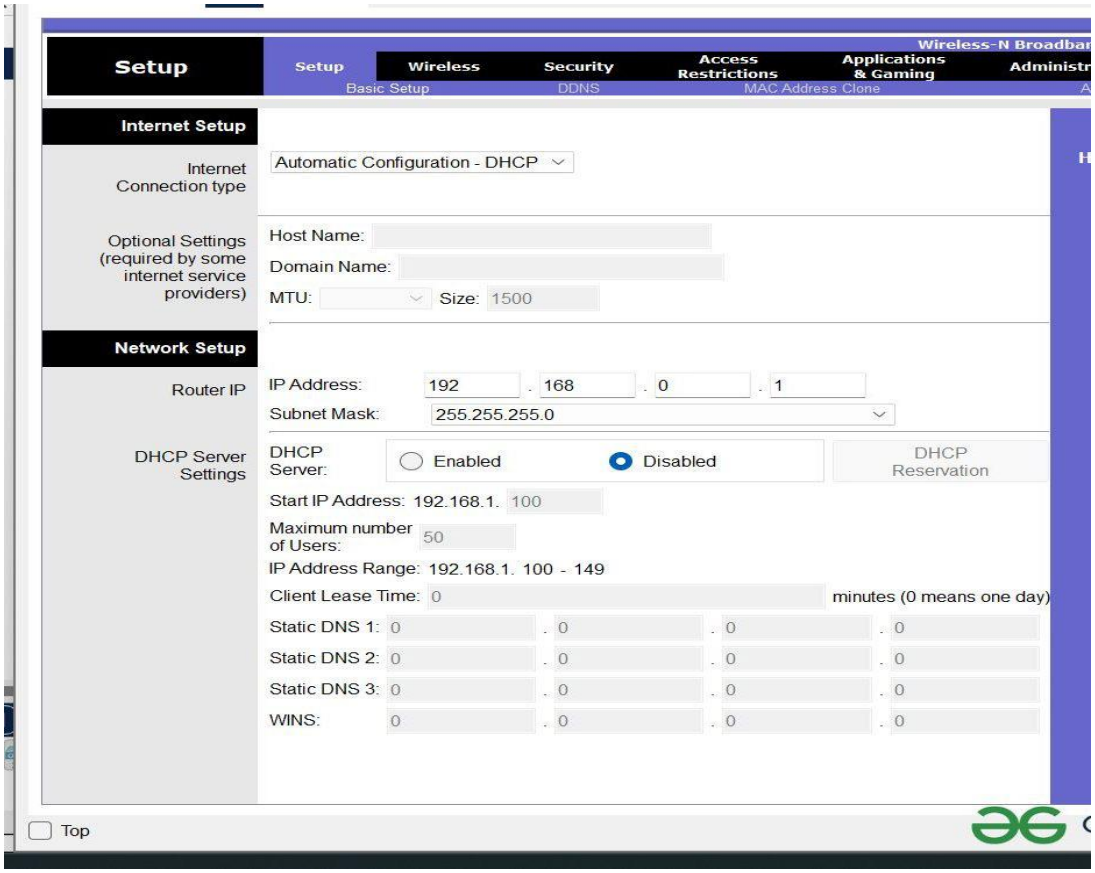

Laptop0
Physical
Config
Desktop
Programming
Attributes
MODULES
WPC300N
PT-LAPTOP-NM-1AM
PT-LAPTOP-NM-1CE
PT-LAPTOP-NM-1CFE
PT-LAPTOP-NM-1CGE
PT-LAPTOP-NM-1FFE
PT-LAPTOP-NM-1FGE
PT-LAPTOP-NM-1W
PT-LAPTOP-NM-1W-A
PT-LAPTOP-NM-1W-AC
PT-LAPTOP-NM-3G/4G
PT-HEADPHONE
PT-MICROPHONE
Physical Device View
Zoom In
Original Size
Zoom Out
Place WPC300N here and turn on the laptop
Customize Icon in Physical View
Customize Icon in Logical View
The Linksys-WPC300N module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module
Top



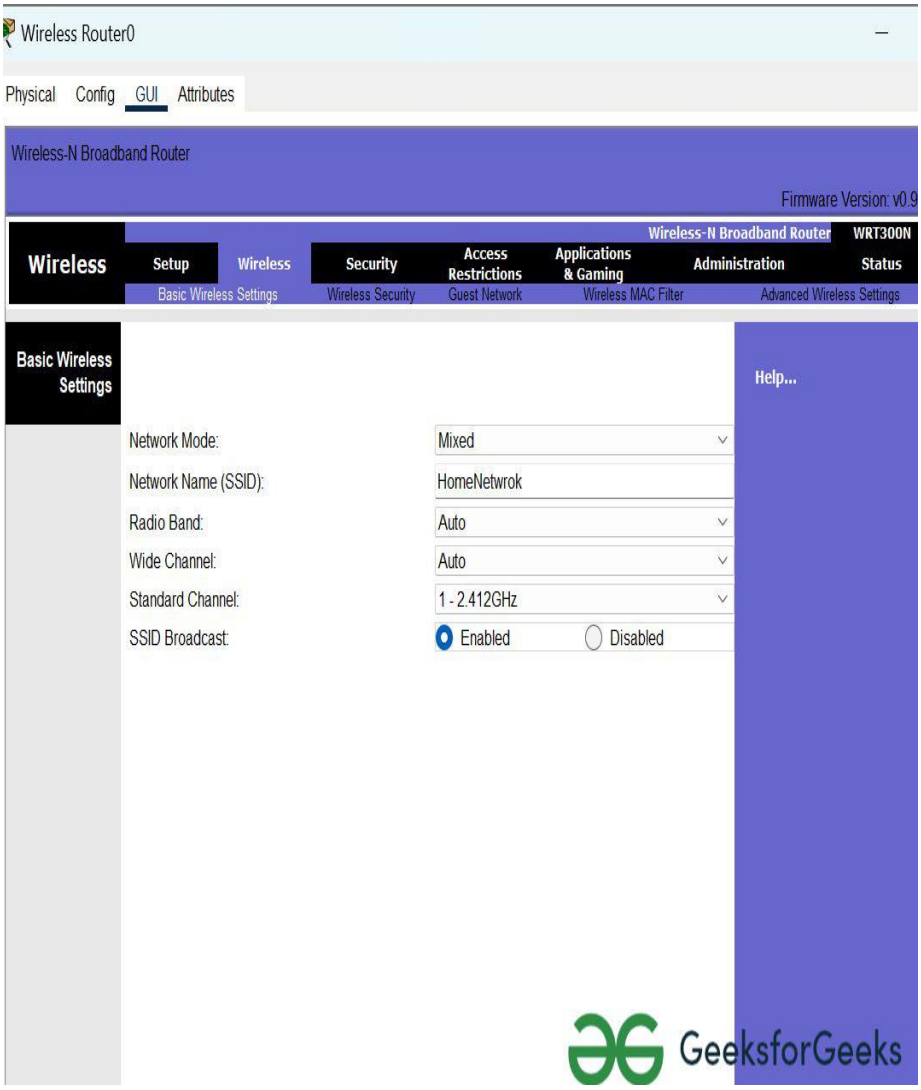

- Replace with **WPC300N** and make sure to turn it ON.
- Repeat the same procedure with **Laptop1** and **Laptop 2.**
- after that, we will assign IP addresses and a *default gateway* to the *laptops*.

**Step 3:** Configure the Router with an IP address and Generate a Security key.

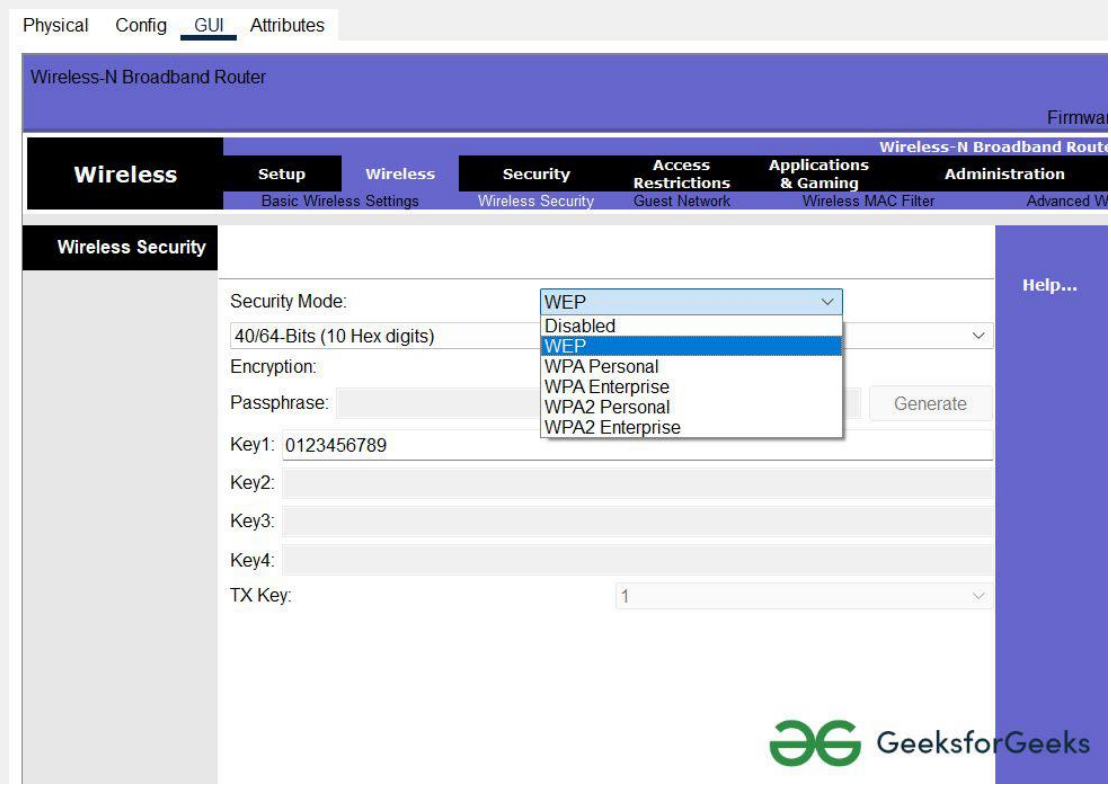
- First, click on Router and Go to GUI.
- Then click on a setup where you will find the IP address assigned to **192.168.0.1** and subnet mask**[255.255.255.0]**.
- Then disable the DHCP server because we have to configure statically.
- Then Save the **settings.**




- Then move to the wireless option.
- set Network Name(SSID) is **Home Network.**
- Save the **settings.**



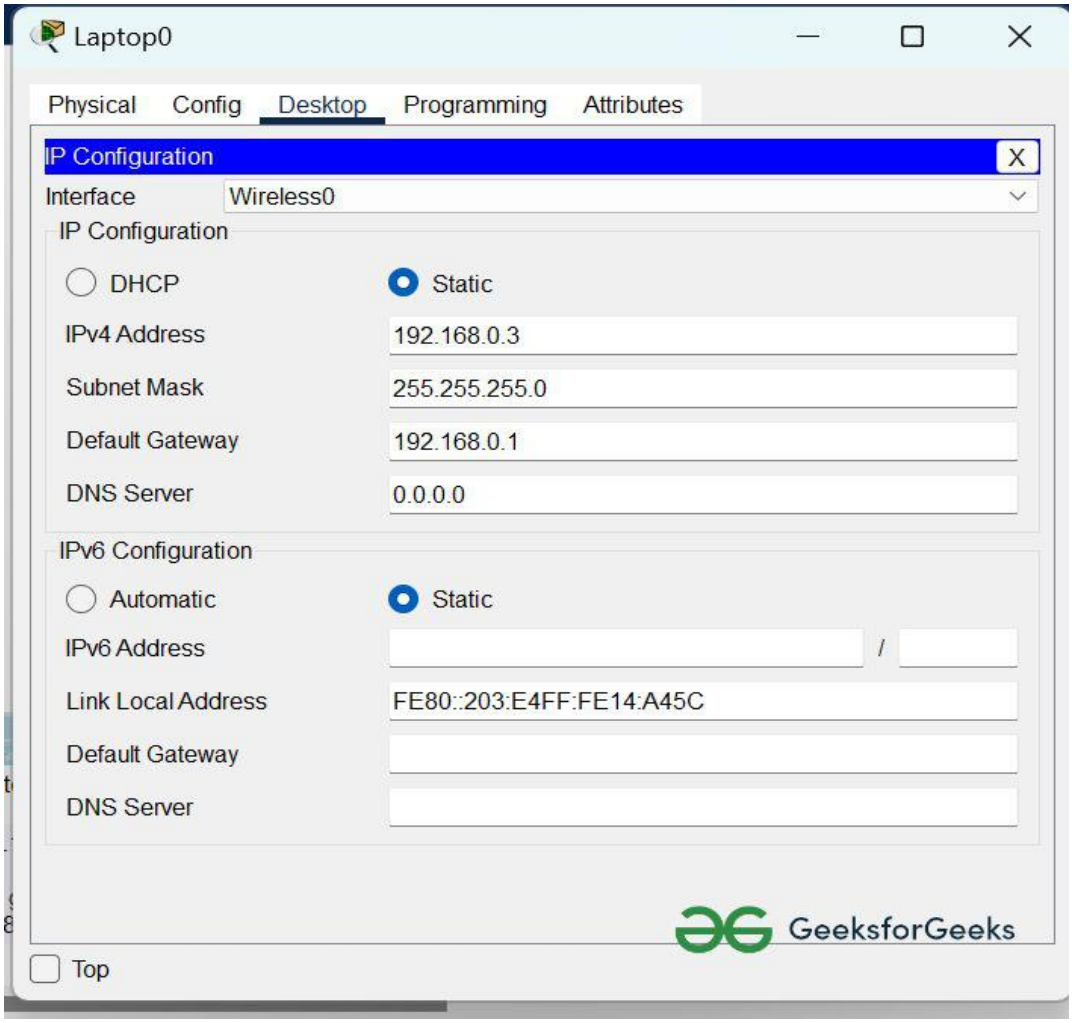

- Then we set the security key.
- Click on **wireless security** and select security mode as **WEP**.
- Then we'll generate KEY by entering 10-digit Hexa-numeric value. eg**: 0123456789.**
- save the settings.




**Step 4:** Now we will configure the laptops using the IP addressing table given below:

| S. NO | Device | IPv4 Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| **1.** | laptop0 | 192.168.0.3 | 255.255.255.0 | 192.168.0.1 |
| **2.** | laptop1 | 192.168.0.4 | 255.255.255.0 | 192.168.0.1 |
| **3.** | laptop2 | 192.168.0.2 | 255.255.255.0 | 192.168.0.1 |

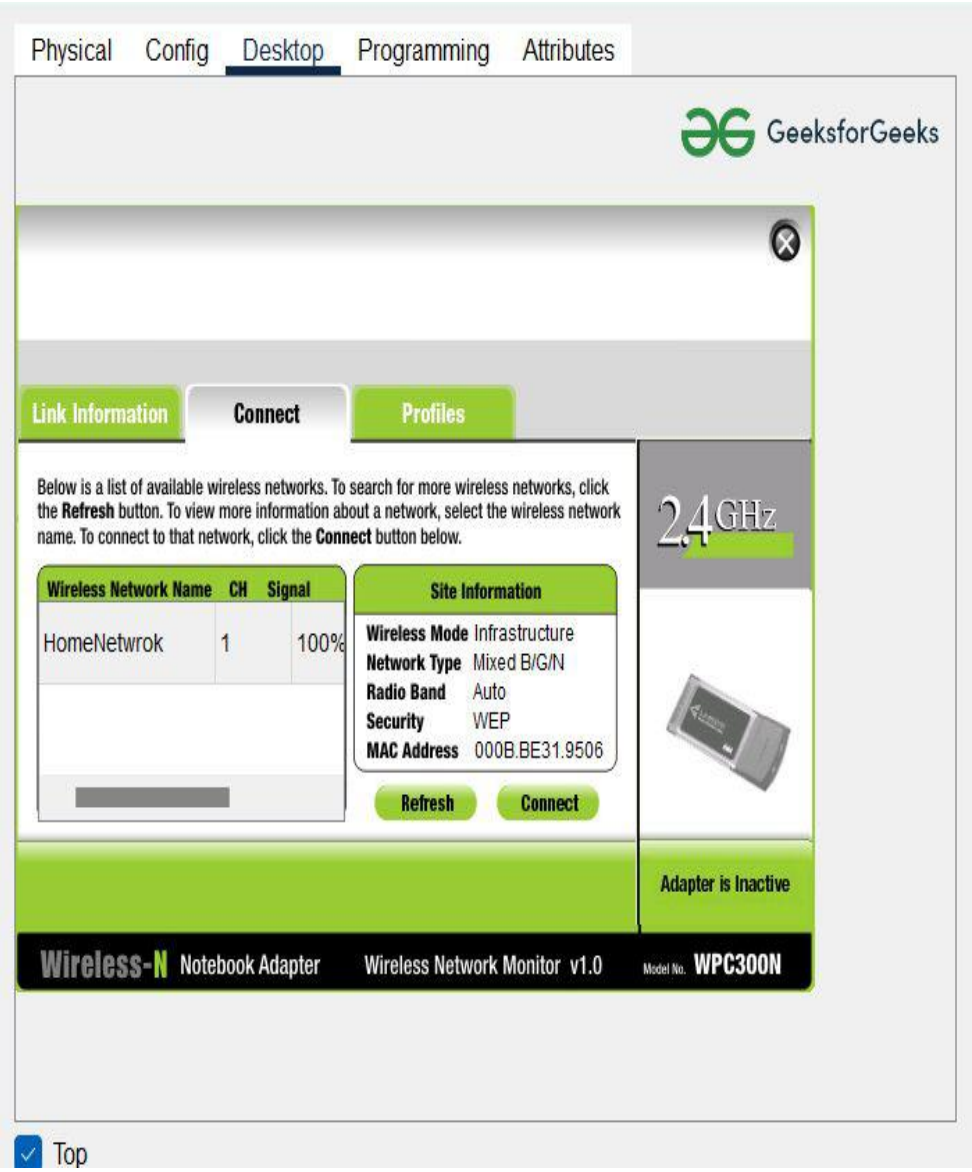
**Configure Laptop0:** To configure the laptop first set the IP configuration as static then add the IPv4 address and default gateway.



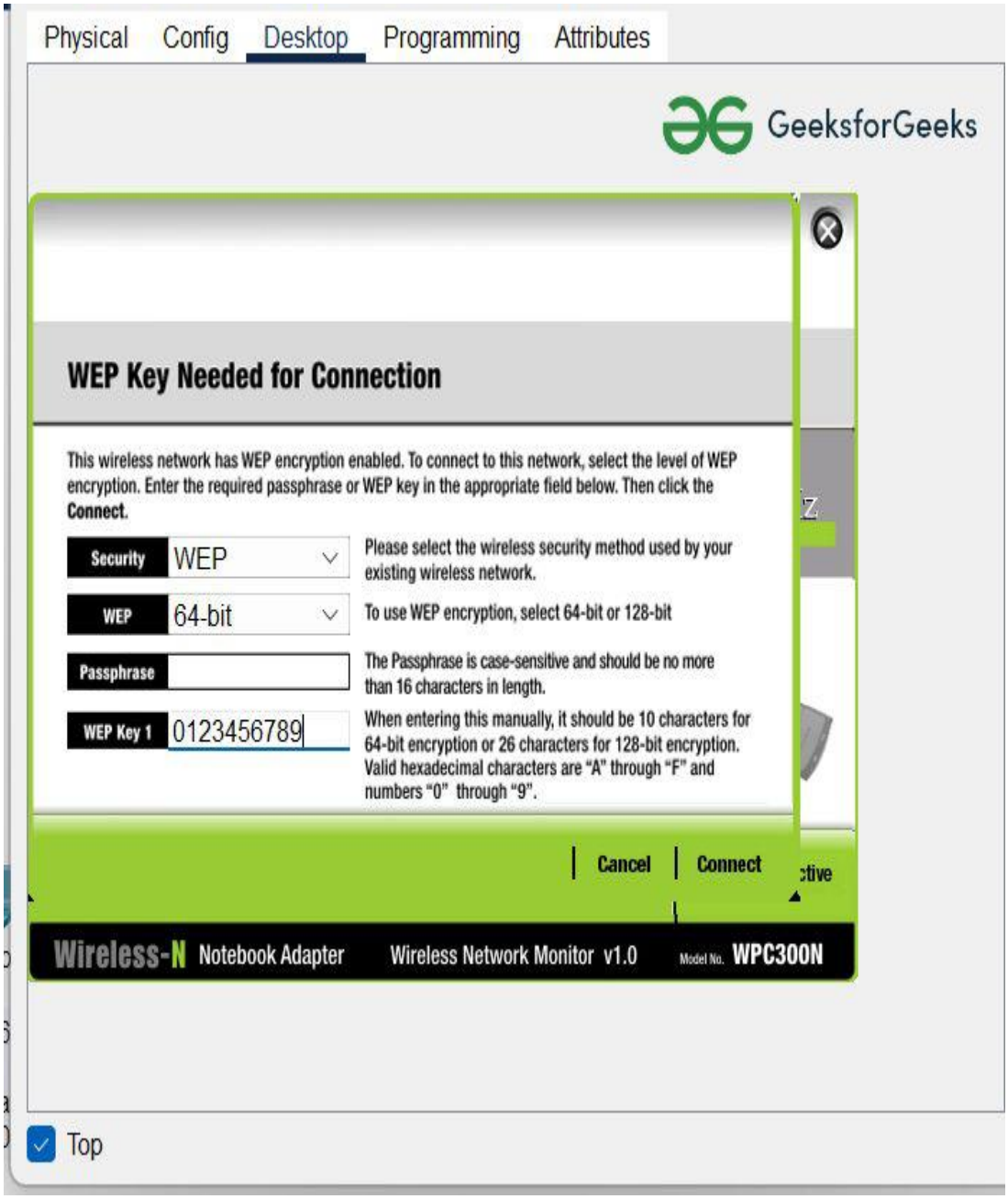

**Note:** Repeat the same process with Laptop1, laptop2 and configure both devices by adding IP configuration.

**Step 5:** Connect the laptop to the router by entering the security key in the laptop.

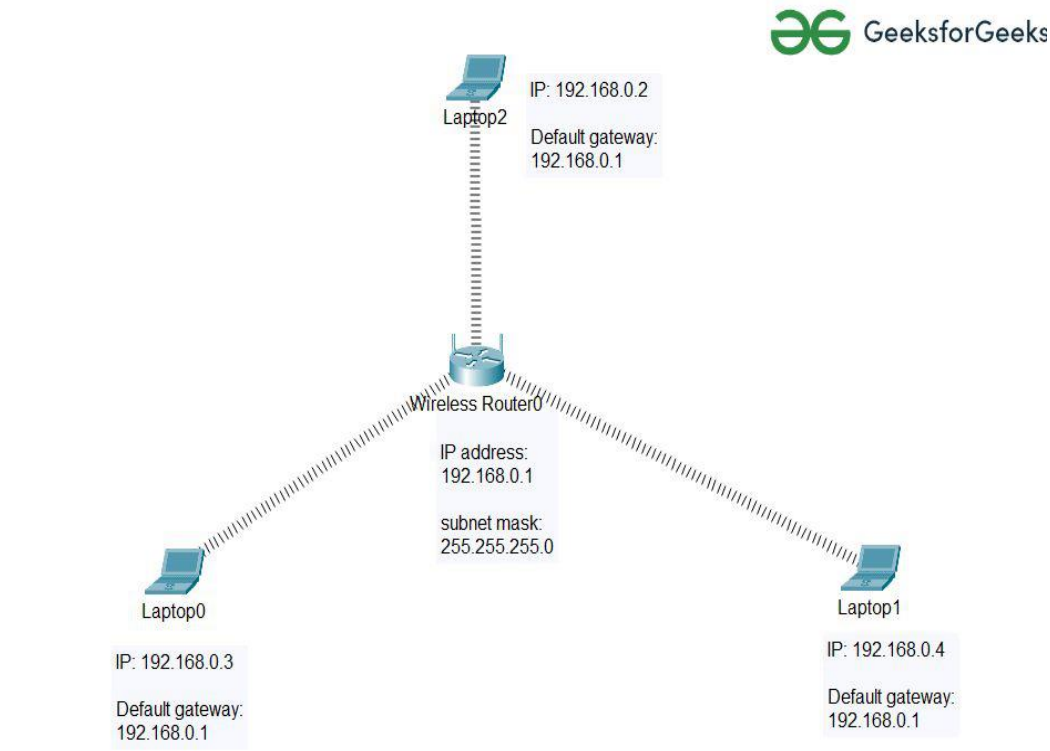
- Click on **laptop0** and go to desktop.
- Click on connect and refresh the network.
- After a few seconds, it will show the name of the network we have assigned.
- Click on Home Network.




- Then enter the security key in WEP key 1 and hit on connect.
- Laptop 0 will connect with the router.

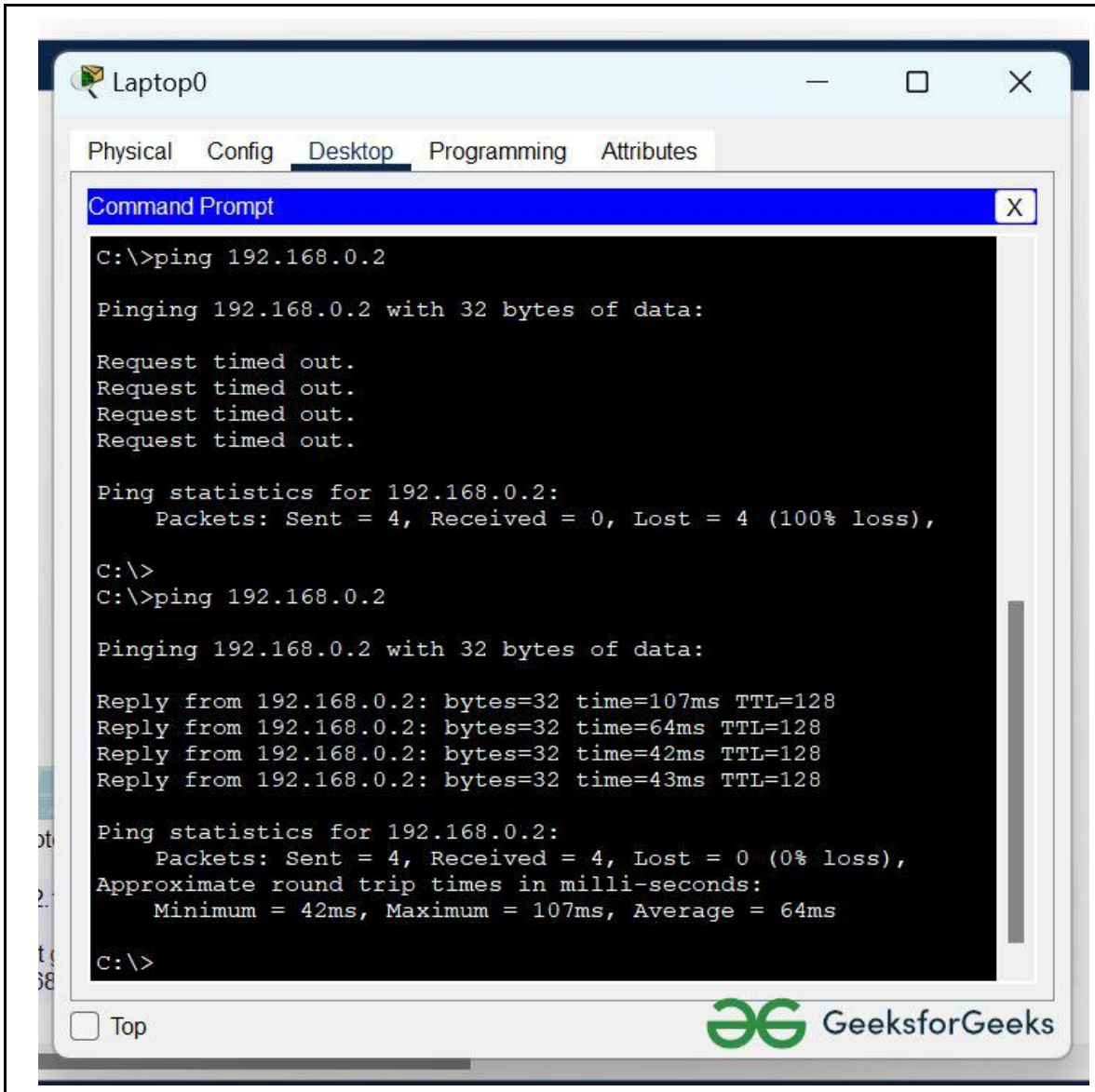



- Repeat the same process with Laptop1 and laptop2 so that they can connect with the router.
- After all of this, all of the hosts will connect with the router




**Step 6:** Then we'll verify the wireless connection by pinging the IP address of any laptop or by sending and receiving data packets. For example: Go to the command prompt of Laptop0 and type the following command:

**command:** ping 192.168.0.3


Laptop0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.0.2
Pinging 192.168.0.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.0.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
C:\>ping 192.168.0.2
Pinging 192.168.0.2 with 32 bytes of data:
Reply from 192.168.0.2: bytes=32 time=107ms TTL=128
Reply from 192.168.0.2: bytes=32 time=64ms TTL=128
Reply from 192.168.0.2: bytes=32 time=42ms TTL=128
Reply from 192.168.0.2: bytes=32 time=43ms TTL=128
Ping statistics for 192.168.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 42ms, Maximum = 107ms, Average = 64ms
C:\>
Top
GeeksforGeeks

**Points to Remember**

**To deploy a wireless network within the indoor office environment for the growing start-up, follow these step-by-step instructions to ensure secure and efficient network management:**

**Step 1:** Prepare for Configuration

**Step 2:** Access the Wireless Router's Administration Interface

**Step 3:** Set Administration Parameters

**Step 4:** Configure the Gateway IP Address

**Step 5:** Set Up Secure Username and Password

**Step 6:** Deploy Wireless Access Points

**Step 7:** Test and Validate the Network

**Step 8:** Finalize and Document

**Application of learning 3.2.**

XYZ Corporation, a growing tech start-up in Kigali city, has recently moved into a new office building. The company needs a robust and secure wireless network to support its team of software developers, project managers, and administrative staff. They have hired you as a network technician, to set up the wireless network. The following activities are required to be performed to ensure a functional and secure wireless environment:

1. Install and connect the wireless access points (APs) to the network infrastructure
2. Set Gateway IP Address, Username, and Password
3. Set up the wireless network settings for optimal performance and user accessibility.
4. Configure Security Mode Settings

As a network technician, you need to perform these activities for setting up a secure and efficient wireless network for XYZ Corporation, ensuring all users can connect and communicate effectively while maintaining high security standards

## Indicative content 3.3: Application of Access Restrictions Parameters

**Duration: 3**

**Theoretical Activity 3.3.1: Identifying Access restrictions Parameters.**

**Tasks:**

1. Answer what you understand by the permissions and security threat within a wireless network.
2. Listen to the trainer's explanation about the permissions and security threat within a wireless network.
3. In your small group, discuss on the following key terms
   a. Mac address filtering
   b. Parental control
   c. Proxy setting
4. Write your findings on flipchart/paper
5. Discuss with others on provided answers and choose the correct answers
6. Take key points on your notebook while trainer is providing expert view and clarification.
7. Ask questions or concerns if any.
8. Read the key readings 3.3.1 in this manual

**Key readings 3.3.1.: Identifying Access restrictions Parameters**

- **MAC Address Filtering**: It primarily involves filtering devices based on their MAC (Media Access Control) addresses. Each device has a unique MAC address, and the router can be configured to allow or deny specific addresses. MAC address filtering allows you to block traffic coming from certain known machines or devices.
- **Configuration Flexibility:** Most routers allow users to easily configure device filtering settings through the router's web interface. Users can add or remove devices from the allowed list as needed.
- **WLAN clients (Wireless Local Area Network)** client, refers to a device that connects to a wireless network. It could be any device such as a laptop, smartphone, tablet, or any other device with Wi-Fi capabilities. Wireless Local Area Network client, refers to a device that connects to a wireless network. It could be any device such as a laptop, smartphone, tablet, or any other device with Wi-Fi capabilities.
- Parental control function allows you to set up unique restrictions on internet access for each member of your family. You can block inappropriate content,

set daily limits for the total time spent online and restrict internet access to certain times of the day.

- Proxy Settings: A proxy server is an intermediate server that acts as a gateway between a user's device and the internet. When a user makes a request to access a web resource, the request is sent to the proxy server first, which then forwards the request to the destination server.



**Practical Activity 3.3.2: Apply Access restrictions Parameters**




**Task:**

1. Go in computer Lab and perform this activity:

Connect to the wireless network provided by the Access point in your computer lab and set up MAC address filtering to restrict access to the network based on device MAC addresses.

Then after, create a guest network that isolates guests from the main network

2. Demonstrate the trainees on how to apply Access restrictions parameters by performing these activities:
   - ✓ Set up Mac address filtering
   - ✓ Apply Parental control
   - ✓ Changing Proxy setting
   - ✓ Creating guest network
3. Repeat the above steps with respect to your task in order to perform activity and ask clarification if necessary.
4. Perform the applications of learning 3.3 and read the key readings 3.3.2



**Key readings 3.3.2: Apply Access restrictions Parameters**

- **To log in to the router/ wireless router/access point:** To open your router's web interface, open a web browser and enter the router's IP address in the address bar. Check the user's manual or manufacturer's web page to find the exact IP address for your router. The most common default router IP addresses include "192.168.8.1", "192.168.1.1", "192.268.0.1", and "10.0.0.1".
- To Log in to the admin website. You will need to enter the admin username and password to log into your router. If you have not set the admin username and password, use the default information. It may be printed on the router itself, in the user's manual, or on the manufacturer's website.

- For MAC Address Filtering:
  - ✓ Each internet-connected device has a unique MAC address. You can find the MAC address on a computer or other mobile devices. You can usually find the MAC address for a device in the "About" section in the Settings menu.
  - ✓ A MAC address looks something similar to this: 08:00:27:0E:25:B8. It may be listed as "MAC address", "Wi-Fi Address" or something similar.
- For Proxy settings:
  - ✓ Navigate to Proxy or Filtering Settings: Look for a section related to proxy settings, content filtering, or access control. The exact location may vary depending on your router model.
  - ✓ Proxy Server Configuration: If your router supports proxy-related features, you may find fields to enter the IP address and port number of the proxy server. Enter the details provided by your network administrator or service provider.
- Filtering and Access Control, configure these settings as needed to achieve similar results to proxy-based filtering.
- Apply Changes: Once you've configured the proxy or filtering settings, apply or save the changes. The router may need to restart or apply the new settings.
- Test the Configuration: Test the proxy or filtering configuration by attempting to access blocked content or accessing the internet through the configured proxy settings.
- For Parental Control: You can block inappropriate content, set daily limits for the total time spent online and restrict internet access to certain times of the day so that your children cannot have internet access during the Internet Access Time you set.

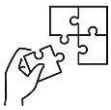


**Points to Remember**

- To log in to the router/ wireless router/access point:
  - ✓ open your router's web interface
  - ✓ Log in to the admin website
- For MAC Address Filtering: Each internet-connected device has a unique MAC address. You can find the MAC address on a computer or other mobile devices. You can usually find the MAC address for a device in the "About" section in the Settings menu.
- For Proxy settings:
  - ✓ Navigate to Proxy or Filtering Settings
  - ✓ Look for a section related to proxy settings, content filtering, or access control.

- ✓ Proxy Server Configuration: If your router supports proxy-related features, you may find fields to enter the IP address and port number of the proxy server
- ✓ Filtering and Access Control: Configure these settings as needed to achieve similar results to proxy-based filtering
- ✓ Apply Changes
- ✓ Test the Configuration

- For Parental Control: You can block inappropriate content, set daily limits for the to tal time spent online and restrict internet access to certain times of the day so that your children cannot have internet access during the Internet Access Time you set



**Application of learning 3.3.**

- Company ABC, a financial services firm located in Kigali city, is setting up a new office in a multi-story building. The company handles sensitive financial data and requires a secure and robust wireless network for its employees to access corporate resources. Due to the nature of their business, access restrictions are critical to ensuring data security and regulatory compliance.
- Here is the guidelines for setting access restrictions:

✓ Guest Network: Visitors and temporary contractors should only have internet access and be isolated from internal resources.

✓ Employee Network: General employees need access to standard corporate resources like email, file servers, and internet.

✓ Sensitive Areas: Finance, HR, and executive teams require access to confidential data and systems.

✓ IT and Admin Network: IT staff need access to all network resources for maintenance and troubleshooting.

- As a network technician you are tasked to identify and apply access restriction parameters to ensure a secure and efficient wireless network in this building of the company.

## Indicative content 3.4: Testing the Deployed Wireless Network Indoor

**Duration: 3**

**Practical Activity 3.4.1: Test the deployed wireless network indoor**

**Task:**

1. Go in computer lab and perform the stated activity:
   Connect to the wireless network of your school then after, test and validate the performance of that network within the school or indoor space. The network should be tested for consistency, especially when moving between areas (roaming), and for potential interference sources
2. Receive these wireless network indoor testing requirements to the groups respectively:
   Those requirements are:
   - Testing tools material and equipment.
   - Site survey document
   - Deployment document besides its design
   - Test cases documents
3. Follow the demonstration of trainer on testing the configuration of wireless network indoor
4. Repeat this activity of testing the configuration of wireless network indoor as demonstrated by trainer.
5. After that, read the key readings 3.4.2 and perform the task provided in application of learning 3.4.

**Key readings 3.4.1.: Test the deployed wireless network indoor**

- Testing a deployed wireless network is essential to ensure that it meets the desired standards in terms of connectivity, configuration, and performance.
- When testing a wireless network, various tools, materials, and equipment are used to ensure optimal performance, security, and coverage. Below is a list of essential items

**Wireless Network Testing Tools:**

✓ **Wi-Fi Analyzers:**

- Ekahau Site Survey: Professional tool for Wi-Fi planning, design, and site surveys.

- NetSpot: User-friendly Wi-Fi analysis and troubleshooting tool.
- Acrylic Wi-Fi: Tool for Wi-Fi spectrum analysis, network planning, and troubleshooting.
- InSSIDer: Tool for network discovery and Wi-Fi performance analysis.

✓ **Network Performance Testers:**

- iPerf: Tool for measuring network bandwidth and testing TCP/UDP throughput.
- Wireshark: Packet analyzer for network troubleshooting and analysis.
- PingPlotter: Tool for visualizing network performance over time.

✓ **Spectrum Analyzers:**

- MetaGeek Wi-Spy: Portable spectrum analyzer for identifying RF interference.
- AirMagnet Spectrum XT: Tool for detecting and identifying RF interference in the Wi-Fi spectrum.

✓ **Wireless Sniffers:**

- Kismet: Wireless network detector, sniffer, and intrusion detection system.
- Aircrack-ng: Suite of tools for assessing Wi-Fi network security, including packet capturing.

✓ **Heatmap Tools:**

- Ekahau HeatMapper: Free tool for visualizing Wi-Fi coverage.
- NetSpot: Allows creation of Wi-Fi heatmaps for analyzing signal strength across different areas.

✓ **Signal Strength Meters:**

- AirCheck G2 (Fluke Networks): Handheld device for quick Wi-Fi network validation and troubleshooting.
- Wi-Fi Explorer: Mac-based tool for assessing signal strength and network performance.

**Materials and Accessories:**

✓ **Cables:**

- Ethernet Cables (Cat5e, Cat6, etc.): For connecting wired devices to the network during testing.
- Coaxial Cables: For connections involving certain types of wireless networking equipment.

✓ **Attenuators: Devices used to reduce signal strength for testing purposes.**

✓ **Adapters:**

- USB Wi-Fi Adapters: For connecting test devices to the wireless network.
- PoE (Power over Ethernet) Adapters: For powering wireless access points and other network devices.

- ✓ **Antenna:**

  - Directional Antennas: For focused signal testing.
  - Omni-Directional Antennas: For general signal distribution testing.
- ✓ **AP Placement and Mounting Kits:**
  - AP Brackets: For mounting access points in various locations during testing.
  - Tripods and Poles: For temporary AP placement during surveys.

**Equipment:**

- ✓ **Laptops/Tablets:** Used for running testing software and monitoring network performance.
- ✓ **Smartphones:** For testing mobile device connectivity and performance on the wireless network**.**
- ✓ **Access Points (APs):** Test APs: For setting up temporary wireless networks or testing new configurations.
- ✓ **Router: Portable Routers:** For setting up a temporary network environment during testing.
- ✓ **Network Switches:** Managed Switches: For configuring VLANs and monitoring traffic during testing.
- ✓ **Battery Packs/UPS:**

  - Uninterruptible Power Supplies (UPS): For ensuring network devices remain powered during testing.
  - Portable Battery Packs: For powering devices in locations without easy access to power.

**Additional Items:**

- ✓ **Documentation Tools:**
  - **Survey Forms:** For documenting site survey results and test configurations.
  - **Laptops/Smartphones**: For capturing data and running tests.
- ✓ **Labels and Markers:** For labelling cables, devices, and testing areas.
- ✓ **Safety Gear:** Ladders, Gloves, Safety Goggles: If testing requires working at heights or in hazardous areas.

- **The methods and tools you can use to conduct testing:**
- **Connectivity Testing:**
  - ✓ Signal Strength:

- ✓ Use a device (e.g., laptop, smartphone) to move around the coverage area and check the signal strength. Low signal strength in certain areas might indicate dead zones or interference.
- ✓ Device Connection:
- ✓ Confirm that devices can connect to the network using the correct security credentials (e.g., Wi-Fi password). Test various types of devices to ensure compatibility.

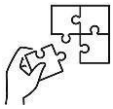
- **Configuration Testing:**
  - ✓ SSID and Security Settings:
  - ✓ Verify that the correct SSID is being broadcast, and the security settings (e.g., WPA2 or WPA3) are configured correctly. Confirm that the encryption settings align with your security policies.
  - ✓ DHCP Functionality:
  - ✓ Ensure that devices connected to the network receive valid IP addresses through DHCP. Test IP address allocation to prevent address conflicts.
  - ✓ Access Control Lists (ACLs):
  - ✓ If using ACLs or MAC filtering, confirm that the configured devices are allowed and unauthorized devices are blocked.
- **Performance Testing:**
  - ✓ **Speed Tests:**
    Use online speed testing tools to measure the actual speed of the internet connection provided by the wireless network. Compare the results with the expected speeds.
  - ✓ **Latency and Ping Tests:**
    Use tools like Ping or Traceroute to measure latency and packet loss. Low latency is crucial for real-time applications like video conferencing and online gaming.
  - ✓ **Load Testing:**
    Simulate a high number of devices connecting to the network simultaneously to assess how well it handles load and whether there are any performance bottlenecks.



**Points to Remember**

- Testing a deployed wireless network is essential to ensure that it meets the desired standards in terms of connectivity, configuration, and performance. Here are some methods and tools you can use to conduct testing:

- **Connectivity Testing:**
  - ✓ Signal Strength:
  - ✓ Use a device (e.g., laptop, smartphone) to move around the coverage area and check the signal strength. Low signal strength in certain areas might indicate dead zones or interference.
  - ✓ Device Connection:
  - ✓ Confirm that devices can connect to the network using the correct security credentials (e.g., Wi-Fi password). Test various types of devices to ensure compatibility.
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- **Performance Testing:**
  - ✓ Speed Tests:
  - ✓ Use online speed testing tools to measure the actual speed of the internet connection provided by the wireless network. Compare the results with the expected speeds.
  - ✓ Latency and Ping Tests:
  - ✓ Use tools like Ping or Traceroute to measure latency and packet loss. Low latency is crucial for real-time applications like video conferencing and online gaming.
  - ✓ Load Testing:
  - ✓ Simulate a high number of devices connecting to the network simultaneously to assess how well it handles load and whether there are any performance bottlenecks.



**Application of learning 3.4.**

ABC Healthcare Facility in Kigali has recently completed the deployment of a new wireless network throughout their indoor premises. This network is critical for the daily operations of the healthcare staff, including doctors, nurses, administrative personnel, and support staff. It supports a range of applications from patient record management to real-time communications and IoT devices monitoring medical equipment. To ensure the network meets the required standards of performance, reliability, and security, a comprehensive testing phase is necessary.

As a network testing technician, you are hired by the company to perform these activities to ensure that wireless network is robust, secure, and fully meets the needs of their staff and operations:

1. Verify that the wireless signal covers all intended areas and maintains adequate strength
2. Measure the network performance to ensure it meets the requirements for various applications used in the facility.
3. Test devices (laptops, smartphones, and medical tablets) for continuous connectivity without drops.

## Indicative content 3.5: Documentation of Deployed Wireless Network

**Duration: 3**

### Theoretical Activity 3.5.1: Documentation of deployed wireless network

**Tasks:**

1. State what you understand about wireless network deployment report of work done
2. Listen to the further explanation on the report of work done
3. In your group, receive task sheet given by your trainer and discuss about them with your expert team:
   Task 1: Description of topology diagram
   Task 2: IP address allocation table
   Task 3: Maintenance schedules
   Task 4: Software information (current versions, dates, licensing, and support)
   Task 5: Network credentials
4. Share ideas from the expert team with your members
5. Listen to the expert view and clarifications from trainer
6. Ask questions if necessary
7. Read the key readings 3.5.1 in this manual

**Key readings 3.5.1.: Documentation of deployed wireless network**

- Creating documentation for a deployed wireless network is essential for proper maintenance, troubleshooting, and future enhancements.

- **The key elements of deployed wireless network design:**
  - ✓ Topology Diagram: Wireless network topology shows how the computers connect each other when there is no physical connection. The computers communicate using wireless devices.
  - ✓ IP address allocation table: IP addresses are allocated to each TCP/IP services address on a TCP/IP Internet. Each address is a unique 32-bit (an IPv4 Internet Address) or a unique 128-bit (an IPv6 Internet Address) quantity defining the host's network and the particular host.
  - ✓ Maintenance schedules: Scheduled maintenance is any task that is given a deadline and assigned to a technician.
  - ✓ Software information

- current versions
- dates
- licensing and support

- **Below is a comprehensive guide on the documentation of a deployed wireless network:**
  - ✓ **Introduction**
  - ✓ **Purpose of Documentation:** Clearly state the purpose of the documentation, emphasizing its role in aiding network administrators, technicians, and other stakeholders in understanding and maintaining the deployed wireless network.
  - ✓ **Network Overview**
    - Network Architecture: Provide an overview of the network architecture, including the layout, devices, and their interconnections.
    - Network Topology: Present a visual representation of the network topology, illustrating the relationships between access points, routers, switches, and other network components.
  - ✓ **Hardware Inventory**
    - Access Points: List all deployed access points, including their make, model, location, and key configuration details.
    - Network Infrastructure: Document routers, switches, and other networking hardware, specifying their models, roles, and configurations.
  - ✓ Network Configuration
    - SSID Details: Provide information on each SSID, including its purpose, security settings, and any specific configurations.
    - Security Policies: Document security protocols, encryption methods, authentication mechanisms, and access control policies implemented in the network.
    - IP Addressing: Detail the IP address scheme used in the wireless network, specifying subnets, gateways, and DHCP settings.
    - Quality of Service (QoS)
    - QoS Policies: Outline the QoS policies in place, explaining how different types of traffic are prioritized to ensure optimal performance.
  - ✓ Network Monitoring
    - Monitoring Tools: List and describe the tools used for monitoring network performance, security, and any anomalies.

        - Logging and Alerts: Explain the logging mechanisms in place, detailing what events are logged and how alerts are generated and communicated.
    - ✓ Troubleshooting Procedures
        - Problem Identification: Document common network issues, their symptoms, and the procedures for identifying and diagnosing problems.
        - Escalation Process: Define the escalation process for handling complex or critical network issues, specifying roles and responsibilities.
    - ✓ Maintenance Schedules:
        - Routine Maintenance: Outline regular maintenance tasks, including software updates, security patches, and hardware checks.
    - ✓ Disaster Recovery
        - Backup and Restore: Explain the backup strategy for critical network configurations and data, detailing how to perform restores in case of failures.
        - Redundancy and Failover: Document mechanisms in place for redundancy and failover, ensuring network availability in the event of component failures.
    - ✓ User Documentation
        - User Guidelines: Provide documentation for end-users, including instructions on connecting to the network, troubleshooting basic issues, and security guidelines.
    - ✓ Conclusion
        - Summarize the documentation, highlighting key components, procedures, and considerations for the deployed wireless network.
- By creating thorough and organised documentation, you ensure that the deployed wireless network is well-understood, maintainable, and adaptable to future changes. Regularly update the documentation to reflect any changes or upgrades to the network.



**Practical Activity 3.5.1: Document deployed wireless network**




**Task:**

1. Read carefully this activity and perform it accordingly:
   You are tasked with documenting an indoor wireless network of your school computer lab that has already been deployed. The documentation will include network hardware details, configuration settings, signal strength and coverage information, and security policies. This documentation will serve as a reference for troubleshooting and future network expansions.

2. Perform the above tasks by following trainer's guidance, ask questions and clarifications if necessary
3. Read the key readings and perform the task provided in application of learning 3.5



**Key readings 3.5.1: Document deployed wireless network**

- **Documenting a deployed wireless network involves creating a comprehensive set of records that detail the network's configuration, topology, and operational parameters.**
- This documentation is crucial for ongoing network management, troubleshooting, and future upgrades or expansions. It ensures that network administrators and IT staff have a clear understanding of how the network is set up and can efficiently manage and maintain it.
- **Benefits of Documenting a Deployed Wireless Network**
  - ✓ **Efficient Troubleshooting**: Provides a reference for diagnosing and resolving network issues quickly.
  - ✓ **Consistency and Standardization**: Ensures that all configurations follow best practices and organizational standards.
  - ✓ **Future Planning**: Aids in planning for network expansions, upgrades, and capacity increases.
  - ✓ **Compliance and Security**: Helps maintain compliance with industry regulations and enhances security posture by detailing access controls and security measures.
  - ✓ **Knowledge Transfer**: Facilitates the transfer of knowledge to new IT staff or external support teams, ensuring continuity in network management**.**
- **Key Elements of Wireless Network Documentation**

- ✓ **Network Topology Diagrams**
  - **Visual Representation**: Create detailed diagrams that illustrate the physical and logical layout of the network, including the placement of access points (APs), switches, routers, and other network devices.
  - **Labels and Legends**: Include labels for each device, connections, and areas covered. Use legends to explain symbols and color codes.
- ✓ **Device Configuration Details**
  - **Access Points**: Document the model, firmware version, IP address, MAC address, SSID, radio mode, channel settings, transmit power, and location for each AP.
  - **Switches and Routers**: Record configurations such as IP addresses, VLAN settings, routing protocols, firewall rules, and security settings.
  - **Controllers**: If a wireless controller is used, include its configuration details, managed APs, and policies.
- ✓ **Network Settings and Parameters**
  - **SSID Information**: List all SSIDs, their associated VLANs, security settings (e.g., WPA3, 802.1X), and passwords.
  - **IP Addressing**: Document the IP address ranges, DHCP scopes, and static IP assignments.
  - **Security Configurations**: Detail authentication methods (e.g., RADIUS, LDAP), encryption protocols, and access control lists (ACLs).
- ✓ **Coverage Maps and Site Survey Results**
  - **Heatmaps**: Include coverage maps generated from site surveys showing signal strength, coverage areas, and potential dead zones.
  - **Interference Analysis**: Document findings related to interference sources and mitigation strategies.
- ✓ **Performance Metrics**
  - **Throughput and Latency**: Record baseline performance metrics such as average throughput, latency, and jitter for key areas.
  - **Capacity Planning**: Include details on user density, expected peak usage, and capacity planning assumptions.
- ✓ **Security and Compliance Documentation**
  - **Policies and Procedures**: Document security policies, access control measures, and compliance requirements (e.g., GDPR, HIPAA).
  - **Audit Logs and Monitoring**: Describe logging and monitoring setups, including what is logged, where logs are stored, and how they are reviewed.
- ✓ **Maintenance and Support Information**
  - **Firmware and Software Updates**: Record the current firmware and

software versions for all devices, along with the update schedule.

- **Vendor Support Contacts**: Provide contact information for hardware and software vendors.
- **Troubleshooting Guides**: Include step-by-step troubleshooting procedures for common issues.

✓ **Backup and Recovery Plans**

- **Configuration Backups**: Document the process for backing up device configurations and the location of backup files.
- **Recovery Procedures**: Outline procedures for restoring configurations in case of device failure.

✓ **User and Device Management**

- **User Access Levels:** Detail user roles, access permissions, and authentication methods.
- **Device Inventory**: Maintain an inventory of all devices connected to the network, including their roles and configurations.

Change Management Records

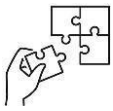
- **Change Logs:** Document all changes made to the network configuration, including the date, the person who made the change, and a description of the change.
- **Approval Processes**: Outline the process for approving and documenting network changes.

- **Document Content**: Provide details on the key elements of the wireless network-related documents. This might include information on network topology, equipment specifications, standards mentioned, security protocols, coverage requirements, and any specific design considerations.
- **Goals and Objectives:** What are the goals and objectives outlined in the documents? Understanding the purpose of the wireless network and the specific outcomes desired will help in interpreting the documents effectively.
- **Requirements and Constraints:** Highlight any requirements or constraints mentioned in the documents. This could include user capacity, coverage area, security measures, or any limitations that need to be considered during the network design.
- **Components and Technologies:** Identify the wireless network components and technologies discussed in the documents. This could include access points, controllers, antennas, wireless standards, security protocols, and other relevant technologies



**Points to Remember**

- Below is a comprehensive guide on the documentation of a deployed wireless network:
  - ✓ Introduction
  - ✓ Purpose of Documentation
  - ✓ Network Overview
  - ✓ Network Architecture
  - ✓ Network Topology
  - ✓ Hardware Inventory
    - Access Points
    - Network Infrastructure
  - ✓ Network Configuration
    - SSID Details
    - Security Policies
    - IP Addressing
  - ✓ Quality of Service (QoS)
    - QoS Policies
  - ✓ Network Monitoring
    - Monitoring Tools
    - Logging and Alerts
  - ✓ Troubleshooting Procedures
    - Problem Identification
    - Escalation Process
  - ✓ Maintenance Schedules
  - ✓ Disaster Recovery
  - ✓ User Documentation
  - ✓ Conclusion
- By creating thorough and organized documentation, you ensure that the deployed wireless network is well-understood, maintainable, and adaptable to future changes. Regularly update the documentation to reflect any changes or upgrades to the network.



**Application of learning 3.5.**

You have been assigned to document a recently deployed wireless network in multiple floors building. The wireless network was implemented to support the connectivity needs of employees and guests across various departments and workspaces. The network includes multiple access points, routers, switches, and employs various security measures to ensure

the confidentiality and integrity of the data transmitted. The document should be submitted with 3 days.

- These are the information you need use:
- Network design and network architecture diagrams
- Devices configurations file.
- Software information

**Objectives:**

Your task is to thoroughly document the deployed wireless network to facilitate ongoing maintenance, troubleshooting, and future expansions. Ensure that your documentation is clear, organised, and comprehensive.

## Learning outcome 3 end assessment

**Theoretical assessment**

1. **Which of the following is true about a MAC address?**
    a. It is a 32-bit address.
    b. It is a 48-bit address.
    c. It is used at the Network Layer of the OSI model.
    d. It can be changed dynamically by the network administrator.
2. **A MAC address is typically represented in which of the following formats?**
    a. Decimal
    b. Binary
    c. Hexadecimal
    d. Octal
3. **Answer by true or false on the following questions**
    a. A site survey is unnecessary for testing a wireless network if the access points are correctly configured.
    b. Throughput testing measures the actual data transfer rate that users experience on the network.
    c. Interference from other wireless devices can affect the performance of a wireless network.
    d. Heatmaps are used to visualize wireless coverage and signal strength across a physical location.
    e. A network performance baseline should be established during initial testing to compare against future performance metrics.
4. **Which of the following is the first step when configuring a new router?**
    a. Update the firmware
    b. Connect the router to a power source
    c. Configure the SSID and password
    d. Set the IP address
5. **Which setting is used to ensure that a router assigns IP addresses to devices on the network automatically?**
    a. DNS
    b. DHCP
    c. NAT
    d. VPN
6. **To secure a wireless network, which of the following encryption methods is considered the most secure?**
    **a.** WEP
    **b.** WPA
    **c.** WPA2

**d.** WPA3

7. **Which setting should be configured to limit the access of specific devices to a wireless network?**
    a. MAC filtering
    b. Port forwarding
    c. QoS
    d. UPnP
8. **To improve wireless network performance, which feature should be enabled to prioritize bandwidth for critical applications?**
    A. QoS
    B. NAT
    C. DMZ
    D. SSID broadcasting
9. **Which of the following is recommended to change immediately after installing a new router?**
    a. Firmware version
    b. MAC address
    c. Default username and password
    d. Subnet mask
10. **How do you access the router's web-based management interface for configuration?**

**Practical assessment**

XYZ is a Telecommunication Company located in Kigali city in the IMBUTO Plaza building of 3 floors, it has 50 staff dispatched in their offices.

The staffs to be connected to the rapid internet connection wirelessly within the building are dispatched as follow:

- 1st floor: 20 staffs
- 2nd floor: 20 staffs
- 3rd floor: 10 staffs

As a wireless network technician, you are tasked to deploy a wireless network where all tasks like designing a topology, installation of devices and cabling, configuring, testing and its documentation are your responsibilities.

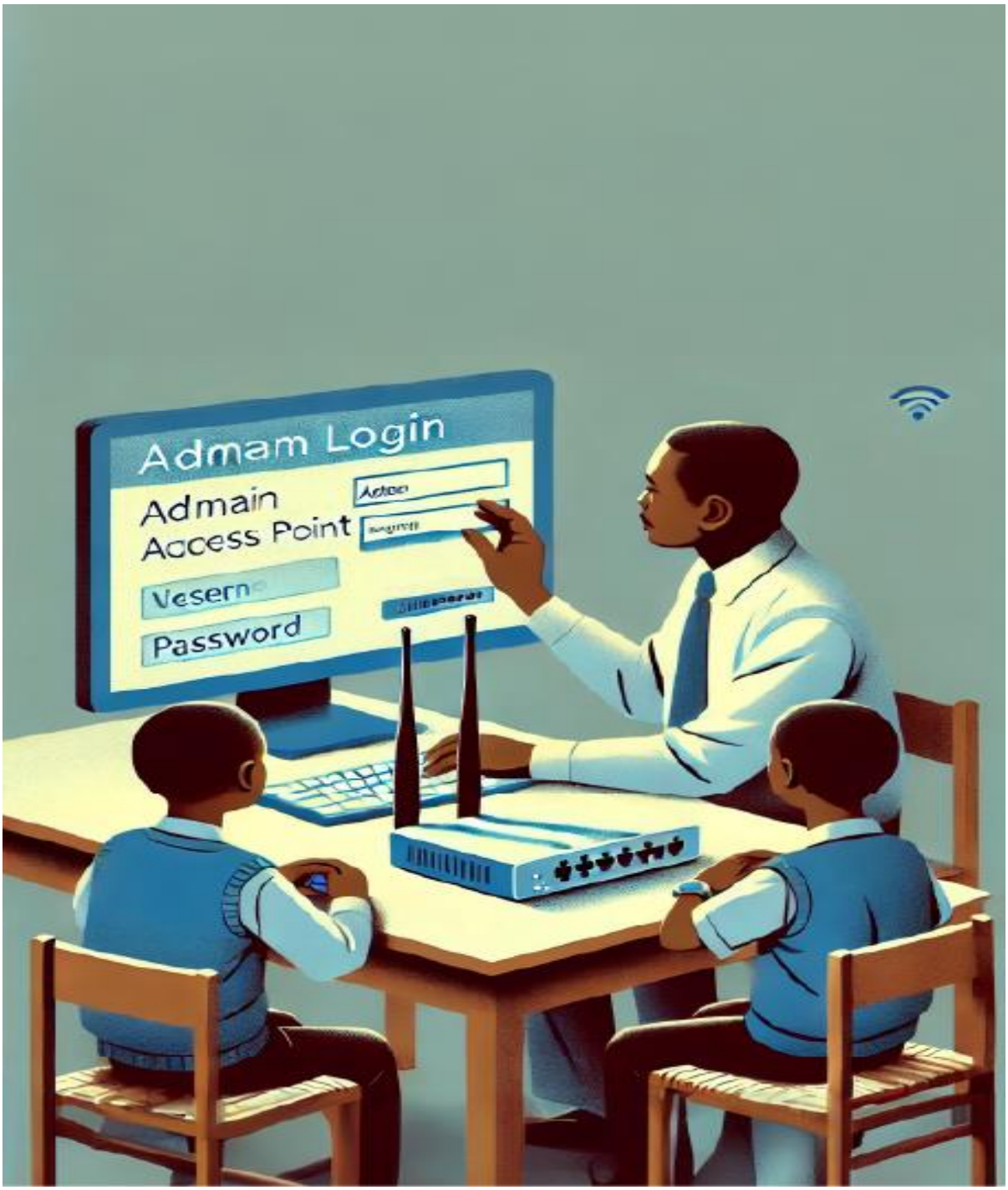
After successful login to the wireless access points, the following features must be enabled to wireless networks within a building respectively.

- Each floor might be having its own wireless network name and secured.
- No one from outside can have access to the wireless network except the one with permanent known devices and guests.
- Parental control must be enabled between 10:00 PM and 05:00AM



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Admam Login
Admain
Access Point
Password

**Indicative contents**

**4.1 Performing Preventive maintenance**

**4.2 Performing Corrective maintenance**

**4.3 Elaboration of maintenance report**

**Key Competencies for Learning Outcome 4: Maintain wireless network indoor**

| Knowledge | Skills | Attitudes |
|---|---|---|
| • Identification of difference between preventive and corrective maintenance<br>• Definition of maintenance report<br>• Identification of element of maintenance report<br>• Identification of common wireless network problems and their causes | • Updating and upgrading of network monitoring software and device firmware<br>• Fixing identified wireless network indoor<br>• replacing or repairing damaged wireless network indoor<br>• scheduling regular cleaning<br>• Checking Physical devices condition and cables in good condition<br>• changing wireless network access credentials<br>• Checking and reducing interference and obstacle<br>• Adding a wireless repeater<br>• writing maintenance report | • being a good observer<br>• having Self confidence<br>• being a Critical thinker |



**Duration:10 hrs**

**Learning outcome 4 objectives**:



By the end of the learning outcome, the trainees will be able to:

1. apply correctly preventive maintenance of wireless network indoor as per manufacturer's guidelines
2. apply correctly corrective maintenance of wireless network indoor as per manufacturer's guidelines
3. Elaborate clearly maintenance report based on the work done



**Resources**

| Equipment | Tools | Materials |
|---|---|---|
| • Wireless Router<br>• Computer<br>• Access point<br>• PPEs | • Cable tester<br>• Cleaning foams<br>• Soft clothes<br>• Multimeter<br>• Electrical Brower | • Wire rope spray<br>• cable ties<br>• internet bundles<br>• Nails<br>• Screws<br>• cleaning foams |

## Indicative content 4.1: Performing Preventive Maintenance

**Duration: 4**

### Theoretical Activity 4.1.1: Description preventive maintenance

**Tasks:**

1. Answer the following questions:
    a. Define preventive software maintenance
    b. Why use preventive maintenance
2. Write your findings on reserved place
3. Discuss on the provided answers and choose the correct answers
4. Listen to the expert view, take key notes
5. Ask questions if necessary
6. Read the key readings 4.1.1 in this manual

**Key readings 4.1.1.: Description preventive maintenance**

- **Preventive Maintenance - Software**
    - ✓ Verify that installed software is current and follow the policies of the organization when installing security updates, operating system, and program updates.
- **Software maintenance tasks:**
    - ✓ Regular change of network access credentials
    - ✓ Wireless network monitoring software Licensing
    - ✓ Updating and upgrading of network monitoring software and device firmware
- Why use preventive maintenance:
    - ✓ Preventive maintenance is a proactive approach to maintaining equipment, machinery, and systems to prevent failures before they occur.
- **Here are some reasons why preventive maintenance is important in various industries:**
    - ✓ **Minimizes Equipment Downtime:** Regular maintenance helps identify and address potential issues before they lead to equipment failure. This reduces unexpected breakdowns and downtime, ensuring that equipment is available and operational when needed.
    - ✓ **Increases Equipment Reliability:** Preventive maintenance helps in keeping equipment in optimal condition, reducing the likelihood of unexpected breakdowns. This, in turn, improves the overall reliability of the equipment and extends its lifespan.

- ✓ **Improves Safety:** Regular maintenance can uncover and rectify safety issues before they become serious problems. This helps create a safer working environment for employees by minimizing the risk of accidents or injuries caused by equipment failures.
- ✓ **Enhances Equipment Efficiency:** Well-maintained equipment tends to operate more efficiently. Preventive maintenance includes tasks such as cleaning, lubrication, and calibration, which can help equipment perform at its best, leading to energy savings and improved productivity.
- ✓ Preventive maintenance is a proactive and systematic approach that offers numerous benefits, including increased reliability, reduced downtime, improved safety, and long-term cost savings.
- ✓ Verify that installed software is current and follow the policies of the organization when installing security updates, operating system, and program updates.



**Practical Activity 4.1.2: Change wireless network access credential**




**Task:**

1. Read carefully this task:
   XCB Company located in Kigali city has a running wireless network. A key employee with access to the company's wireless network leaves the organization. The risk of former employees using their knowledge of the network credentials to access company resources could be a concern. In that case you are needed to reset the wireless network credentials to ensure that only current employees have access to the network. New credentials are issued to current staff.

2. Receive the task sheet containing steps for resetting wireless network credentials and listen to the explanations and guidelines
3. Follow the demonstrations and write the steps followed in resetting the network credentials
4. Connect a computer to one of LAN port of wireless router or access point and open any web browser.
5. Type default IP address of the wireless router in that browser and change SSID and password
6. Ask questions if necessary
7. Read the key readings 4.1.2 and do the application of learning 4.1



**Key readings 4.1.2: Change wireless network access credential**

- **Network access credentials:** Wireless network credentials include a wireless network name (SSID) and password.
  - ✓ SSID refers to a network's name.
  - ✓ The Password is the secret word or phrase you will enter when first connecting a device to your wireless network. Each computer or device you connect will need to use this password.
- Here are general steps to guide you through the process of changing wireless network access credentials:
  - ✓ **Access the Router's Web Interface:** Open a web browser and enter the IP address of your router in the address bar. Common addresses include "192.168.1.1" or "192.168.0.1." Consult your router's manual or check the router itself for the correct address.
  - ✓ **Enter Login Credentials**: Log in to the router's web interface using the administrator credentials. This information is often found on a sticker on the router or in the router's manual. If you've changed the credentials before and can't remember them, you may need to reset the router to its factory settings.
  - ✓ **Locate Wireless Settings**: Navigate to the wireless settings section of the router's configuration interface. This is often labeled as "Wireless," "Wireless Settings," or something similar.
  - ✓ **Change the SSID (Network Name):** Locate the field that contains the SSID (Service Set Identifier) or network name. Change it to your desired name.
  - ✓ **Change the Wireless Security Key (Password):** Find the security settings, often labelled as "Security," "Wireless Security," or similar. Look for the field containing the current wireless security key or password (WPA2, WPA3, etc.). Enter a new, strong password.
  - ✓ **Save Changes**: After making the necessary changes, find the option to save or apply the settings. This may be a "Save," "Apply," or "Submit" button. Click on it to save your new wireless network credentials.
  - ✓ **Reconnect Devices**: Once the changes are saved, you'll need to reconnect your devices to the wireless network using the new SSID and password.
  - ✓ **Verify Connectivity**: Ensure that your devices can connect to the newly configured wireless network using the updated credentials. Test the connection to confirm that everything is working as expected.
  - ✓ **Update Other Devices**: If you have other devices that automatically connect to the wireless network (smartphones, tablets, etc.), update their network settings with the new credentials.



**Points to Remember**

- **Network access credentials:** Wireless network credentials include a wireless network name (SSID) and password.
- **Here are general steps to guide you through the process of wireless network credentials:**
  - ✓ Access the Router's Web Interface
  - ✓ Enter Login Credentials
  - ✓ Locate Wireless Settings
  - ✓ Change the SSID (Network Name)
  - ✓ Change the Wireless Security Key (Password)
  - ✓ Save Changes
  - ✓ Reconnect Devices
  - ✓ Verify Connectivity
  - ✓ Update Other Devices






**Theoretical Activity 4.1.3: Description of factors affecting wireless signal stability**

**Tasks:**

**1.** In small groups, you are requested answer the following questions
   1. what do you understand by Signals stability?
   2. what are factors that can affect signal stability
   3. list main signal stability maintenance tasks

**2.** Write your findings on provided space

**3.** Present your findings in front of a whole class

**4.** Listen to the expert view from trainer

**5.** Ask questions or clarifications

**6.** Read the key readings 4.1.3 in this manual



**Key readings 4.1.3.: Description of factors affecting wireless signal stability**

- Signal stability refers to the consistency and reliability of a wireless signal over time. In the context of wireless networks, signal stability means that the strength and quality of the signal remain steady and do not fluctuate significantly, ensuring reliable communication between devices
- Wireless signal stability can be affected by various factors, and understanding these factors is crucial for optimizing and maintaining a reliable wireless connection. Here are some key factors that can influence wireless signal stability:
  - Distance from the Router:
    - Wireless signals weaken as you move farther away from the router. Walls and other obstacles between your device and the router can also attenuate the signal. Ensuring that your device is within a reasonable range of the router can improve signal stability.
  - Physical Obstacles:
    - Walls, floors, furniture, and other physical obstructions can interfere with wireless signals. Thick walls or materials like concrete and metal can significantly attenuate or block signals, leading to reduced stability.
  - Interference from Other Devices:
    - Other electronic devices, such as cordless phones, microwave ovens, and Bluetooth devices, operate on frequencies that can interfere with Wi-Fi signals. Using devices that operate on the same frequency band as your Wi-Fi (2.4 GHz or 5 GHz) can cause signal instability.
  - Network Overload:
    - Too many devices connected to a single Wi-Fi network can lead to network congestion and reduced stability. Quality of Service (QoS) settings on the router can help prioritize certain types of traffic and improve overall network performance.
  - Router Placement:
    - The placement of the router within your home or office can impact signal stability. Placing the router in a central location and avoiding placement in enclosed spaces or near large metal objects can help distribute the signal more effectively.
  - Router Settings and Configuration:
    - Incorrect router settings, outdated firmware, or misconfigurations can lead to signal instability. Ensure that your router is running the latest firmware, and review and optimize its settings for your specific needs.
  - Device-Specific Issues:

- The quality and capabilities of the wireless adapter in your device can affect signal stability. Older or low-quality adapters may struggle to maintain a stable connection.

**Main Tasks Involved In Maintaining Signal Stability**

- Regular Monitoring of Signal Strength:
  - ✓ Continuously monitor the signal strength across different areas of the network using tools like Wi-Fi analyzers or network management software.
  - ✓ **Purpose**: To identify weak spots, signal drops, or areas with potential interference.
- Adjusting Access Point (AP) Placement:
  - ✓ Evaluate and optimize the placement of access points to ensure even coverage and minimize dead zones
  - ✓ Purpose: To maintain strong signal strength throughout the coverage area and avoid signal overlap or interference.
- Channel Management:
  - ✓ Regularly check and adjust the Wi-Fi channels used by access points to minimize interference from other nearby networks or devices.
  - ✓ Purpose: To prevent co-channel interference and improve overall signal stability and network performance.
- Firmware Updates:
  - ✓ Keep the firmware of wireless routers, access points, and network adapters up to date.
  - ✓ Purpose: Firmware updates often include performance improvements, bug fixes, and enhanced security features that contribute to better signal stability.
- Antenna Positioning and Configuration:
  - ✓ Adjust and optimize the positioning and configuration of antennas on access points and routers
  - ✓ Purpose: To maximize coverage, improve signal quality, and reduce interference.
- Reducing Interference from Other Devices:
  - ✓ Task: Identify and minimize interference from non-Wi-Fi devices such as microwaves, cordless phones, Bluetooth devices, and other electronics.
  - ✓ Purpose: To maintain clear Wi-Fi signals and avoid disruptions caused by electromagnetic interference.
- Signal Boosting:

- ✓ Task: Use signal boosters or repeaters in areas where the signal strength is weak to extend coverage and enhance signal stability.
- ✓ Purpose: To ensure consistent connectivity in areas that are far from the main access point or obstructed by walls or other barriers.

- Implementing Quality of Service (QoS):
  - ✓ Task: Configure QoS settings on the router to prioritize critical applications, such as VoIP or video conferencing, over less important traffic.
  - ✓ Purpose: To maintain stable and high-quality connections for priority tasks, even during high network usage.
- Regular Site Surveys:
  - ✓ Task: Conduct periodic site surveys to assess signal coverage, interference, and performance in different areas of the network.
  - ✓ Purpose: To identify and address potential issues before they impact users, ensuring stable and reliable signals.
- Cable and Connector Maintenance:
  - ✓ Task: Regularly inspect and maintain network cables, connectors, and related hardware.
  - ✓ Purpose: To prevent signal loss or degradation caused by damaged or poorly connected cables.
- Load Balancing:
  - ✓ Task: Distribute network traffic evenly across multiple access points or frequency bands to prevent any single AP from becoming overloaded.
  - ✓ Purpose: To maintain consistent signal performance and avoid overloading individual access points, which can lead to instability.
- Power Level Adjustment:
  - ✓ Task: Adjust the transmit power level of access points to balance coverage and reduce interference.
  - ✓ Purpose: To maintain an optimal balance between coverage area and signal stability, avoiding signal overlap and interference.
- Environmental Adjustments:
  - ✓ Task: Assess and mitigate environmental factors such as physical obstructions, reflective surfaces, and temperature variations.
  - ✓ Purpose: To minimize the impact of environmental factors on signal propagation and stability.
- Implementing Dual-Band Networks:
  - ✓ Task: Use both 2.4 GHz and 5 GHz bands in your network to balance load and reduce congestion.

- ✓ Purpose: To provide more stable connections by spreading out devices across different frequency bands, reducing interference and congestion.

- Security Maintenance:
  - ✓ Task: Regularly update network security settings, including encryption methods and passwords.
  - ✓ Purpose: To prevent unauthorized access that could disrupt the network and affect signal stability.
- By addressing these factors and optimizing your wireless network setup, you can improve the stability and performance of your Wi-Fi connection



**Practical Activity 4.1.3: Setting up wireless network repeater**




**Task:**

1. Read carefully the stated activity and perform it accordingly:
   You are tasked with expanding the wireless network coverage in your school to areas where the signal is weak, such as certain classrooms or hallways. By setting up a wireless repeater, the network coverage will be extended without running additional cables. The exercise involves identifying weak signal areas, configuring the repeater, and testing the extended network.
2. Follow the demonstration for setting up wireless network repeater
3. Repeat the activity by yourself, starting from connecting a computer or laptop to the repeater using an Ethernet cable, then setup wireless network repeater
4. During the process, make sure that you follow each step as demonstrated by trainer, ask questions if necessary.
5. Perform the task provided in application of learning 4.1
6. Read the key readings for more clarification



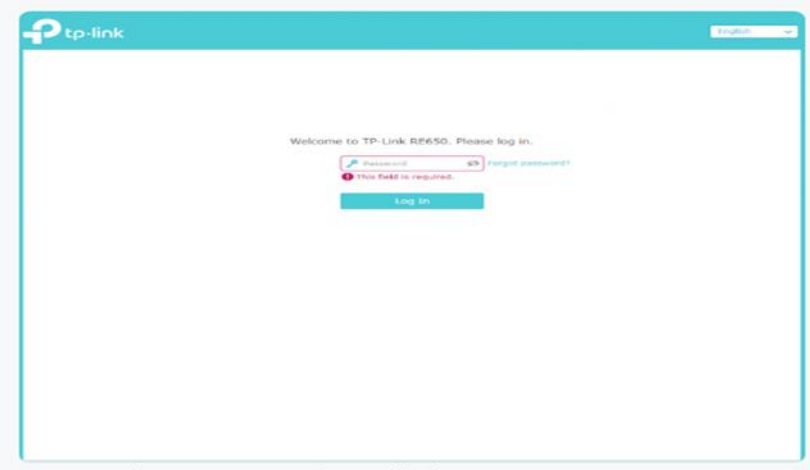
**Key readings 4.1.3: Setting up wireless network repeater**

- A Wireless repeater is a networking device that extends the coverage area of a wireless network by amplifying and retransmitting the existing Wi-Fi signal. It is a solution to improve wireless connectivity in areas where the signal strength from the main wireless router may be weak or limited.
- Steps to Setting up wireless network repeater
  - ✓ Choose the Right Location: Place the extended coverage to the desired area.

- ✓ Power On the Repeater: Plug in the repeater to a power source and turn it on.
- ✓ Connect to the Repeater: On your computer or mobile device, open the list of available Wi-Fi networks.
  - Look for the new network created by the repeater (it might have a name like "YourNetworkName_EXT" or similar).
  - Connect to this network using the default password provided in the repeater's documentation.

✓ Access the Repeater's Configuration Page: Open a web browser and enter the IP address of the repeater. The IP address should be mentioned in the user manual. Common addresses include 192.168.0.1 or 192.168.1.1.

✓ Login to the Repeater: Enter the default username and password for the repeater. This information is usually provided in the user manual.



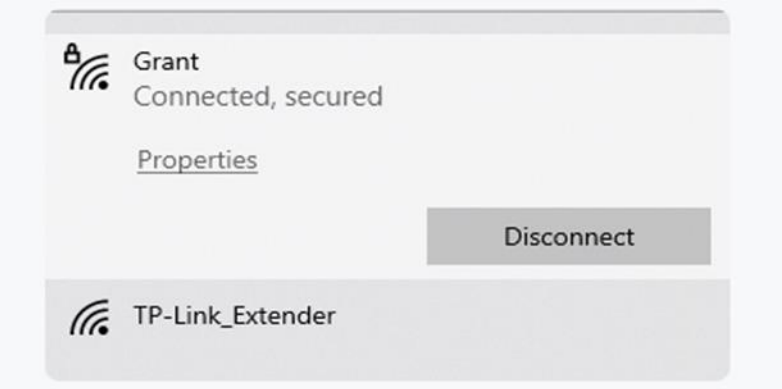

✓ Configure the Repeater: Once logged in, navigate to the wireless settings section.
- Look for an option like "Wireless Repeater," "Range Extender," or similar.
- Choose the option to scan for available Wi-Fi networks.
- Select the main Wi-Fi network (SSID) that you want to extend and enter the password.

✓ Configure SSID and Security Settings: Set the SSID (network name) and security settings for the extended network. You can choose to keep it the same as the main network or give it a different name.

✓ Apply Changes: Save the changes and wait for the repeater to restart.

✓ Reconnect Devices: Once the repeater has restarted, reconnect your devices to the extended network using the new SSID and password.

✓ Check Connection: Test the connection in the extended area to ensure that devices can connect to the internet without issues.




**Points to Remember**

- A Wireless repeater is a networking device that extends the coverage area of a wireless network by amplifying and retransmitting the existing Wi-Fi signal
- Steps to Setting up wireless network repeater:
  - ✓ Choose the Right Location
  - ✓ Power On the Repeater
  - ✓ Connect to the Repeater
  - ✓ Access the Repeater's Configuration Page
  - ✓ Login and configure repeater

**Application of learning 4.1.**

XCV TSS located in RUHANGO District has a wireless network with SSID named bwiza and password (bwiza@123). This network has capacity for providing wireless signal within their computers LAB's room. After building the second LAB near the first one, they have seen that computers in that LAB can't be connected to the existing wireless network. You are hired as a network technician to perform the following tasks:

1. Change Wireless network name to bwiza2023 and password to bwiza@2023
2. Setup a wireless repeater to extends the coverage area of a wireless network.

## Indicative content 4.2: Performing Corrective Maintenance

**Duration: 4 hrs**

**Theoretical Activity 4.2.1: Identification of Corrective maintenance**

**Tasks:**

1. In small groups, you are requested answer the following question

    a. identify some common wireless network problems and their potential causes
    b. List steps you can take to address common wireless network challenges.
2. write your answers on provided space
3. discuss on provided answers and choose correct answers
4. listen to the clarifications from trainer, ask questions if necessary
5. read the key readings 4.2.1

**Key readings 4.2.1.: Identification of Corrective maintenance**

- There are Common problems and their causes
    - ✓ Slow Connection:
        - Cause: Signal interference from other devices, obstacles, or router placement issues.
    - ✓ Weak Signal Strength:
        - Cause: Distance from the router, obstacles, or signal interference.
    - ✓ Connection Drops
        - Cause: Outdated router firmware, conflicting settings, or signal interference.
- Wireless network problems refer to issues and challenges that can occur in a wireless networking environment, leading to disruptions, slow performance,
- There are some immediate steps you can take to address common wireless network challenges.
    - ✓ Physical Relocation: If environmental factors are affecting your wireless connectivity, adjusting the actual location of WLAN hardware might reduce issues.
    - ✓ WLAN Upgrade: Ensure hardware is set to the latest standards and the hardware is up-to-date.
    - ✓ Deploy Network Monitoring Tools: Sometimes visibility into network activity can provide insights into the actual causes of the most common

wireless network challenges. Cisco Prime is one great tool for improving network visibility and management.

- Common Wireless Network Challenges:
  - Signal Interference:
    - Challenge: Interference from other Wi-Fi networks, electronic devices, or physical obstacles can degrade the wireless signal.
    - Solution: Choose optimal Wi-Fi channels, use dual-band or tri-band routers, and position the router away from potential interference sources.
  - Limited Range and Coverage:
    - Challenge: The Wi-Fi signal may not cover the entire area, leading to dead zones or weak connectivity.
    - Solution: Use range extenders, mesh networks, or additional access points to extend coverage and strategically position routers for optimal reach.
  - Bandwidth Limitations:
    - Challenge: High demand for bandwidth in crowded areas or homes with numerous devices can result in slow speeds.
    - Solution: Upgrade to a higher bandwidth plan, use Quality of Service (QoS) settings to prioritize traffic, and limit the number of connected devices if necessary.
  - security Vulnerabilities:
    - Challenge: Wireless networks are susceptible to unauthorized access, data interception, and other security threats.
    - Solution: Implement strong encryption (WPA3), regularly update passwords, enable network firewalls, and use secure authentication protocols.
  - Device Compatibility:
    - Challenge: Older devices may not support modern wireless standards or security protocols, causing connectivity issues.
    - Solution: Upgrade device firmware or hardware, and maintain a balance between legacy and modern devices on the network.



**Practical Activity 4.2.2: Replacing wireless adapter card**




**Task:**

1. Read carefully this task:

   Ramba Company located in Nyanza District has a wireless network with the capacity for providing wireless signal to their employers. After engaging a new employer, they have seen their computer can't be connected to the existing wireless network due to the defective wireless network card. You are requested to solve that problem by replacing the existing wireless network adapter to the new one.
2. Receive the task sheet containing instruction and steps for replacing wireless adapter card
3. Listen to the explanation and clear work instruction of the task
4. Follow the demonstrations for installing wireless adapter card
5. Repeat the previous task as demonstrated by the trainer
6. Ask questions or clarification if necessary to accomplish the given task.
7. Read the key readings 4.2.2
8. Perform the application of learning 4.2



**Key readings 4.2.2: Replacing wireless adapter card**

- Network adapter card: It is an expansion card that enables a computer to connect to a network.
- Description of how to install wireless adapter card:
  - Choose the Right Wireless Adapter: Ensure the wireless adapter card is compatible with your computer's hardware and operating system.
  - Power Off Your Computer: Before making any hardware changes, turn off your computer and disconnect it from the power source.
  - Open the Computer Case: If your computer is a desktop, you'll need to open the case. Consult your computer's manual for instructions on how to do this safely.
  - Identify an Appropriate PCI Slot:
    - Most wireless adapter cards are installed in PCI (Peripheral Component Interconnect) slots on the motherboard. Identify an available PCI slot that matches the card's interface (PCI, PCIe).
  - ✓ Install the Wireless Adapter:

- Gently insert the wireless adapter into the chosen PCI slot. Make sure it is properly aligned with the slot and press it down firmly until it clicks into place.

✓ Secure the Wireless Adapter: Use the screw or clip provided with the wireless adapter to secure it in place. This helps prevent the card from coming loose.

✓ Close the Computer Case: Once the wireless adapter is securely in place, close the computer case and secure it using the appropriate screws.

✓ Connect Antennas (if applicable): If your wireless adapter card requires external antennas, attach them to the designated connectors on the card.

✓ Power On Your Computer: Reconnect the computer to the power source and turn it on.

✓ Install Drivers:

- After booting up, insert the driver installation disc that came with the wireless adapter or download the latest drivers from the manufacturer's website. Follow the installation instructions.

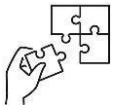
✓ Configure the Wireless Network: Once the drivers are installed, you may need to configure the wireless network settings. This can usually be done through the operating system's network settings.

✓ Test the Connection: Ensure that the wireless adapter is working correctly by connecting to a wireless network and testing the internet connection.



**Points to Remember**

- Network adapter card : It is an expansion card that enables a computer to connect to a network
- To install wireless adapter card:
  - Choose the Right Wireless Adapter
  - Power Off Your Computer
  - open the Computer Case
  - Identify an Appropriate PCI Slot
  - Install the Wireless Adapter
  - Secure the Wireless Adapter
  - Close the Computer Case
  - Connect Antennas (if applicable)
  - Power On Your Computer
  - Install Drivers
  - Configure the Wireless Network
  - Test the Connection



**Application of learning 4.2.**

The TAZNE Company has been experiencing intermittent connectivity issues and performance degradation across its wireless network. In addition to connectivity problems, some users have reported software-related issues, such as difficulty in connecting to the VPN, failure of applications to synchronize data, and unexpected software crashes. The company is also concerned that some hardware components are not functioning as expected, which might be contributing to the network issues.

As a network technician you are tasked with diagnosing and resolving the issues to ensure seamless and secure wireless connectivity for all employees

## Indicative content 4.3: Elaboration of Maintenance Report

**Duration: 2**

### Theoretical Activity 4.3.1: Identification of maintenance report elements

**Tasks:**

1. Answer the following questions:
    a. What is a maintenance report?
    b. Explain the purpose for making a maintenance report
    c. identify the element of a maintenance report
2. Discuss on the above questions and write your answers on the flipchart/paper
3. Present your findings in front of a whole class
4. Listen to the expert view and clarifications from the trainer
5. Ask questions if necessary
6. Read the key readings 4.3.1 in this manual

### Key readings 4.3.1.: Identification of maintenance report elements

- A maintenance report is a detailed document that covers any routine or emergency maintenance services performed on a specific vehicle, piece of equipment, or machine. It serves as a guide outlining the proper care of the equipment in your workplace and can help your company avoid duplicate service requests.
- A maintenance report serves the purpose of documenting and summarising information related to maintenance activities performed on equipment, machinery, facilities, or other assets
- Elements of Maintenance report: Certainly! A maintenance report typically includes several key elements that provide a comprehensive overview of the maintenance activities conducted. Here's an explanation of each element:
    - Title and Date:
        - Explanation: This section simply states the title of the report, such as "Maintenance Report," and includes the date when the maintenance activities took place. It provides a quick reference point for the timeframe of the report.
    - Introduction:
        - Explanation: The introduction provides a brief overview of the maintenance work, including the equipment or systems involved and the

overall objectives. It sets the context for the reader and helps them understand the purpose of the maintenance activities.

- Maintenance Team Information:
  - Explanation: This section lists the names or roles of the individuals or teams involved in the maintenance. Including contact information helps in case further clarification or communication is needed.

✓ Equipment/System Details:
  - Explanation: Specify the details of the equipment or systems that underwent maintenance. This includes identification numbers, model names, and any other relevant information that uniquely identifies the equipment.

✓ Maintenance Schedule:
  - Explanation: Outline the planned maintenance schedule, detailing the start and end times. If there were any deviations from the original plan, they should be noted in this section.

✓ Maintenance Tasks:
  - Explanation: Provide a detailed description of each maintenance task performed. This section outlines the specific activities undertaken during the maintenance, such as inspections, repairs, replacements, or calibrations.

✓ Issues Encountered: Explanation: Document any issues or anomalies discovered during the maintenance process. Describe the nature of the problems, their severity, and the potential impact on the equipment or system.

✓ Actions Taken: Explanation: This section details the actions taken to address the identified issues. It may include descriptions of repairs, part replacements, adjustments, or any other corrective measures undertaken during the maintenance.

✓ Materials and Resources Used:
  - Explanation: List the materials, tools, and resources utilized during the maintenance.
  - This includes spare parts, consumables, and any specialized equipment necessary for the tasks.

✓ Testing and Verification:
  - Explanation: Describe any testing or verification procedures conducted to ensure that the equipment or systems are functioning correctly after maintenance. This ensures that the maintenance activities were successful.

✓ Recommendations: Provide recommendations for future maintenance or improvements based on the findings of the current maintenance activities. This

could include suggestions for preventive measures to avoid similar issues in the future.

- ✓ Conclusion: Summarise the overall outcomes of the maintenance. Acknowledge successful tasks and note any outstanding issues or concerns. This section provides a concise wrap-up of the maintenance activities.
- ✓ Signatures and Approvals: Include signatures of individuals involved in the maintenance, such as technicians, supervisors, or managers. If there are any necessary approvals from relevant stakeholders, they should be documented in this section.



**Points to Remember**

- Maintenance report: is a detailed document that covers any routine or emergency maintenance services performed on a specific vehicle, piece of equipment, or machine. It serves the purpose of documenting and summarising information related to maintenance activities performed on equipment, machinery, facilities, or other assets
- Elements of Maintenance report
  - ✓ Title and Date Introduction
  - ✓ Maintenance Team Information
  - ✓ Equipment/System Details
  - ✓ Maintenance Schedule
  - ✓ Maintenance Tasks
  - ✓ Issues Encountered
  - ✓ Actions Taken
  - ✓ Materials and Resources Used
  - ✓ Testing and Verification
  - ✓ Recommendations
  - ✓ Conclusion
  - ✓ Signatures and Approvals



**Practical Activity 4.3.2: Elaborate maintenance report**




**Task:**

1. Read carefully this task:
   You are supposed to elaborate the maintenance report of the activity performed on practical activity 4.1 and 4.2 using report template given by your trainer.
2. Elaborate report of the activity performed on practical activity 4.1 and 4.2 using the summary and the report template given by your trainer under the guidance of your trainer
3. Check if all needed information is included in your report
4. Read key readings and do the application of learning 4.3



**Key readings 4.3.1.: Elaborate maintenance report**

- To elaborate a maintenance report means to create a detailed and comprehensive document that outlines the process of maintaining, diagnosing, and resolving issues in a system (such as a network).
- The report typically includes:
  - ✓ Problem Identification: A clear description of the issues found during maintenance.
  - ✓ Actions Taken: Detailed steps taken to fix or improve the system, including any troubleshooting, repairs, or configurations.
  - ✓ Findings: The results of diagnostic tests or inspections, including any discovered issues or faults.
  - ✓ Recommendations: Suggestions for future improvements, maintenance schedules, or preventive measures to avoid recurring issues.
  - ✓ Summary: A concise conclusion that summarizes the overall health of the system after maintenance.
- The Network Maintenance Report Template

_____

1. Network Overview

Network Name: _____

Location: _____

Date of Maintenance: _____

Technician Name: _____

Network Components (Routers, Access Points, Switches, etc.):

Router Model: _____

Access Point Model: _____

Other Devices: _____

_____

2. Maintenance Summary

Reason for Maintenance (Routine check, troubleshooting, upgrade, etc.):

_____

Description of Issue(s):

_____

(e.g., slow network speeds, dropped connections, signal interference, etc.)

_____

3. Diagnostics and Findings

Testing Tools Used:

(Wi-Fi analyser, speed test, ping test, etc.)

Tool 1: _____

Tool 2: _____

Test Results:

Signal Strength: _____ (e.g., -70dBm)

Internet Speed: _____ (e.g., 50 Mbps download / 20 Mbps upload)

Ping Results: _____ (e.g., 30ms latency)

Issues Identified:

Issue 1: _____

Issue 2: _____

_____

4. Actions Taken

Step-by-Step Description of Actions:

Action 1: _____

(e.g., changed wireless channel from 6 to 11)

Action 2: _____

(e.g., repositioned router to reduce interference)

Hardware Replaced (if applicable):

Equipment Name: _____

Reason for Replacement: _____

_____

5. Results After Maintenance

Network Performance Improvements:

Signal Strength: _____ (e.g., Improved from -70dBm to -60dBm)

Internet Speed: _____ (e.g., Increased from 50 Mbps to 80 Mbps)

Ping Results: _____ (e.g., Reduced latency from 30ms to 15ms)

_____

6. Recommendations

Short-Term Recommendations (immediate actions to improve performance):

Recommendation 1: _____

Recommendation 2: _____

Long-Term Recommendations (upgrades, replacements, monitoring):

Recommendation 1: _____

Recommendation 2: _____

Next Maintenance Date: _____

_____

7. Additional Comments

Any other observations, challenges, or suggestions:

_____

_____

8. Attachments (if applicable)

Screenshots of test results, network configurations, etc.

Diagrams of the network layout or updated configurations.

Photos of replaced equipment.

_____

Signature:

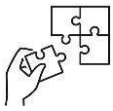
Technician: _____

Date: _____



**Points to Remember**

- While elaborating maintenance report, Use clear and concise language to ensure the report is easily understandable
- Clearly state the problems or issues identified during the maintenance.
- Include all steps taken to address the issues, including configurations, repairs, or adjustments.
- Suggest future actions or preventive measures to enhance system performance and avoid recurring issues
- Schedule a follow-up date or provide recommendations for regular checks to maintain network health.



**Application of learning 4.3.**

The network specialist of BFD ltd is trying to solve network problems by troubleshooting the Wireless station connection to AP, after troubleshooting he found out that an issue is based on signal interference. After finding and correcting the issue he need you to help him in writing a report based on the details given by the company network specialist.



## Learning outcome 4 end assessment

**Theoretical assessment**

1. Which of the following is a common task in preventive maintenance?
   a. Replacing faulty hardware
   b. Performing regular backups
   c. Resolving security breaches
   d. Addressing user complaints
2. What is the primary goal of corrective maintenance?
   a. To prevent potential issues
   b. To improve system performance
   c. To repair and resolve existing issues
   d. To document maintenance activities
3. Which section of a maintenance report includes recommendations for future actions?
   a. Introduction
   b. Preventive Maintenance
   c. Corrective Maintenance
   d. Findings and Recommendations
4. Which of the following is NOT typically included in a preventive maintenance plan?
   a. Regular updates and patches
   b. System performance tuning
   c. Emergency repairs
   d. Hardware inspections
5. Answer by true or false
   a. Preventive maintenance aims to identify and fix problems before they cause significant downtime.
   b. Corrective maintenance involves regular system updates and patches.
   c. A maintenance report should include a summary of both preventive and corrective maintenance activities.
   d. Preventive maintenance is performed only after a system failure occurs.
   e. The conclusion section of a maintenance report is used to reflect on the effectiveness of maintenance activities.

6. **Match the Maintenance Activities with their maintenance type**

| answer | Maintenance activities | Maintenance type |
|---|---|---|
| 1……. | 1.Inspecting and cleaning hardware | a. Preventive Maintenance |
| 2……. | 2.Replacing faulty components | b. Corrective Maintenance |
| 3…….. | 3.Performing data backups | c. Both |
| 4…….. | 4.Documenting resolved issues | |
| 5…….. | 5.Applying security patches | |

**Practical assessment**

XYZ Corporation, a large manufacturing company located in Kigali, has a complex IT infrastructure supporting its production and administrative functions. The network comprises various routers, switches, servers, and wireless access points. Recently, the company has faced unstable network connections and hardware failures. As the network technician the IT department, you have been tasked with ensuring the smooth operation of the network by performing both preventive and corrective maintenance. Additionally, you are required to document and elaborate on the maintenance activities in a comprehensive maintenance report.

The tasks you are requested to perform are as follow:

a. Inspect and clean all network equipment (routers, switches, servers, access points).
b. Update firmware and software on all network devices to the latest versions
c. Ensure all configurations are backed up and documentation is up to date.
d. Perform regular security audits and update security protocols as necessary
e. Identify and rectify existing network issues by Diagnosing intermittent network issues reported by users
f. Address any detected security vulnerabilities.
g. Restore any corrupted or lost data from backups then after you have to document all maintenance activities in a detailed report.

END



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